



Image credit: Marguerite Tonjes

Statistics *speed-run*

Nick Smith

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Goal

To give you plausible familiarity with sentences like:

“An observed (expected) upper limit is placed on the signal strength μ , using the profile likelihood ratio test statistic, following the CL_s criterion, under asymptotic assumptions, and found to be ...”

Plan:

- Probability, likelihood, and inference
 - Bayesian inference
 - Maximum likelihood point estimation
- Frequentist hypothesis tests
 - Likelihood ratio test statistic
 - Under-fluctuation and CLs
 - Asymptotic behavior
- Adding uncertainties
 - Statistical model with auxiliary measurements
 - Profiling and marginalizing nuisance parameters

Probability, likelihood, and inference

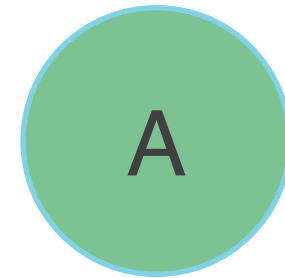
Probability

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- Kolmogorov axioms: for a sample space S , we have

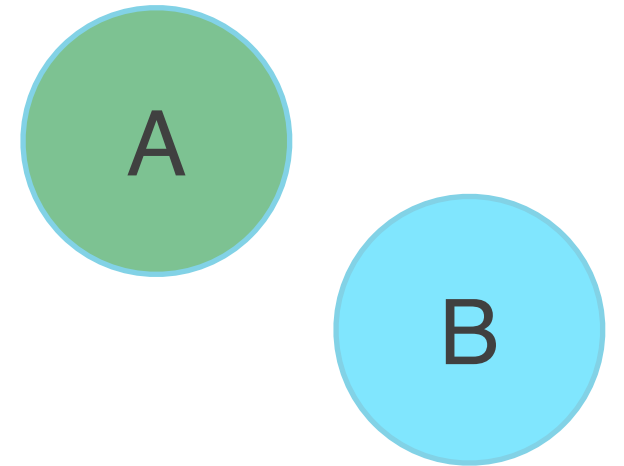
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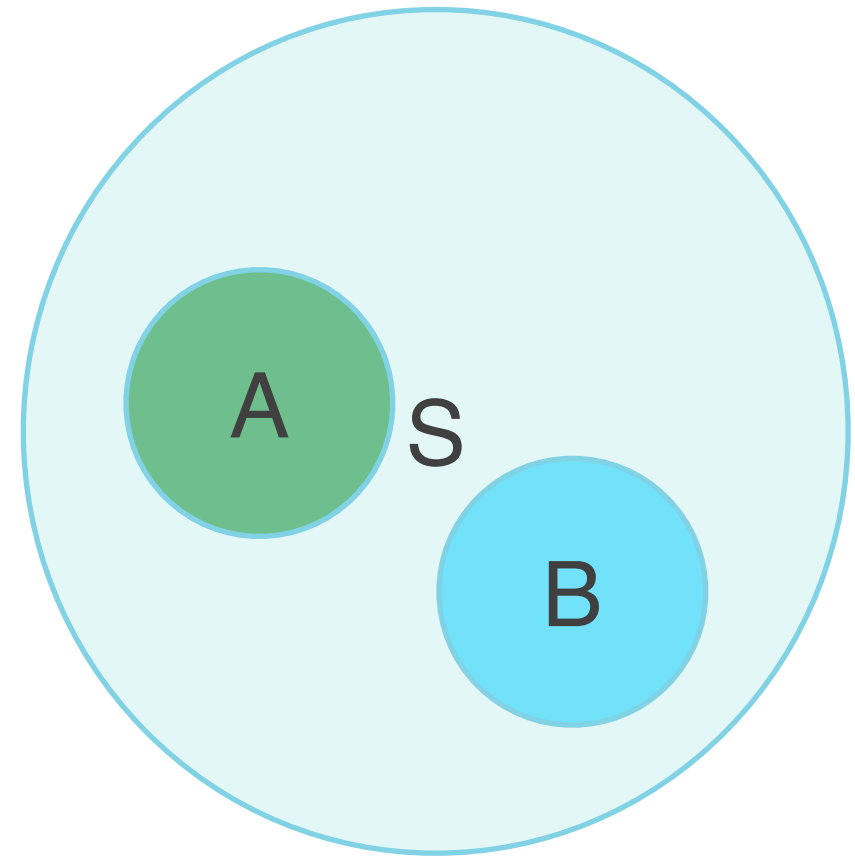
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 - $\forall A, B \subset S, A \cap B = \emptyset \quad P(A \cup B) = P(A) + P(B)$



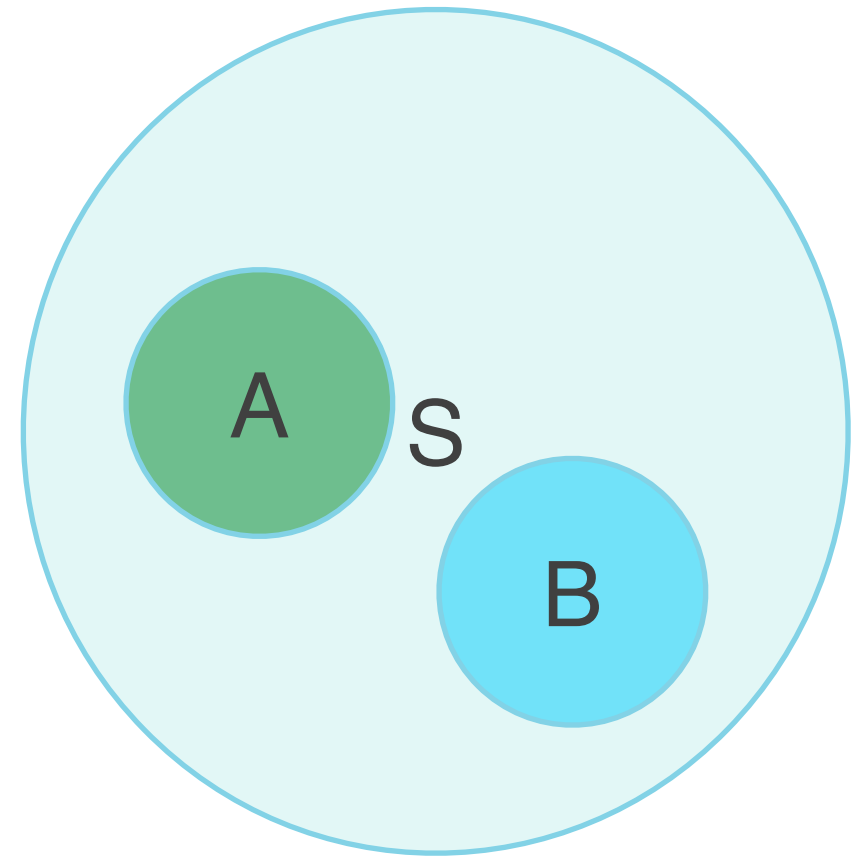
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- Conditional probability



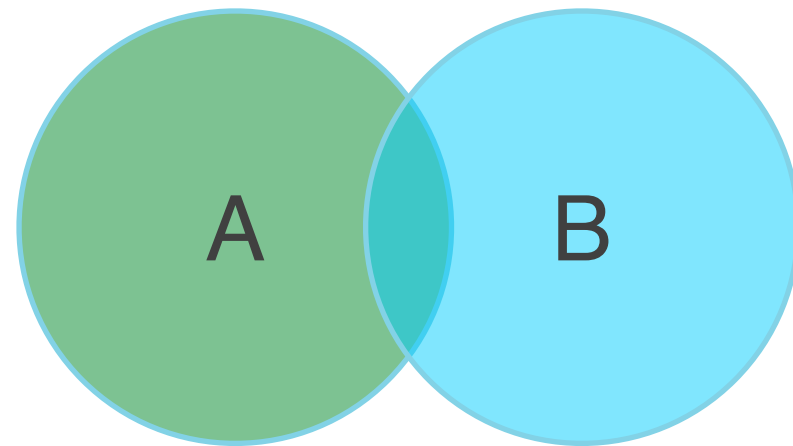
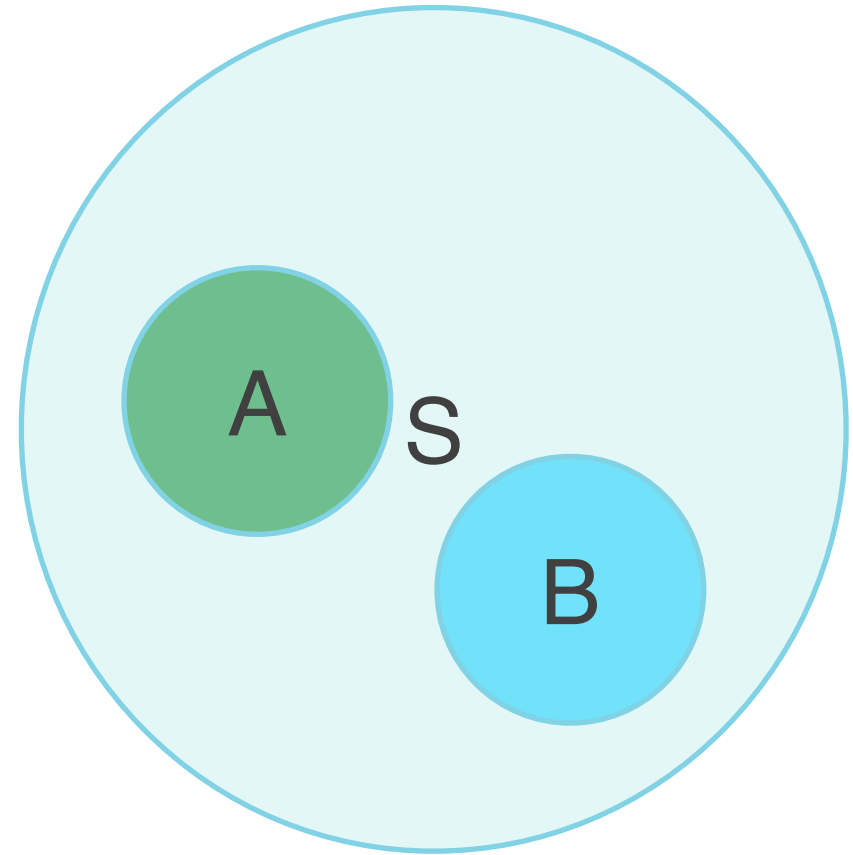
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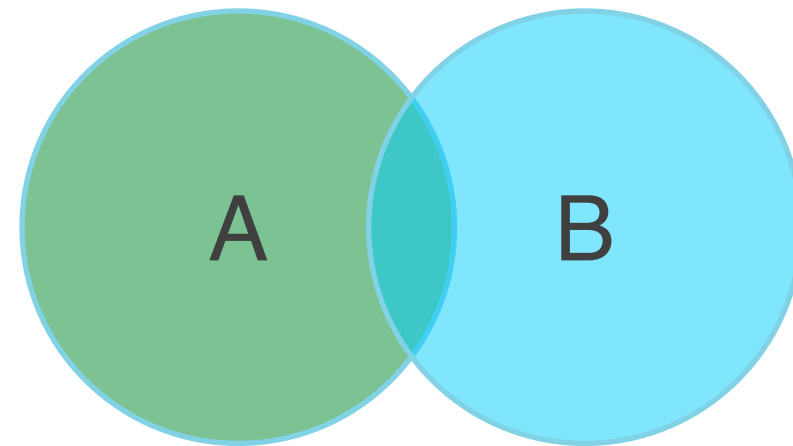
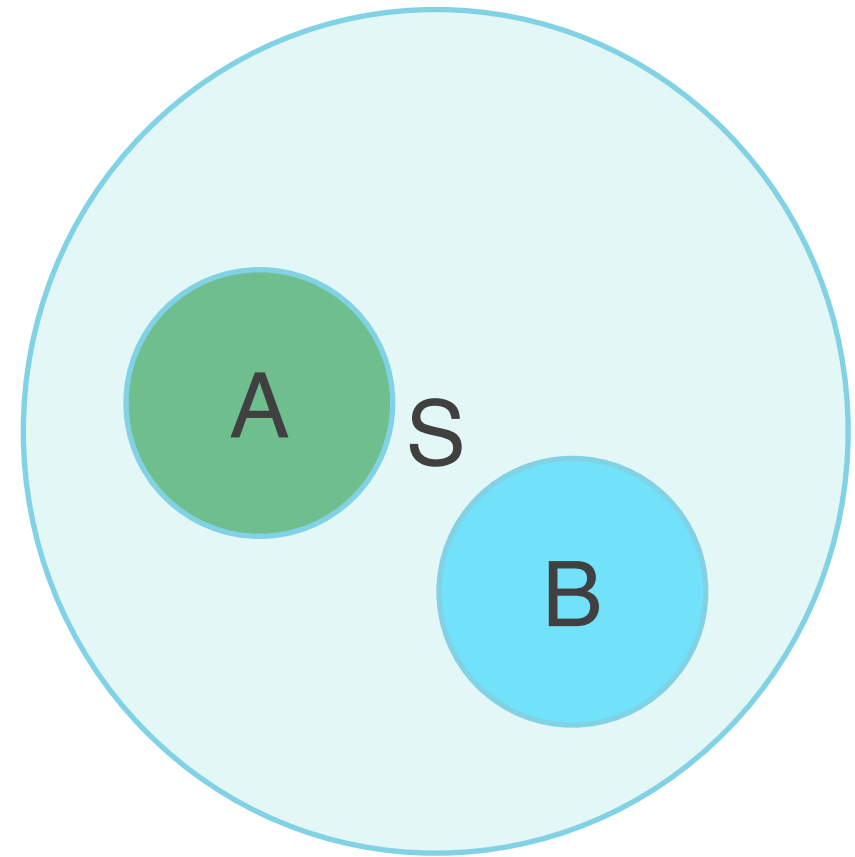
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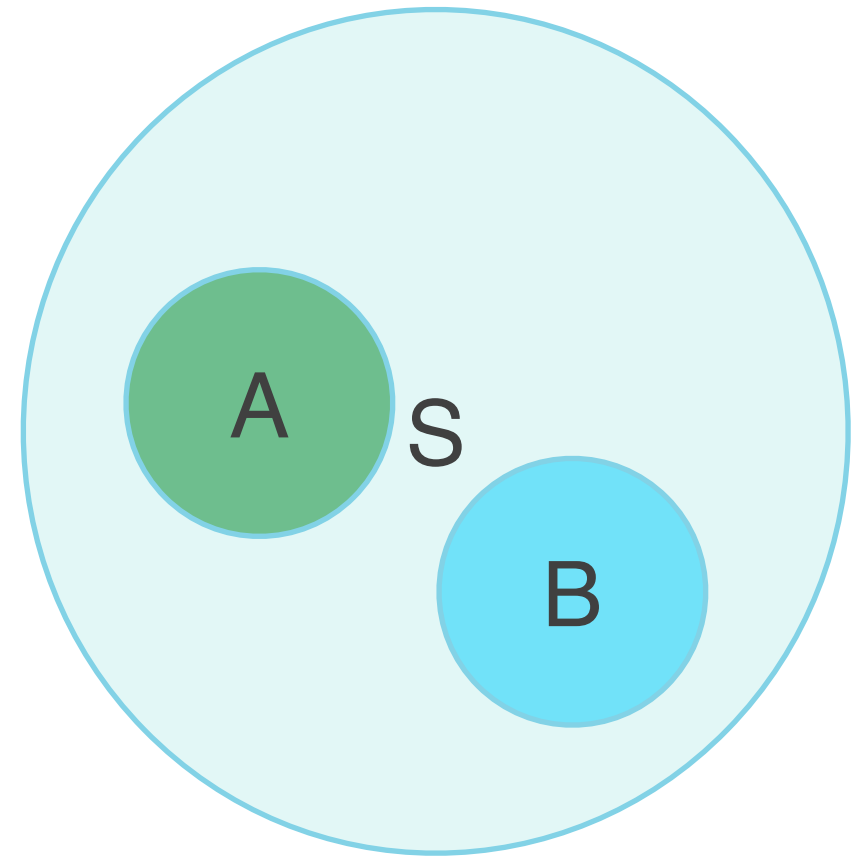
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Probability

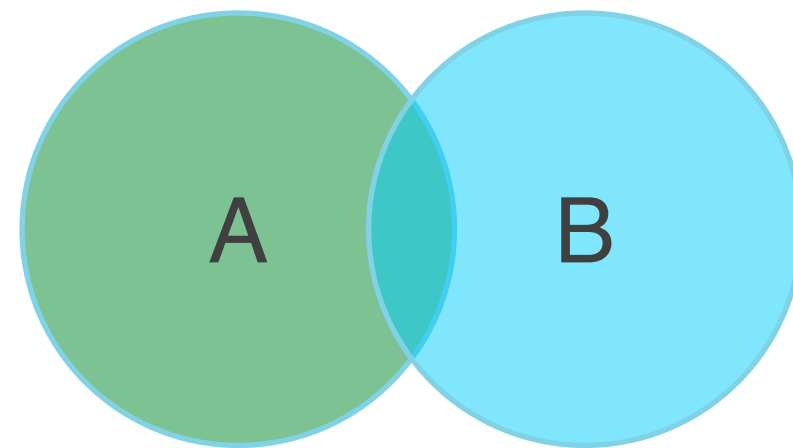
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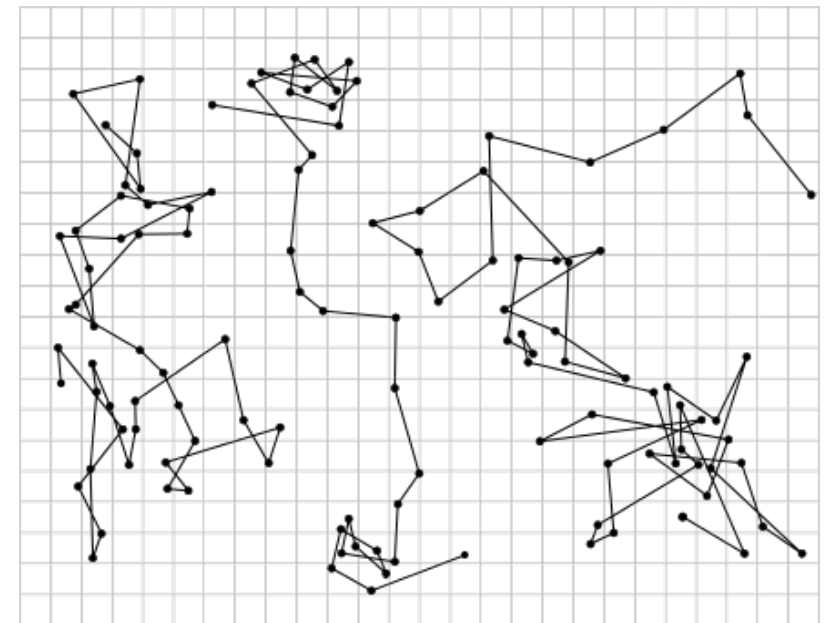
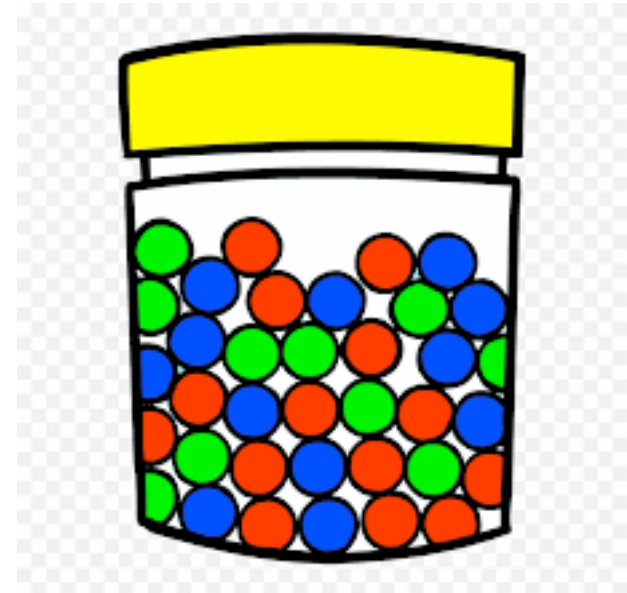
- For a partition $S = \bigcup_i A_i$ where $i \neq j \implies A_i \cap A_j = \emptyset$,
$$P(B) = \sum_i P(B | A_i)P(A_i)$$

Probability density / mass

- Probability mass function (pmf)
 - probability of observing a specific outcome
- Probability density function (pdf), e.g. $P(x)$
 - $P(x) dx = \text{differential probability of observing an outcome}$
- In both cases:
 - defined over a space of outcomes/observables/samples
 - imply a cumulative (cdf), percentile (inverse cdf), etc. in 1D
 - may be parameterized

Examples

- Marbles: $P(\text{draw 2 red, 2 green, 1 blue from jar})$
 - pmf [Multivariate hypergeometric distribution](#)
- Brownian motion: $P(\text{displacement after some time})$
 - pdf [Normal distribution](#)
- Counts in a particle detector after some time
 - pmf [Poisson distribution](#)



Distributions of interest

- Poisson distribution

- $$P(n | \lambda) = \frac{\lambda^n e^{-\lambda}}{n!}$$

- Normal distribution

- $$P(x | \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Central limit theorem:

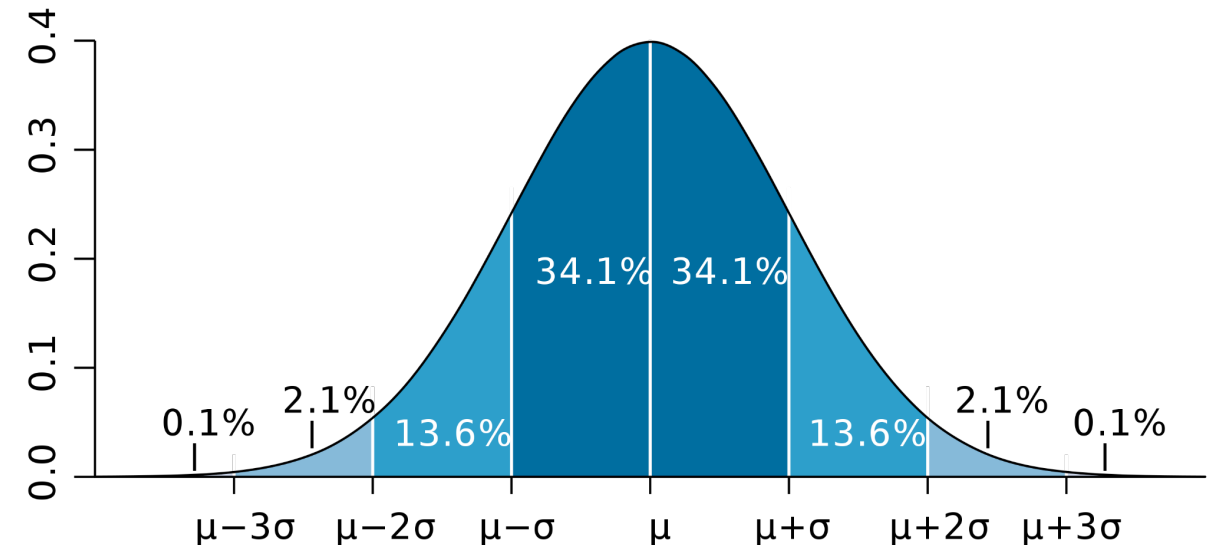
- sums of independent random-distributed variables tend towards a Normal-distributed variable

- Standard (Z) score:

- Convention for interesting percentiles: “1σ” = 0.6827..., “2σ” = 0.9545..., “5σ” = 1-5.7e-7
 - These are 2-sided. Can also define 1-sided (half)
- Often quote 95 %-ile

- Log-normal distribution

- Definition: Normal in log-space (change of variables: $y = \ln(x)$, $dy = x^{-1}dx$)
- Corollary to central limit theorem:
 - *products* of [...] tend towards a Log-normal distributed variable
 - Common model for calibration uncertainties (more later)



Poisson process

- In CMS, collision events occur at a rate $\lambda(x, t) = L(t) \sigma_{pp \rightarrow X}(x) \epsilon(x, t)$
 - Where (for now ignoring the model parameters)
 - $L(t)$ is the instantaneous luminosity
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 - in time and “observable space” (e.g. muon 4-momentum, etc.)

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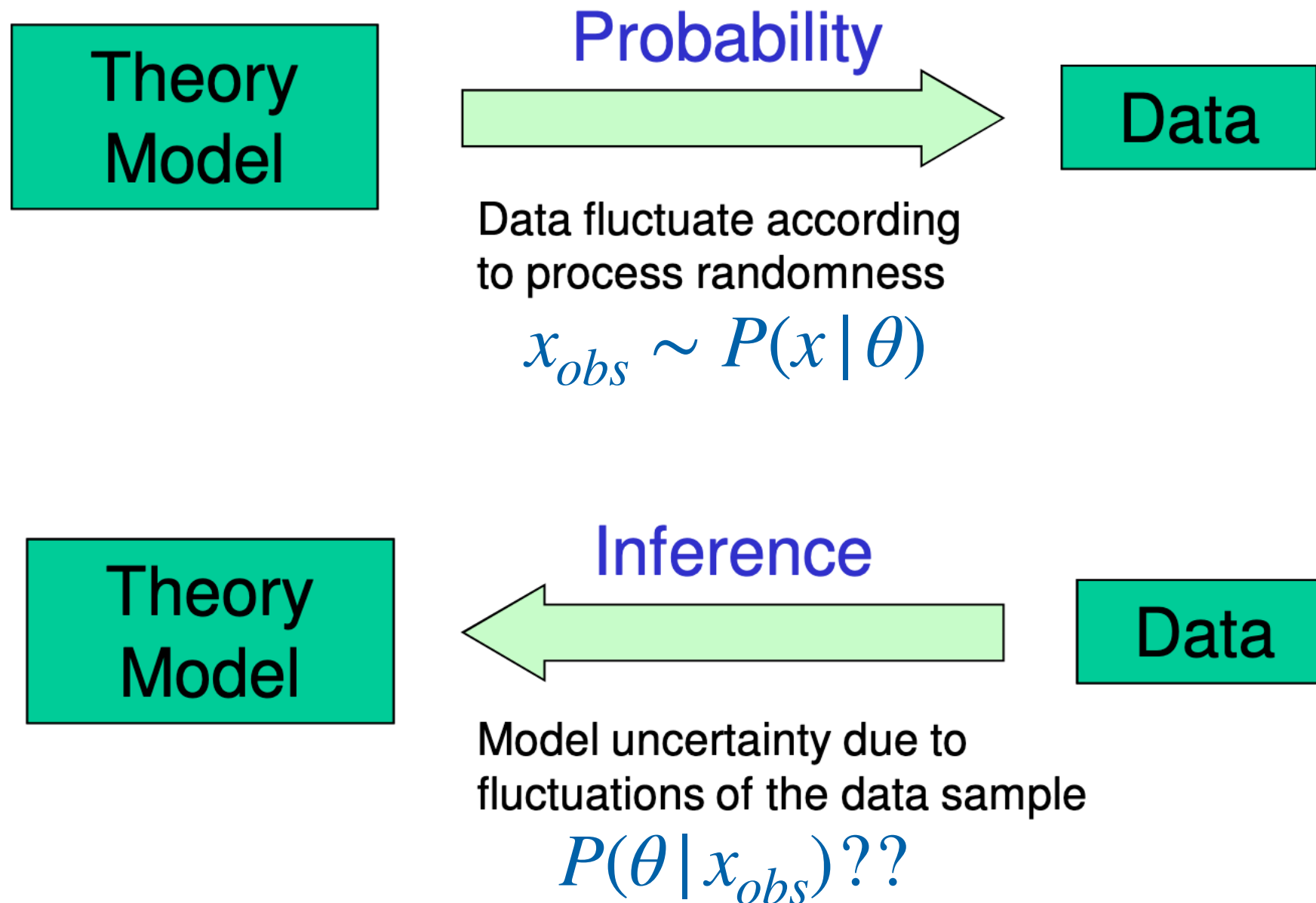
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- This is a Poisson Process
 - Binned model: overall PDF is a joint distribution (product) over disjoint regions
- $$P(\text{data} | \text{model}) = \prod_i P(N_i | \Lambda_i)$$
- Un-binned model: conditional on N, λ can be interpreted as a PDF (integrating t)

$$P(\text{data} | \text{model}) = P(N | \Lambda) \prod_i^N \lambda(x_i) dx_i$$

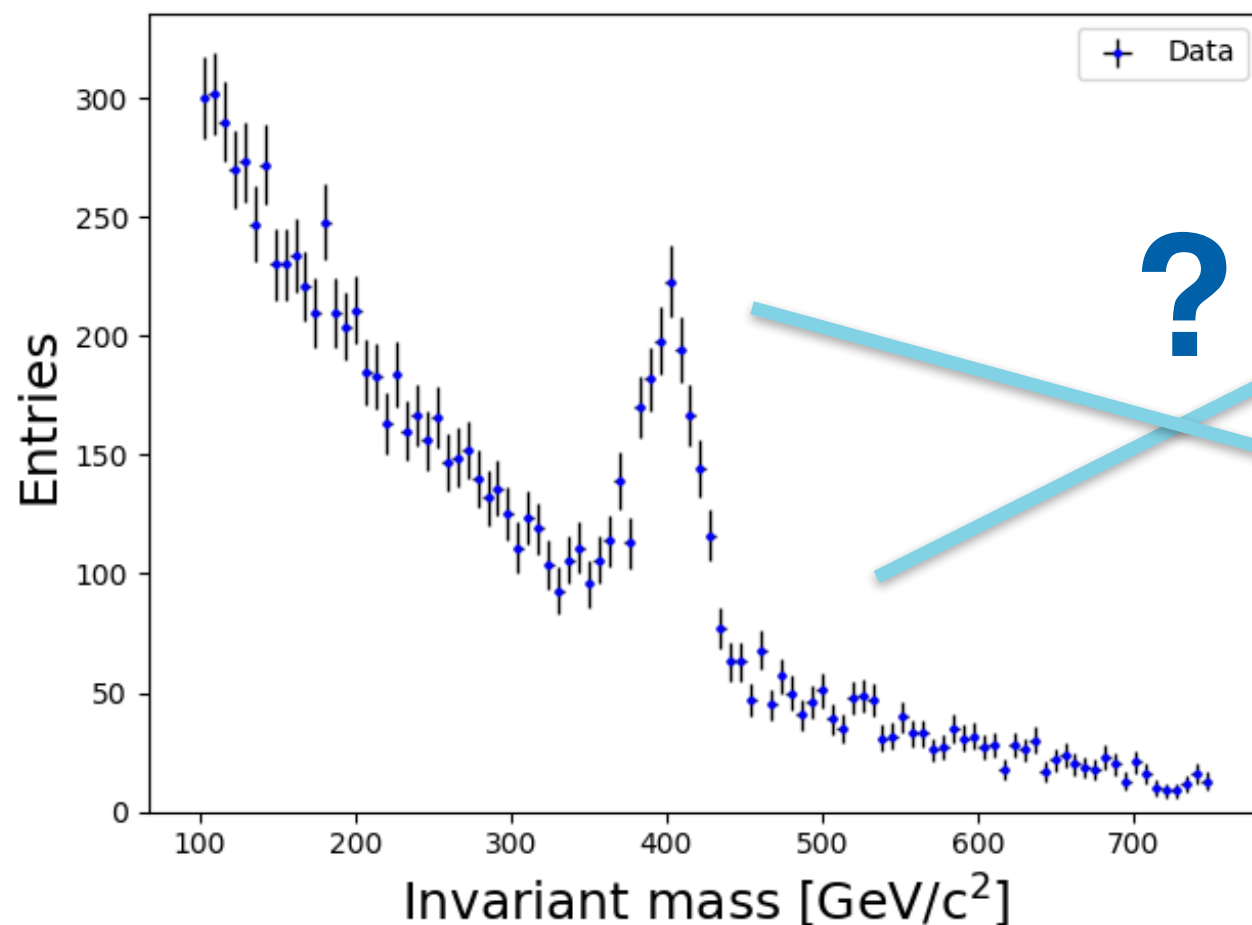
(actually just the limit as bin size goes to zero:
[R. Barlow, "Extended maximum likelihood"](#))

Inference

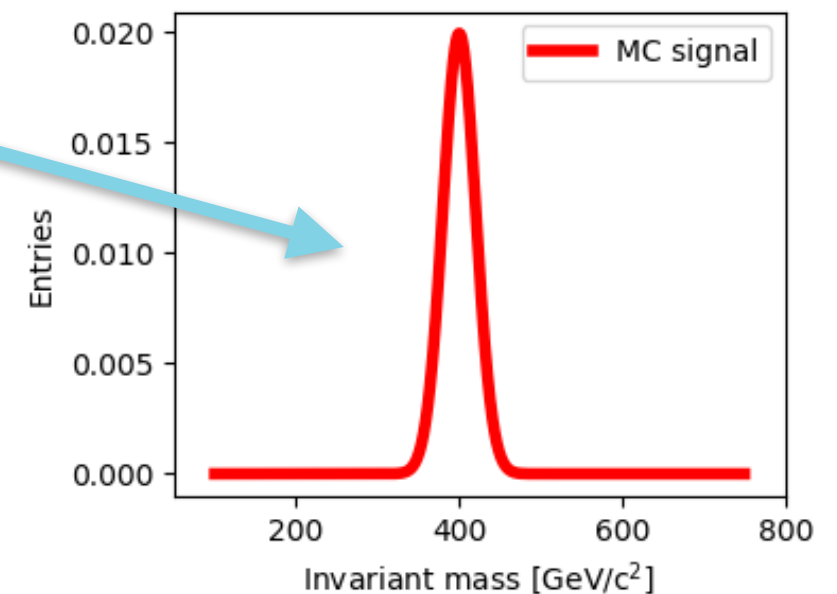
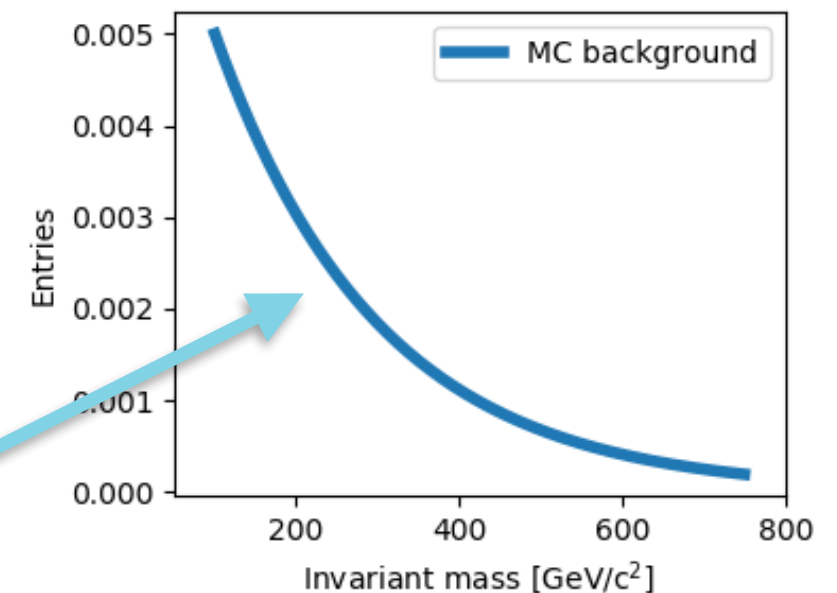


Inference example

- Given this data and a model for signal and background, I might infer:
 - The amount of signal present (a *parameter of interest*, or POI)
 - The functional form of the background, if a-priori unknown
 - Parameterized by *nuisance parameters*
 - We will discuss those later

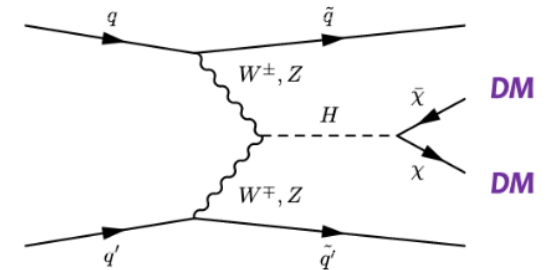
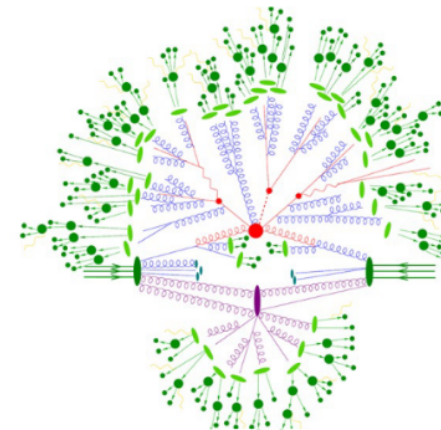
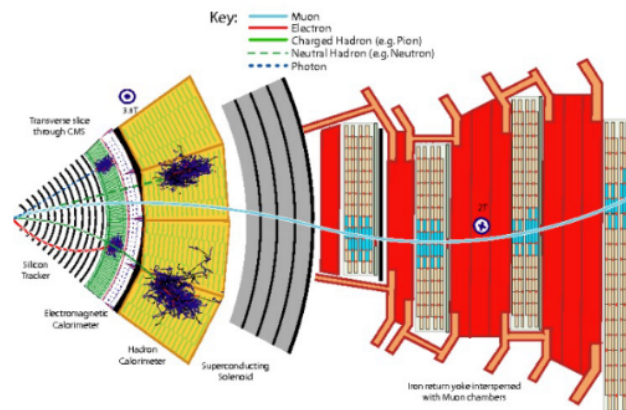
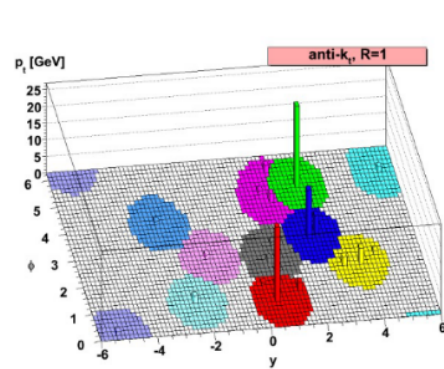
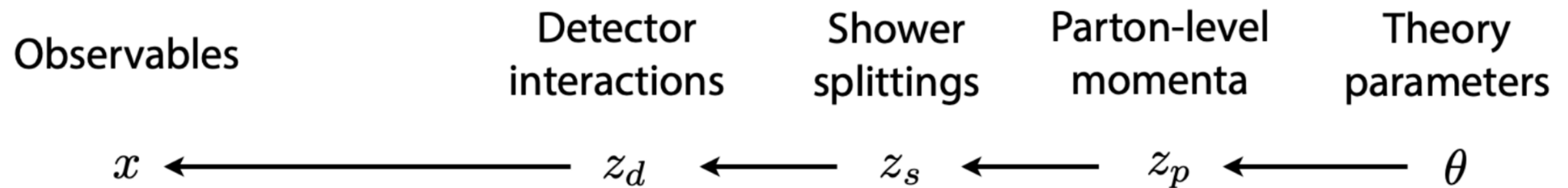


?



Inference for full simulation

- The whole picture is more complex
 - We often cannot compute $P(x | \theta)$, but we can efficiently sample it
 - surrogate model using Monte Carlo (MC) estimates of bin yields



$$p(x|\theta) = \int dz_d \int dz_s \int dz_p p(x|z_d)$$

Features

$$p(z_d|z_s)$$

$$p(z_s|z_p)$$

$$p(z_p|\theta)$$

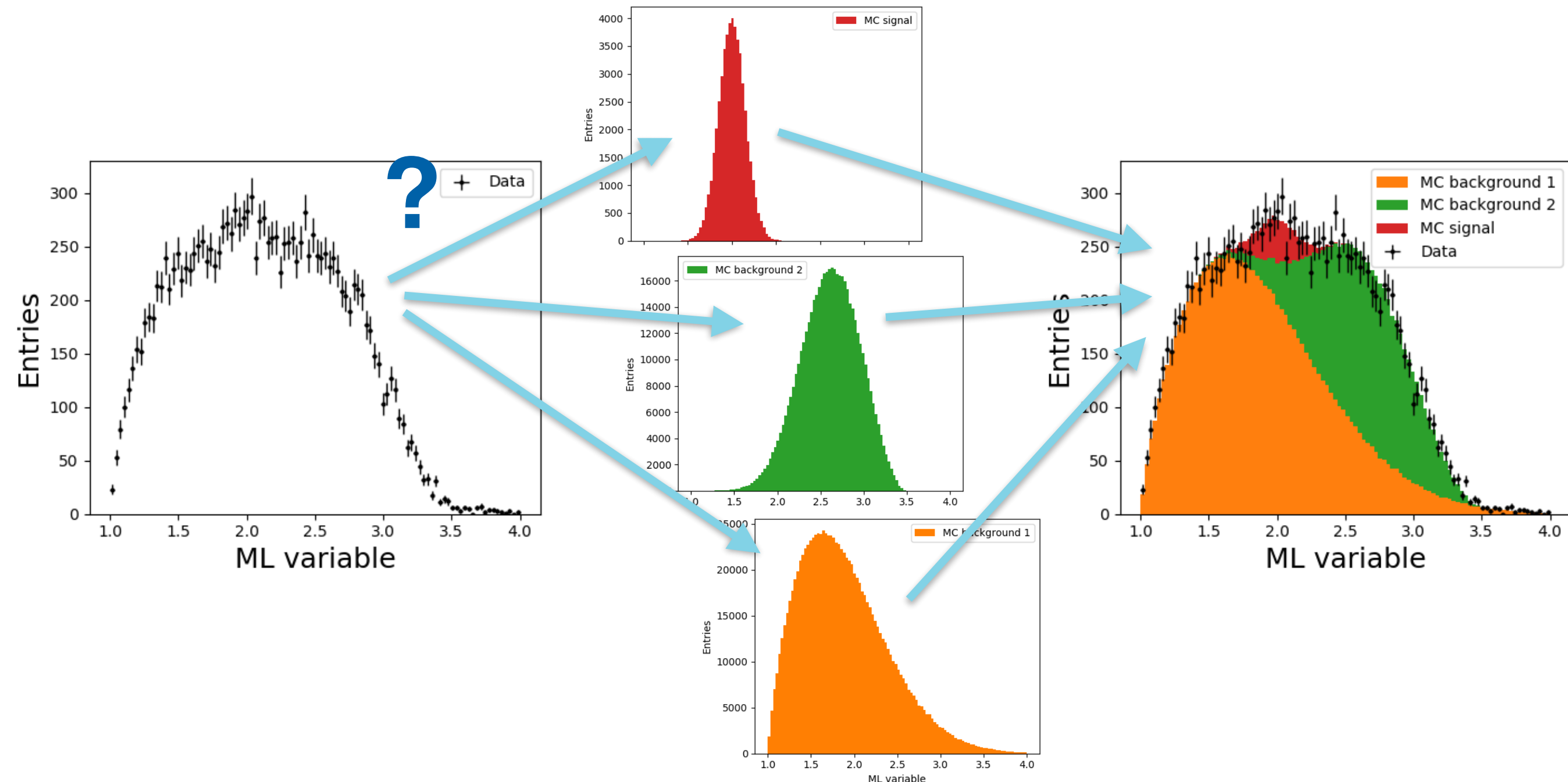
Latent Variables

Model Parameters

diagram: K. Cranmer

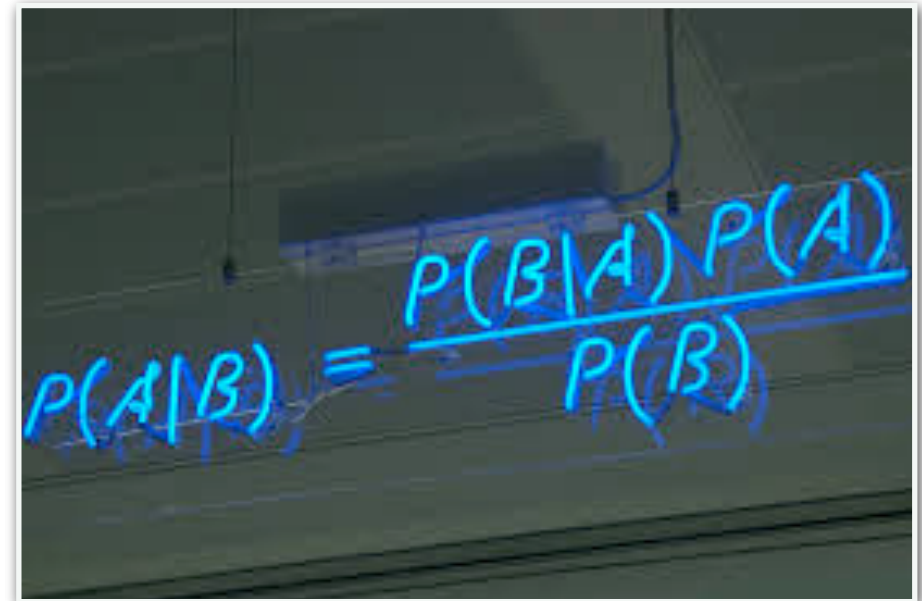
Templates

- We often build models via template histograms derived from MC
 - Typically to infer signal strength μ = normalization of signal template



Bayesian inference

- We would like to infer $P(\theta | x_{obs})$
 - i.e. given our observation x , what is the probability distribution of model parameter θ ?
- Bayes' theorem tells us:
 - $$P(\theta | x_{obs}) = \frac{P(x_{obs} | \theta)P(\theta)}{P(x_{obs})}$$
 - Ok, we have our model $P(x | \theta)$, but what about the other terms?
- $P(\theta)$ is the prior probability distribution for θ
 - Bayesian: we provide this based on our prior belief
 - Or some recipe (objective Bayes, etc.)
 - Frequentist: **no such thing!**
- $P(x_{obs})$ is the evidence
 - Here a normalization:
$$P(x_{obs}) = \int P(x_{obs} | \theta)P(\theta)d\theta$$
 - In practice, can be hard to compute!
 - Better use found in [Bayes' factor](#) to compare models

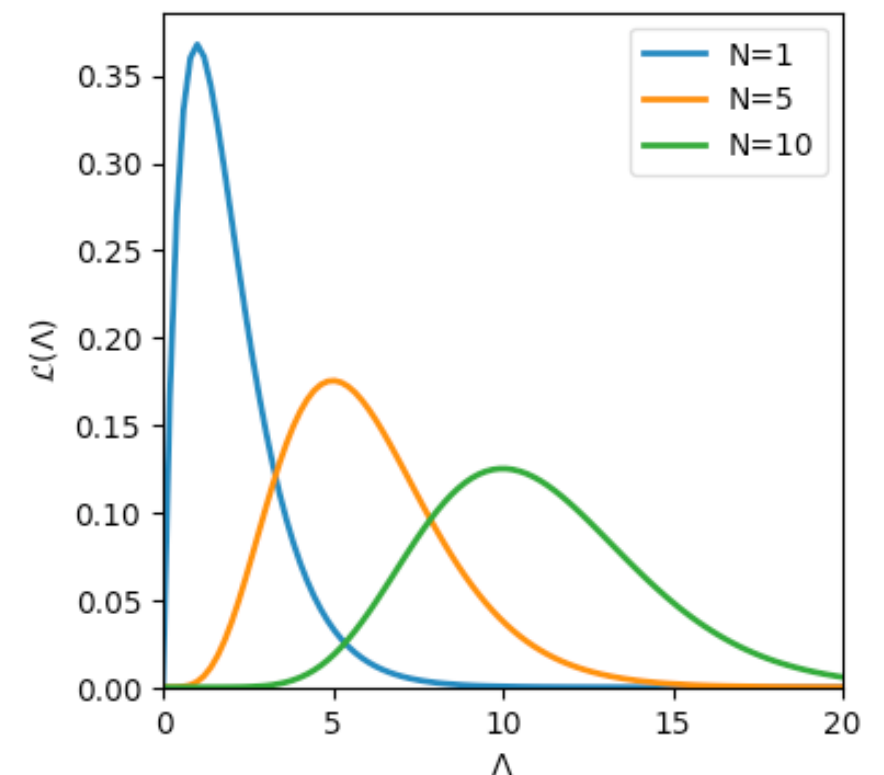


A photograph of a chalkboard with the Bayes' theorem formula written in blue chalk. The formula is $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$. The text is written in a casual, handwritten style.

Maximum likelihood point estimate

- For a fixed *observation*, can define likelihood $\mathcal{L}(\theta) = P(x_{obs} | \theta)$
 - This is a function of θ , but **not** a probability density
 - More exposition in [PDG 40.2](#)
- $\hat{\theta}$ that maximizes this function is the *maximum likelihood estimate* (MLE)
 - $\hat{\theta} = \arg \max_{\theta} [\mathcal{L}(\theta)] = \arg \min_{\theta} [-\ln \mathcal{L}(\theta)]$
 - This is a random variable
 - We usually minimize the negative log-likelihood (NLL) numerically
 - the core job of MINUIT's MIGRAD routine

- Poisson example: $\hat{\Lambda} = N$

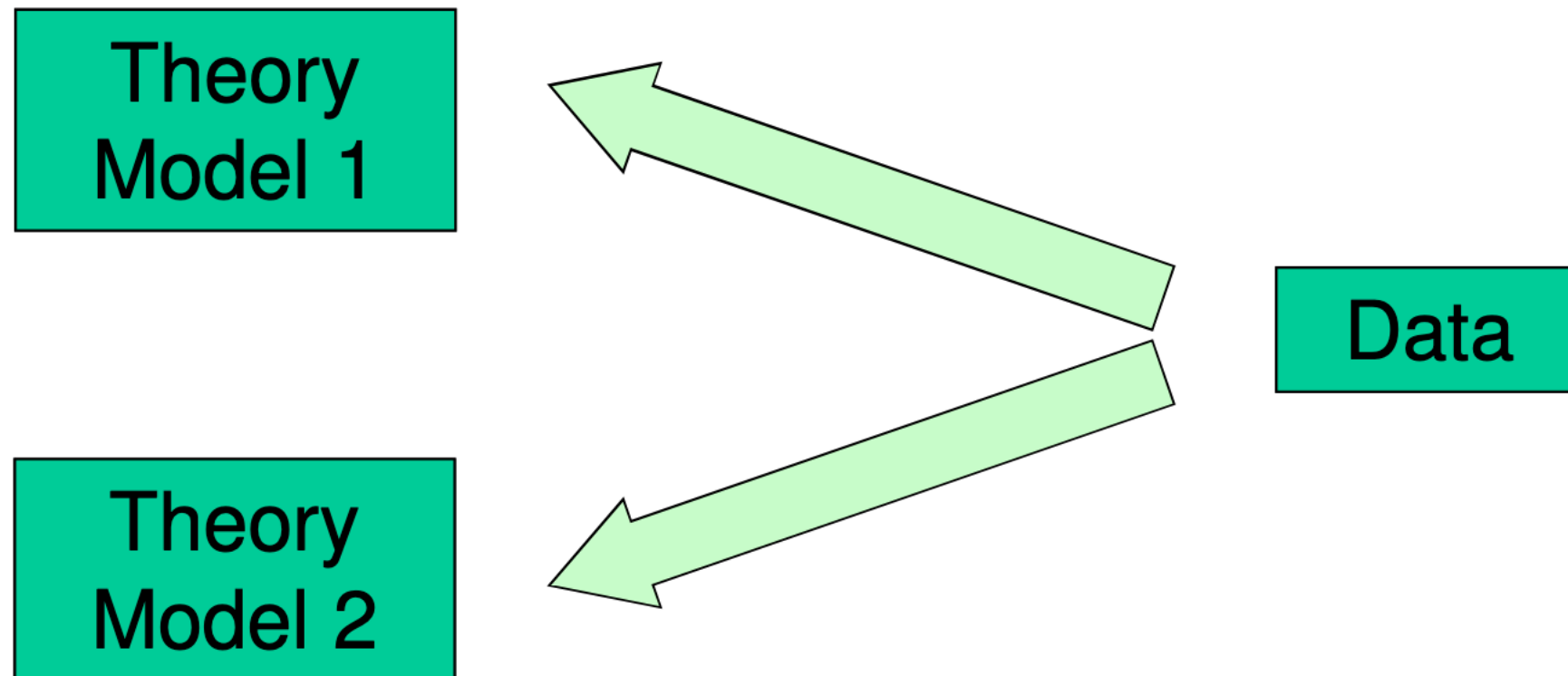


Maximum likelihood trivia

- Absolute value of $\mathcal{L}(\hat{\theta})$ is usually not meaningful
 - Can be calibrated by taking a ratio (chi-square, saturated binned model)
- The MLE has good limiting properties as sample size $\rightarrow \infty$
 - Consistent: sequence of MLEs converges to true value
 - Efficient: variance of MLE saturates the [Cramér–Rao lower bound](#)
 - Distribution of MLE approaches Normal distribution
 - Asymptotically unbiased
 - Bias can exist for finite samples, can be corrected (with increase in variance)
- The likelihood (and it's maximum) is invariant under change of variables
 - Again, it is not a PDF!
- Ok, but this is just a *point*. Can frequentists say more without a prior?

Frequentist hypothesis tests

Hypothesis tests



Which hypothesis is the most consistent with the experimental data?

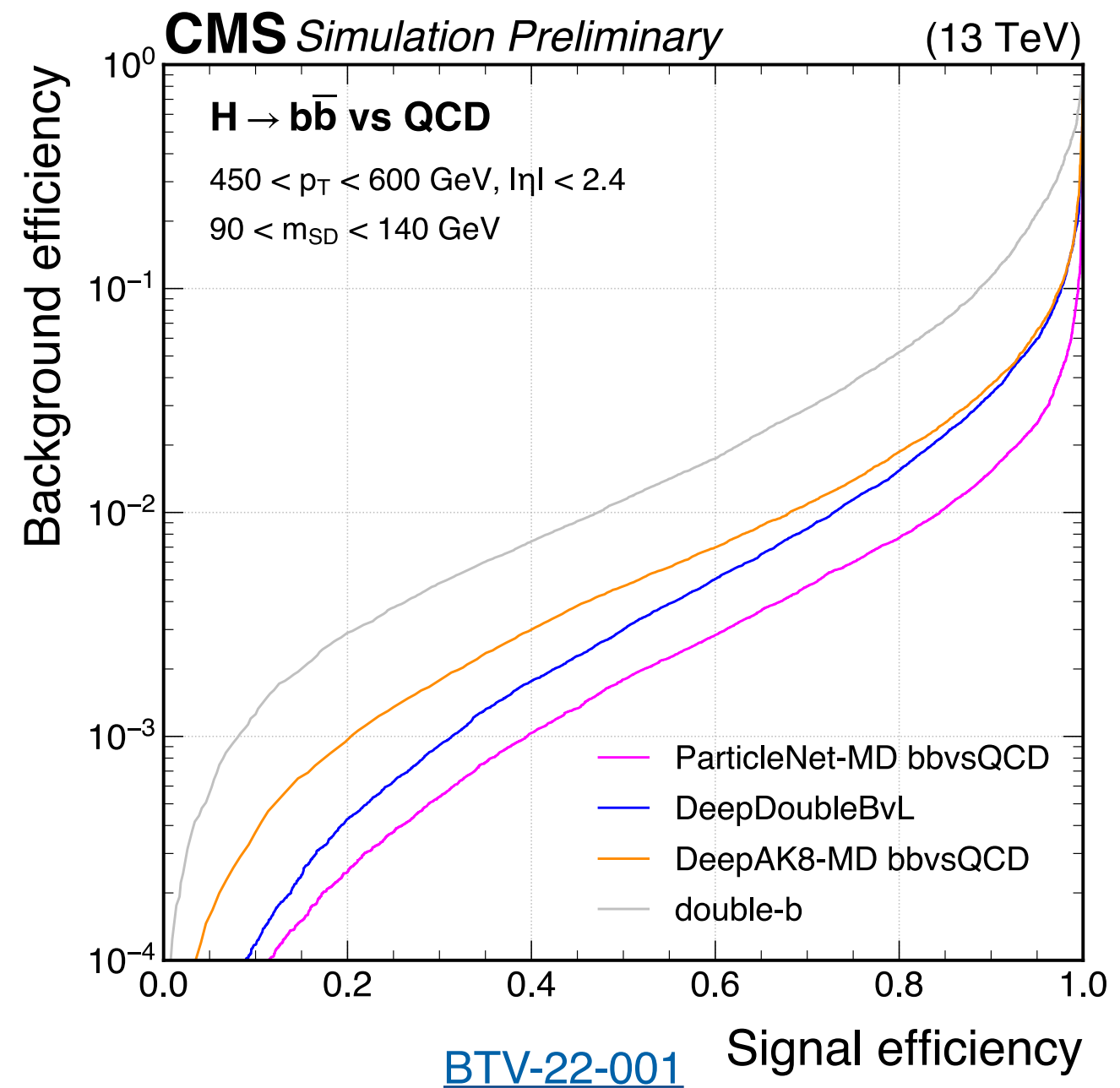
Hypothesis tests

- Simple test parameterized by two probabilities: α , β
 - β with respect to an alternate hypothesis

Table of error types		Null hypothesis (H_0) is	
		TRUE	FALSE
Decision about null hypothesis (H_0)	Don't reject	Correct inference (true negative) (probability = $1-\alpha$)	Type II error (false negative) (probability = β)
	Reject	Type I error (false positive) (probability = α)	Correct inference (true positive) (probability = $1-\beta$)

Hypothesis test example

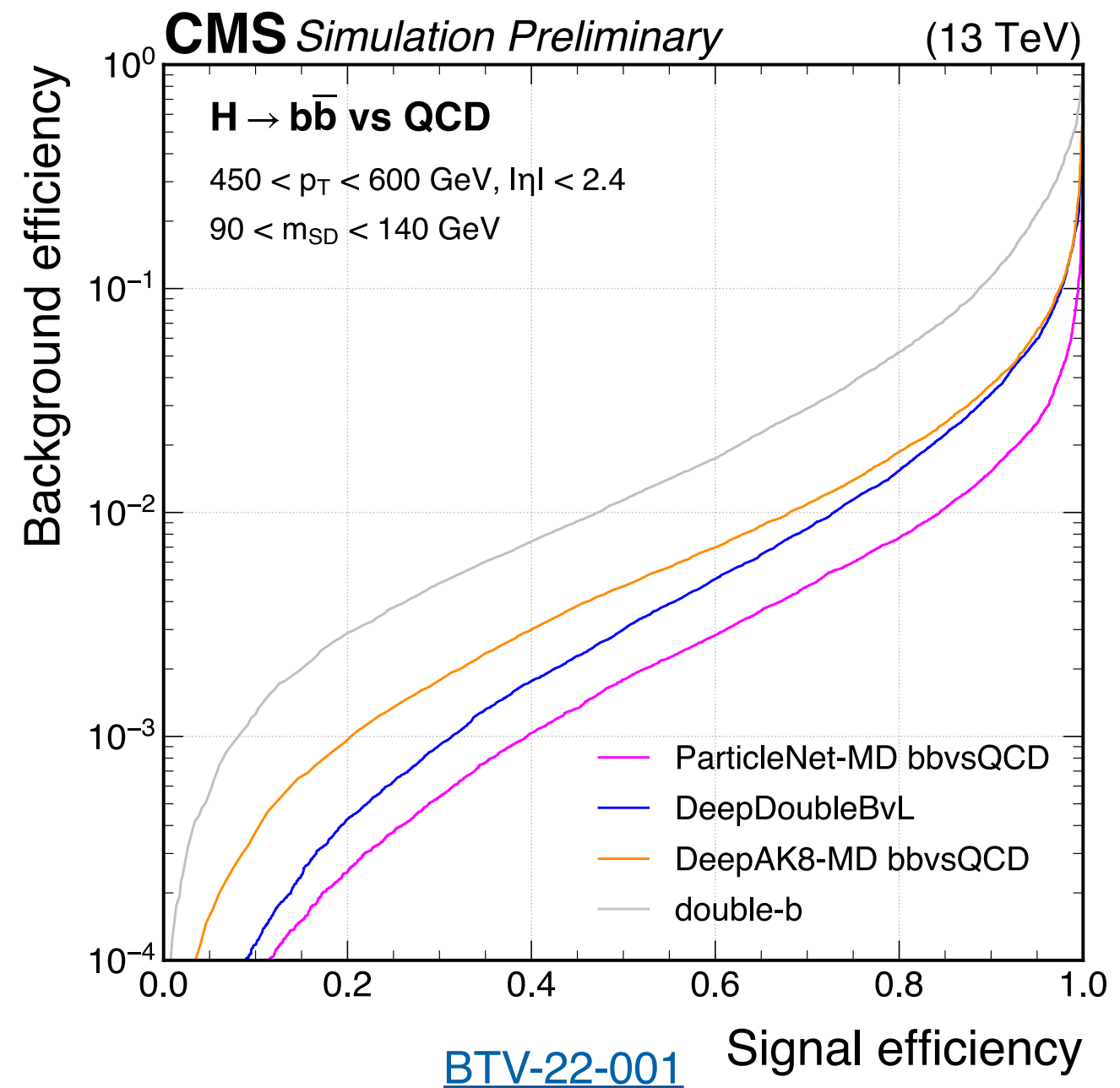
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Hypothesis test example

- ROC plot: let H_0 be QCD jet

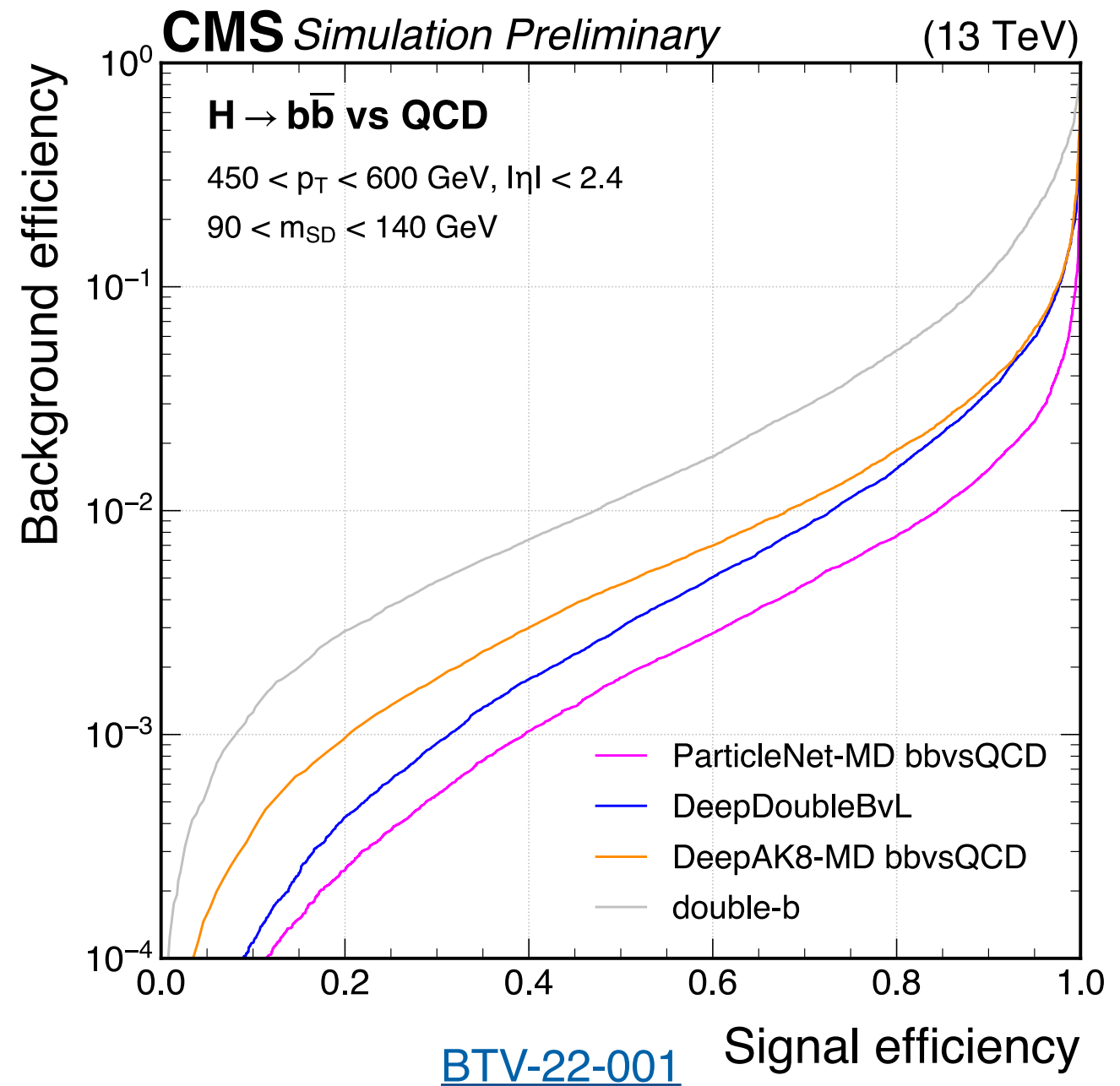
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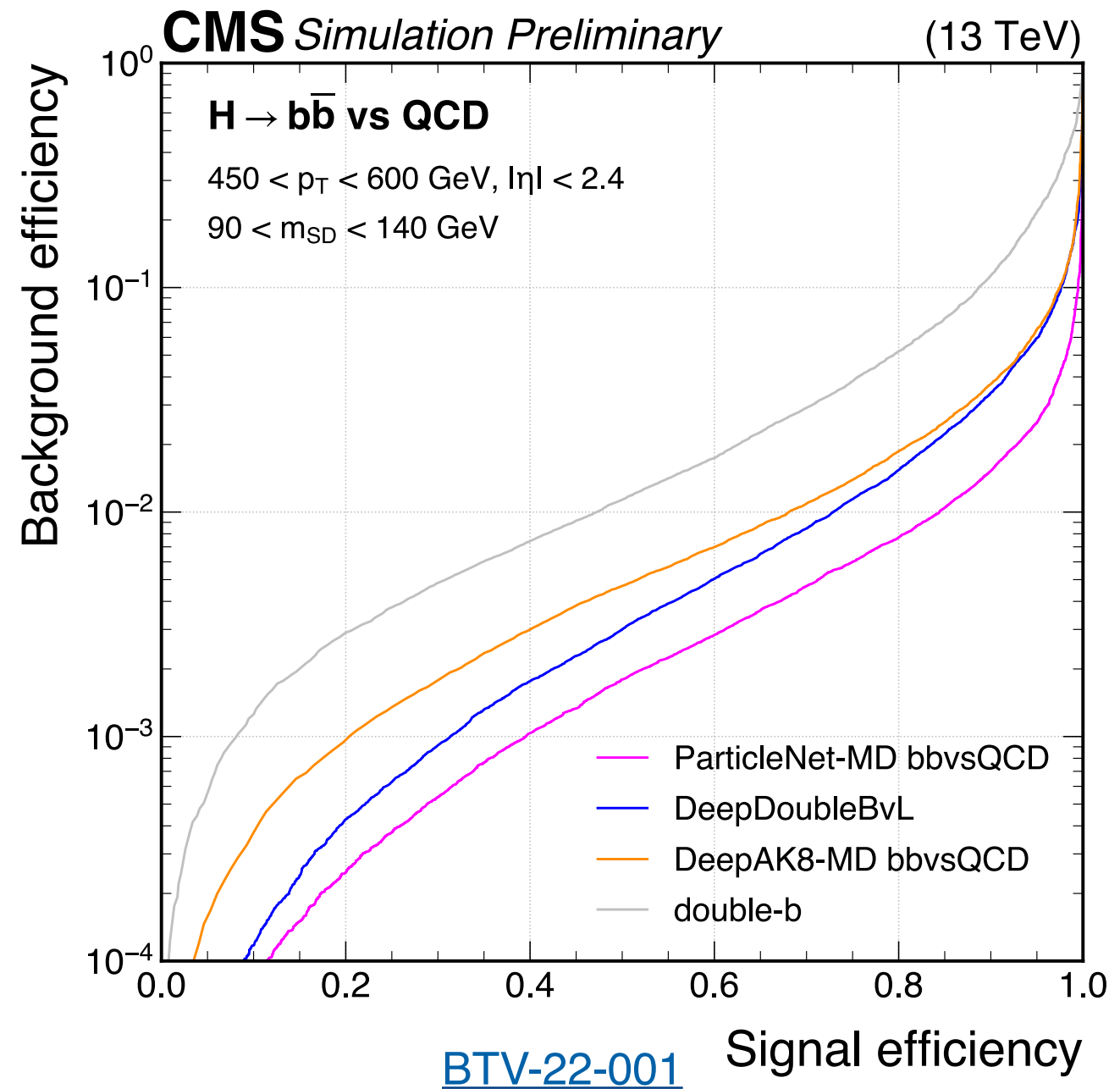
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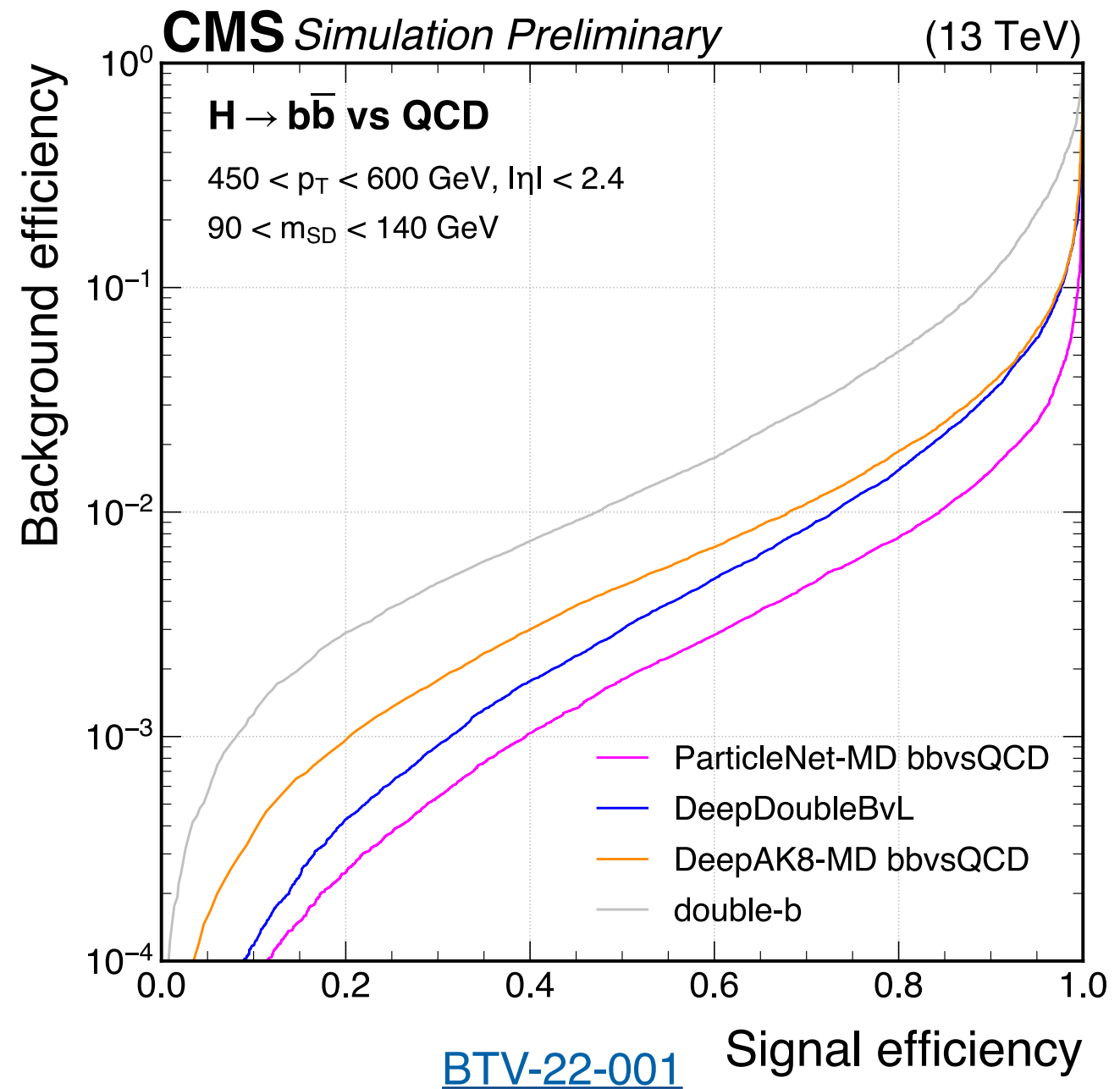
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Hypothesis test example

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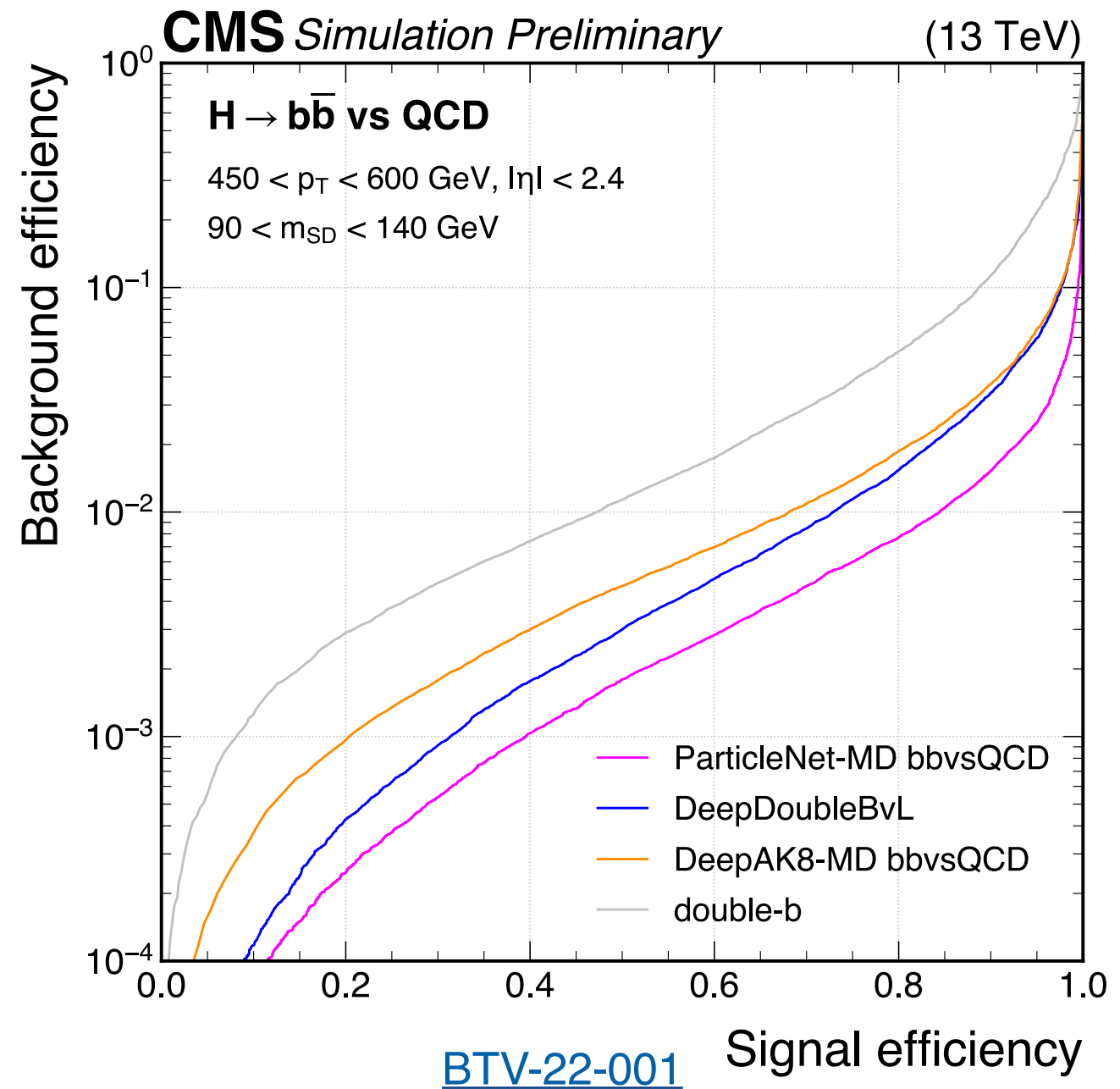
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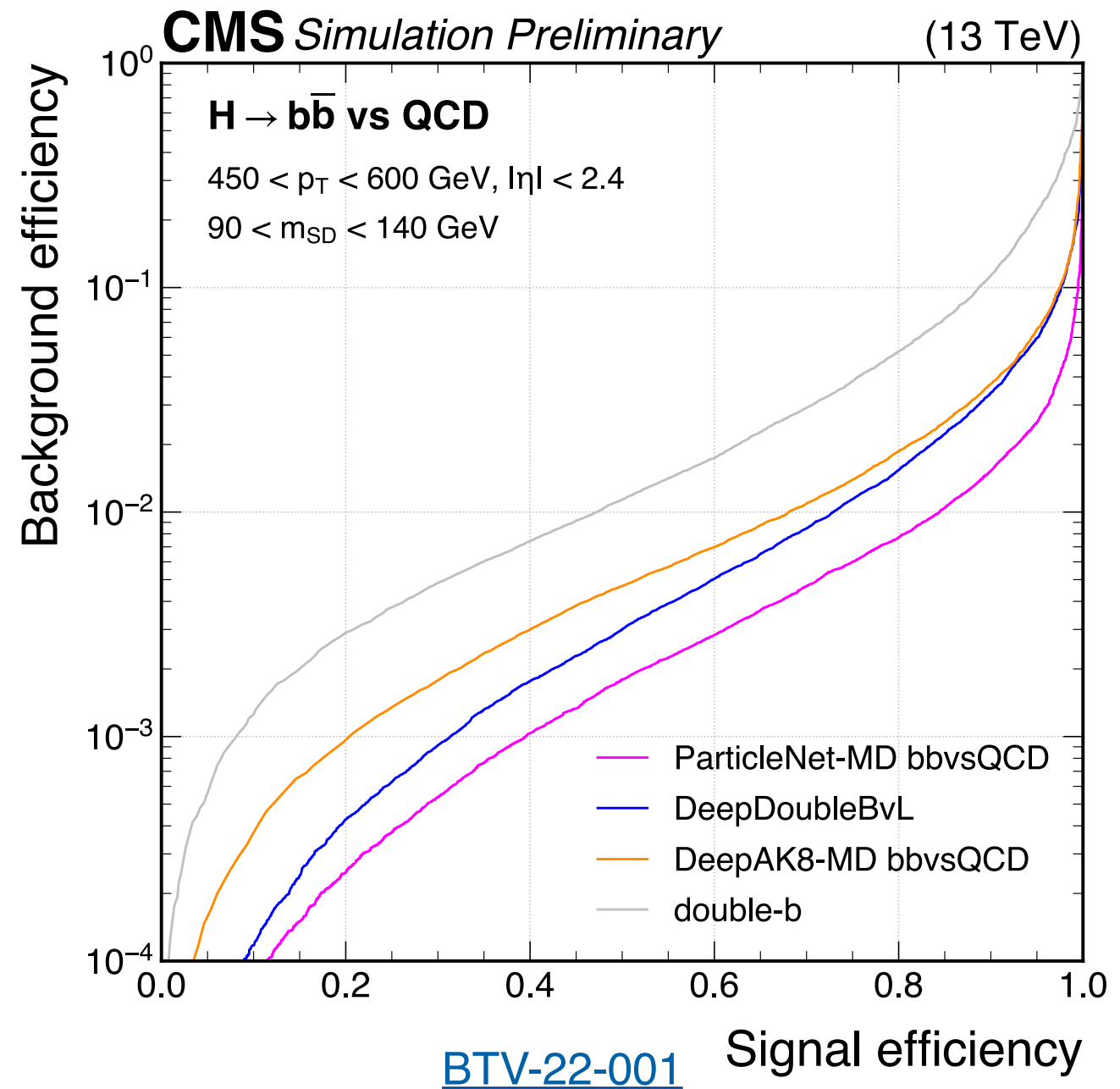
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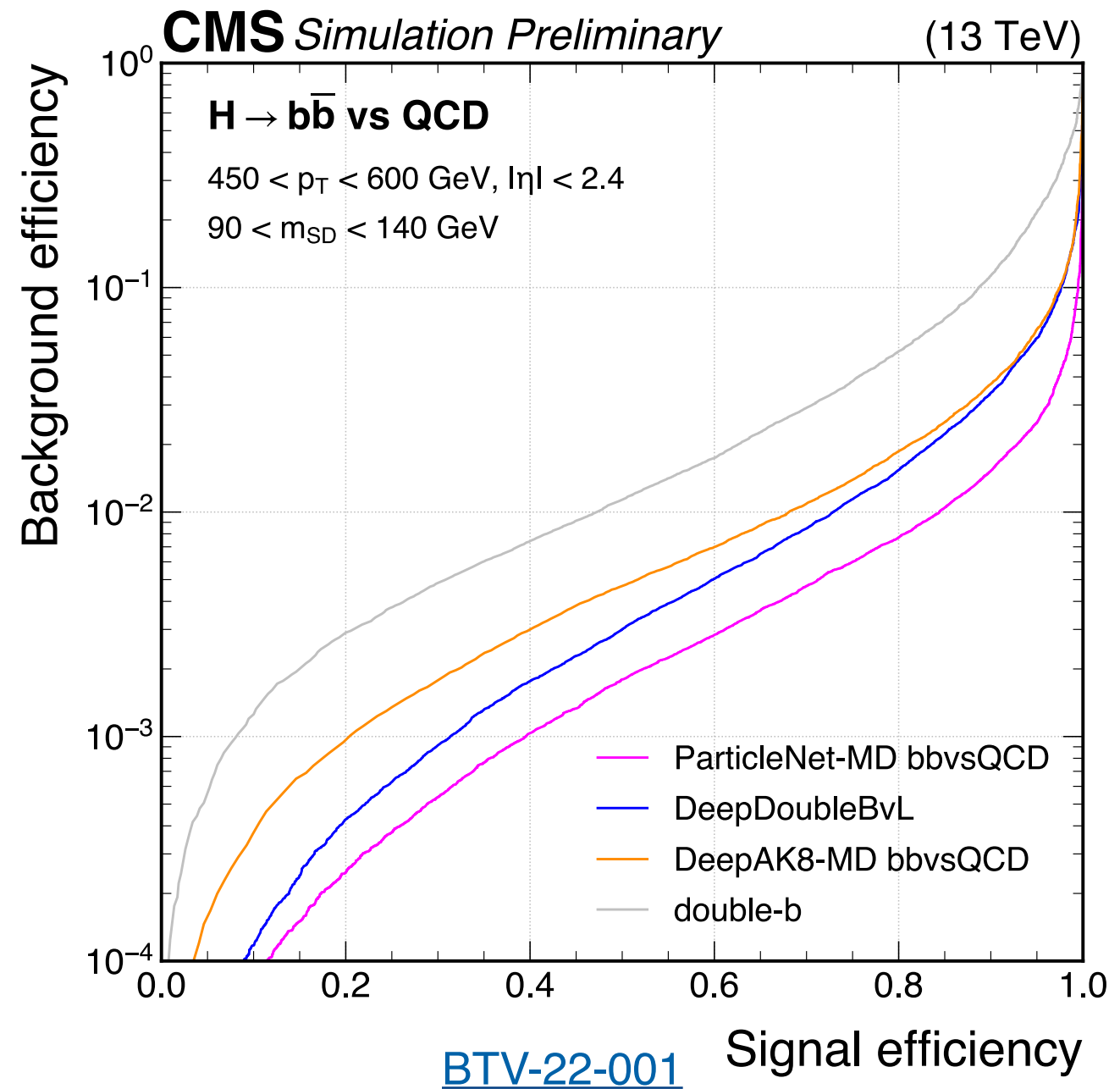
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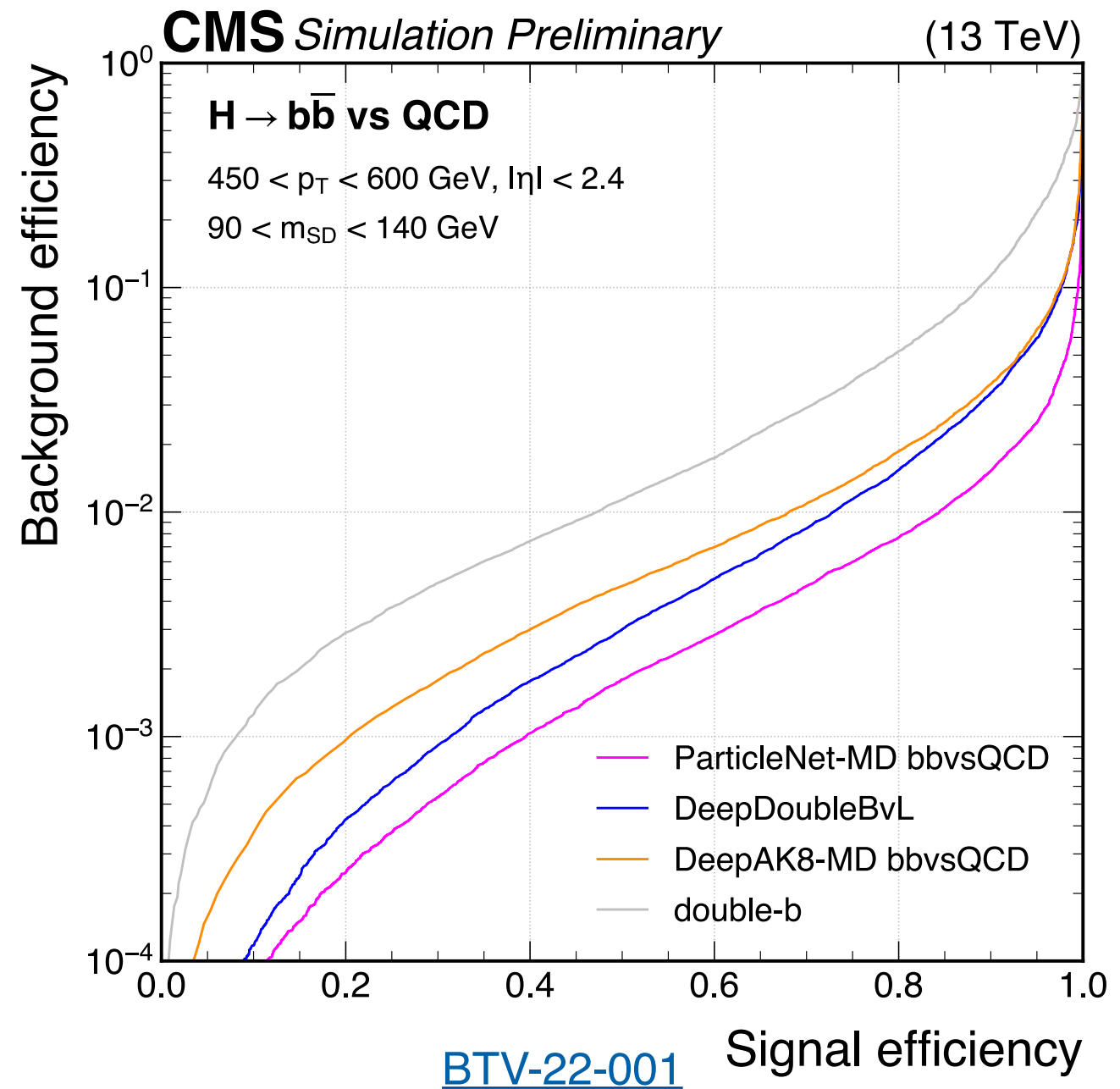
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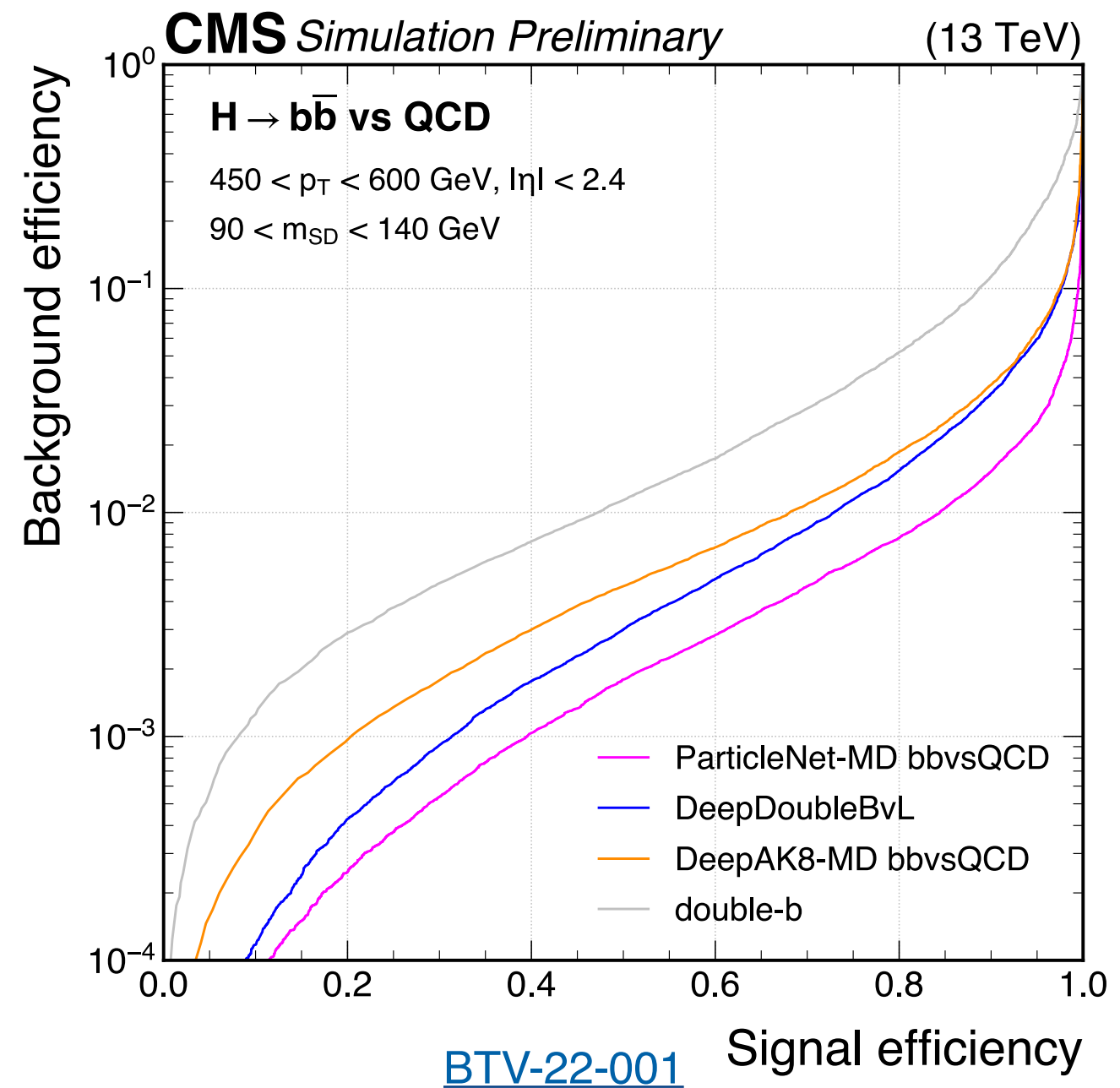
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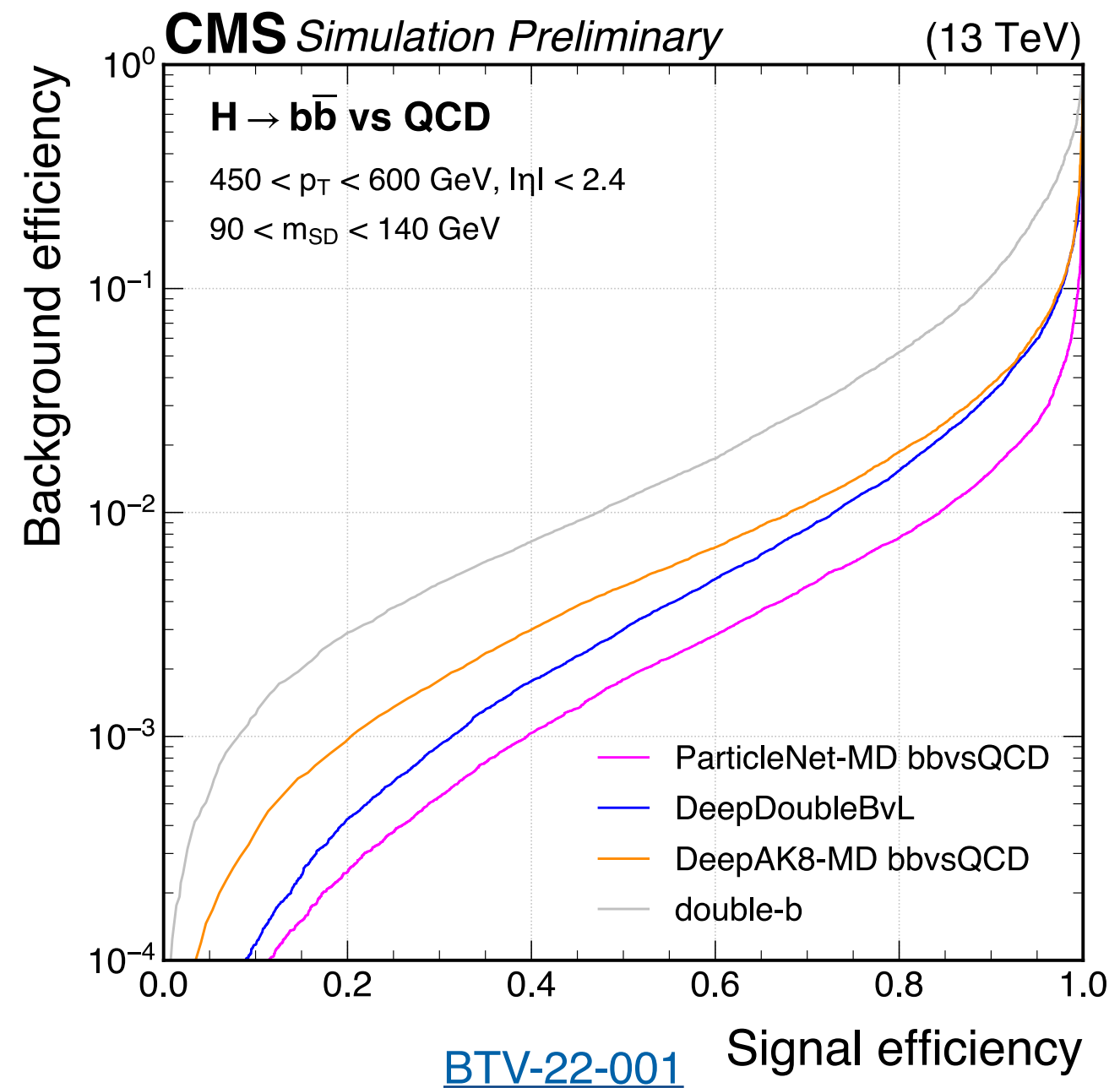
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- Prefer tests with higher *power* for a given *size*

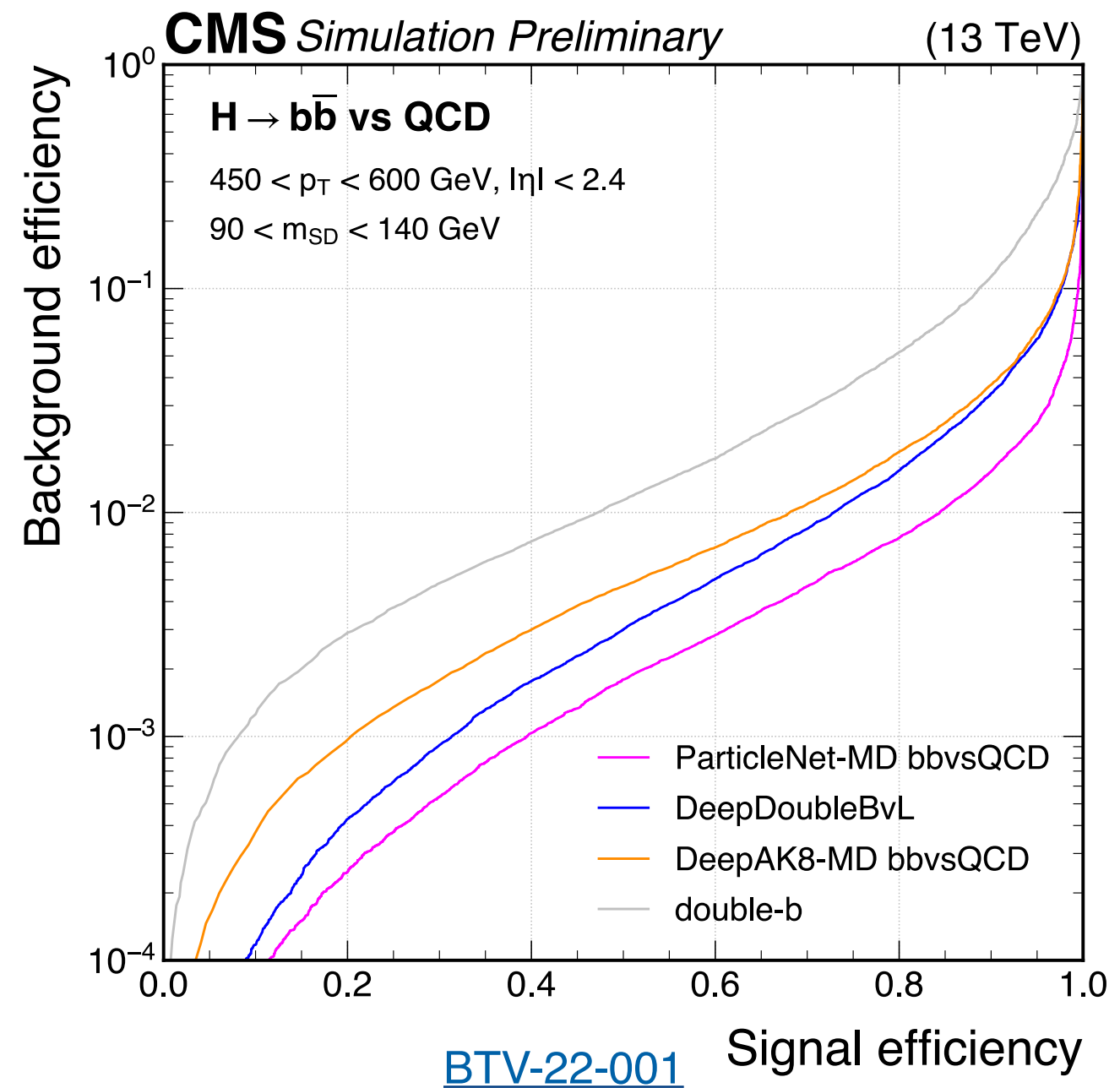
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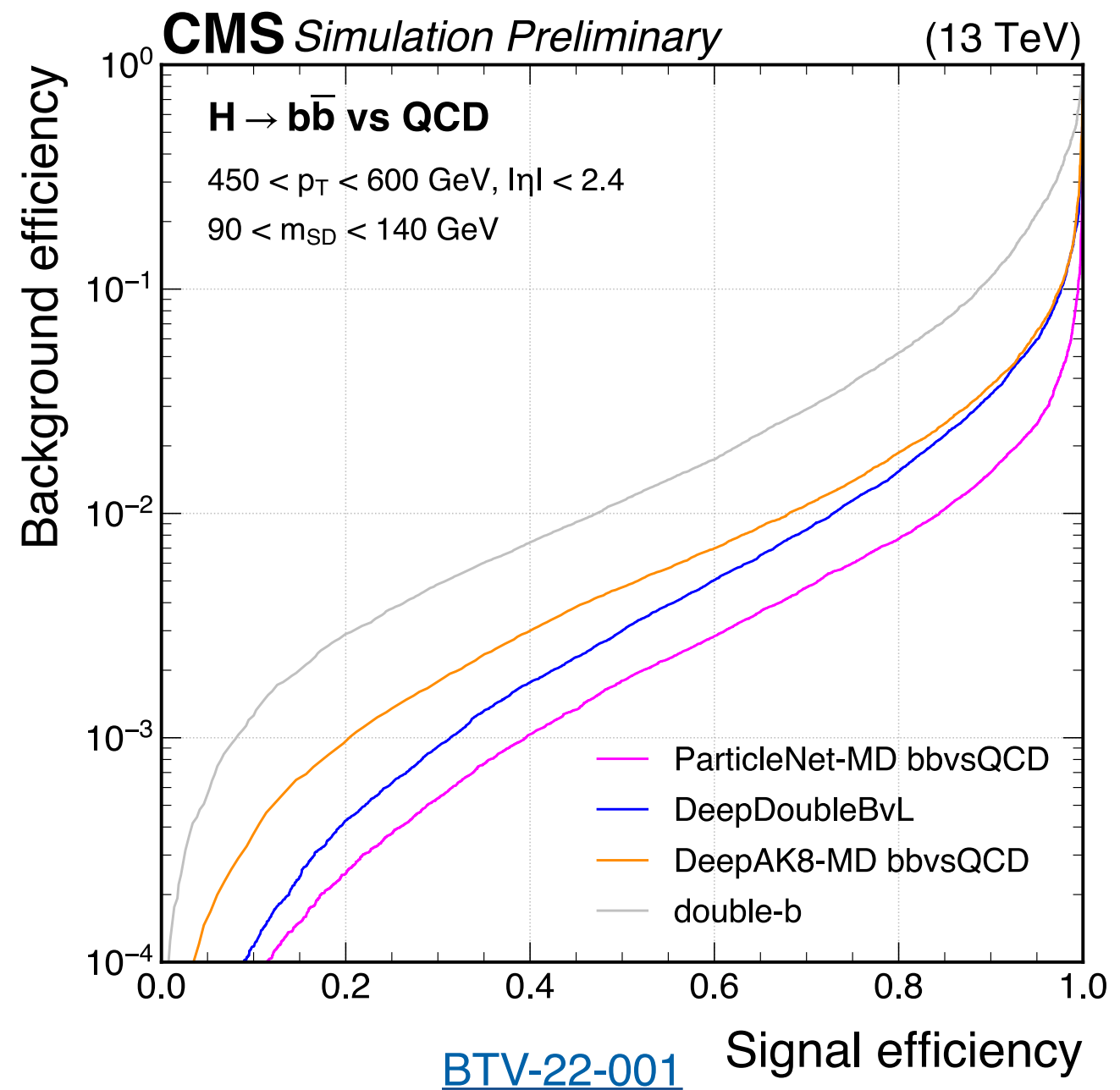
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 - Here, test size is large
 - For discovery, test size is *very small*

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- A good ordering will lead to
 - Good *coverage*: in repeated experiments the (unknown) θ_{true} will be in the set with probability at least $1-\alpha$, though it may over-cover
 - High *power* ($1-\beta$): the set does not contain θ_{alt} for some specified alternative hypothesis

Likelihood ratio test

1. Define $t_{\theta}(x) = -2 \ln \frac{\mathcal{L}(\theta)}{\mathcal{L}(\hat{\theta})} = -2 \ln \frac{P(x|\theta)}{P(x|\hat{\theta})}$

2. Compute associated pdf (change of variables)

$$P(t_{\theta} | \theta') = \int \delta(t_{\theta} - t_{\theta}(x)) P(x | \theta') dx$$

3. For each θ , find the critical value $t_{\theta,c}$ s.t.

$$\int_{t_{\theta,c}}^{\infty} P(t_{\theta} | \theta') dt_{\theta} \leq \alpha$$

4. Perform experiment, get x_{obs} , report confidence set $\{\theta | t_{\theta}(x_{obs}) < t_{\theta,c}\}$

Again, the set is the random variate, and will contain (unknown) θ_{true} with probability $1-\alpha$.

Dimension of θ and x are arbitrary.

If θ is 1-d and t_{θ} is monotone, can make a [Feldman-Cousins](#) interval.

Likelihood ratio examples

$$t_{\mu} = -2 \ln \frac{\mathcal{L}(\mu)}{\mathcal{L}(\hat{\mu})}$$

Likelihood ratio examples

- Poisson example

$$P(x|\mu) = \frac{\mu^x e^{-\mu}}{x!}$$

$$t_\mu = -2 \ln \frac{\mathcal{L}(\mu)}{\mathcal{L}(\hat{\mu})}$$

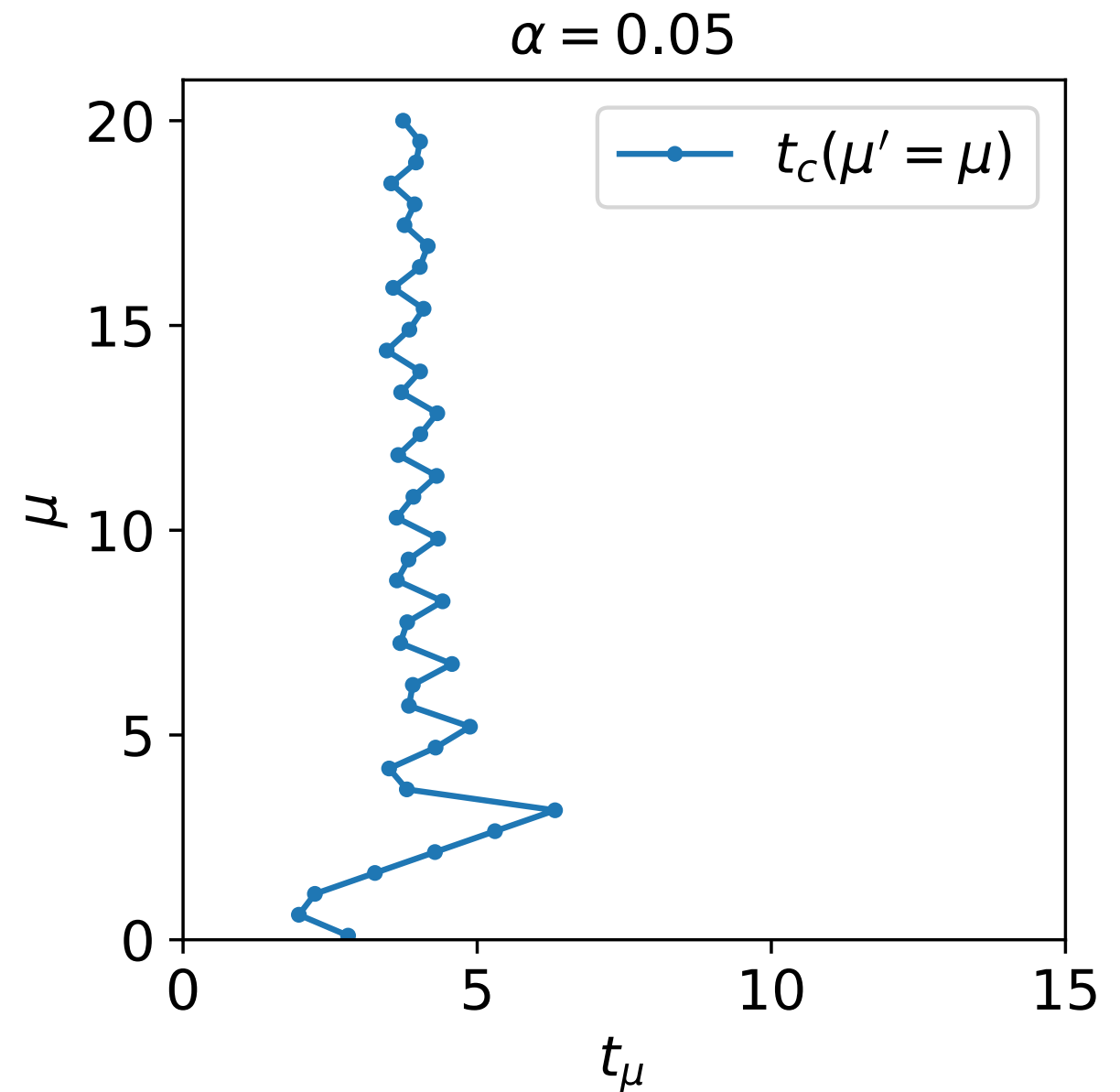
Likelihood ratio examples

- Poisson example

- $P(x|\mu) = \frac{\mu^x e^{-\mu}}{x!}$

- For each μ :

- Throw 10k toys
 - Compute $\hat{\mu}$, t_μ
 - Find 0.95 quantile in distribution



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Likelihood ratio examples

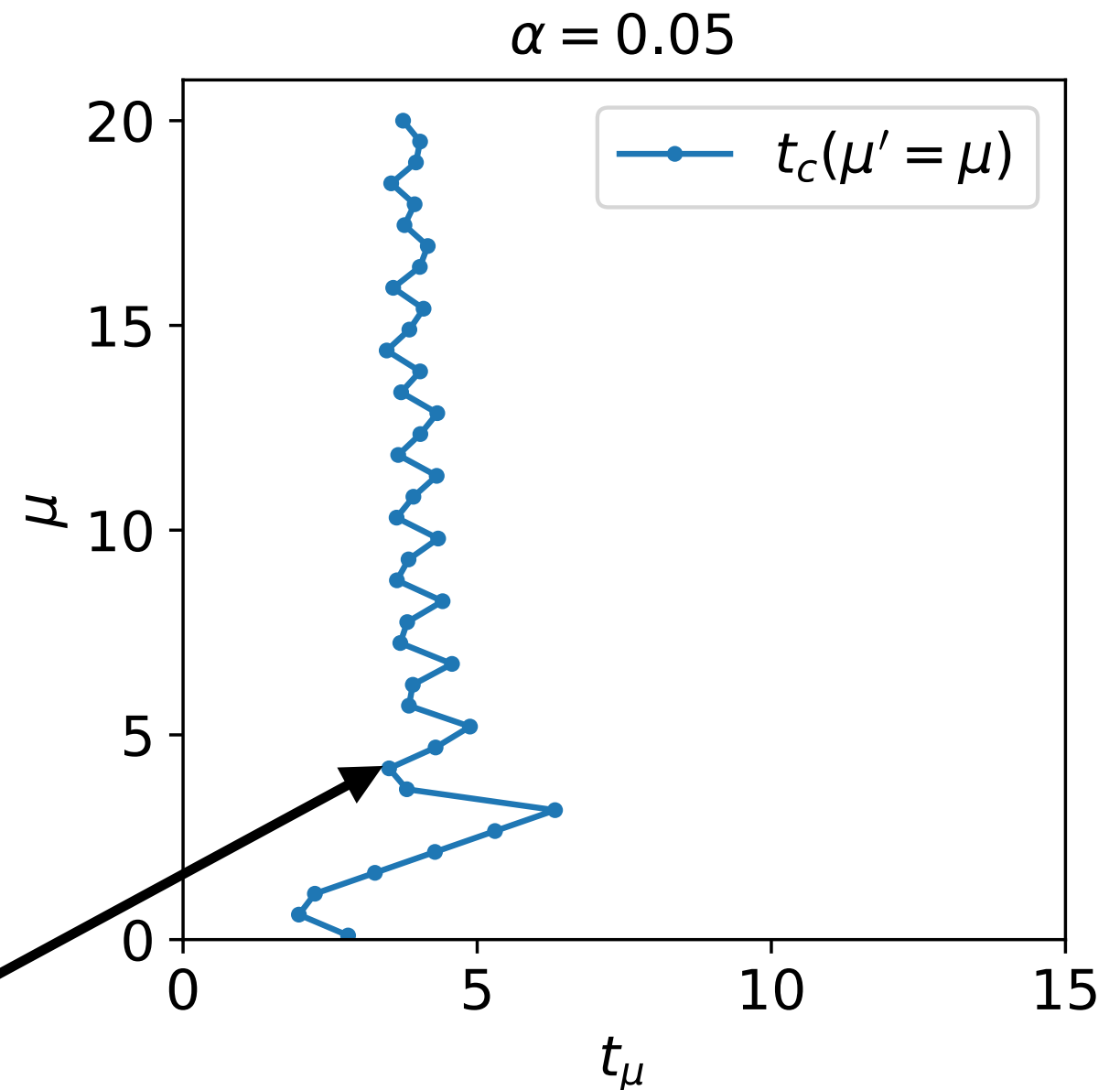
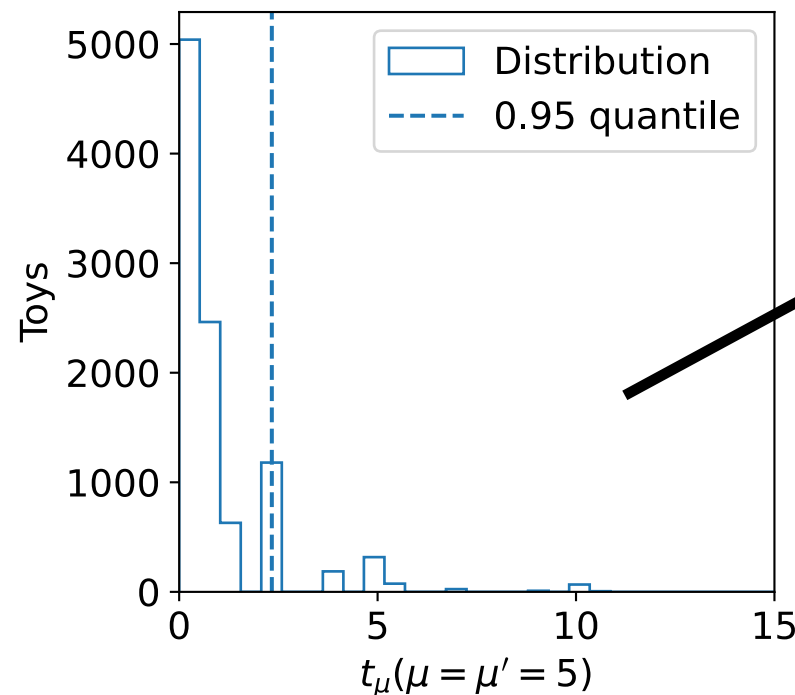
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Note: jagged behavior is due to discrete nature of t , not limited toy statistics



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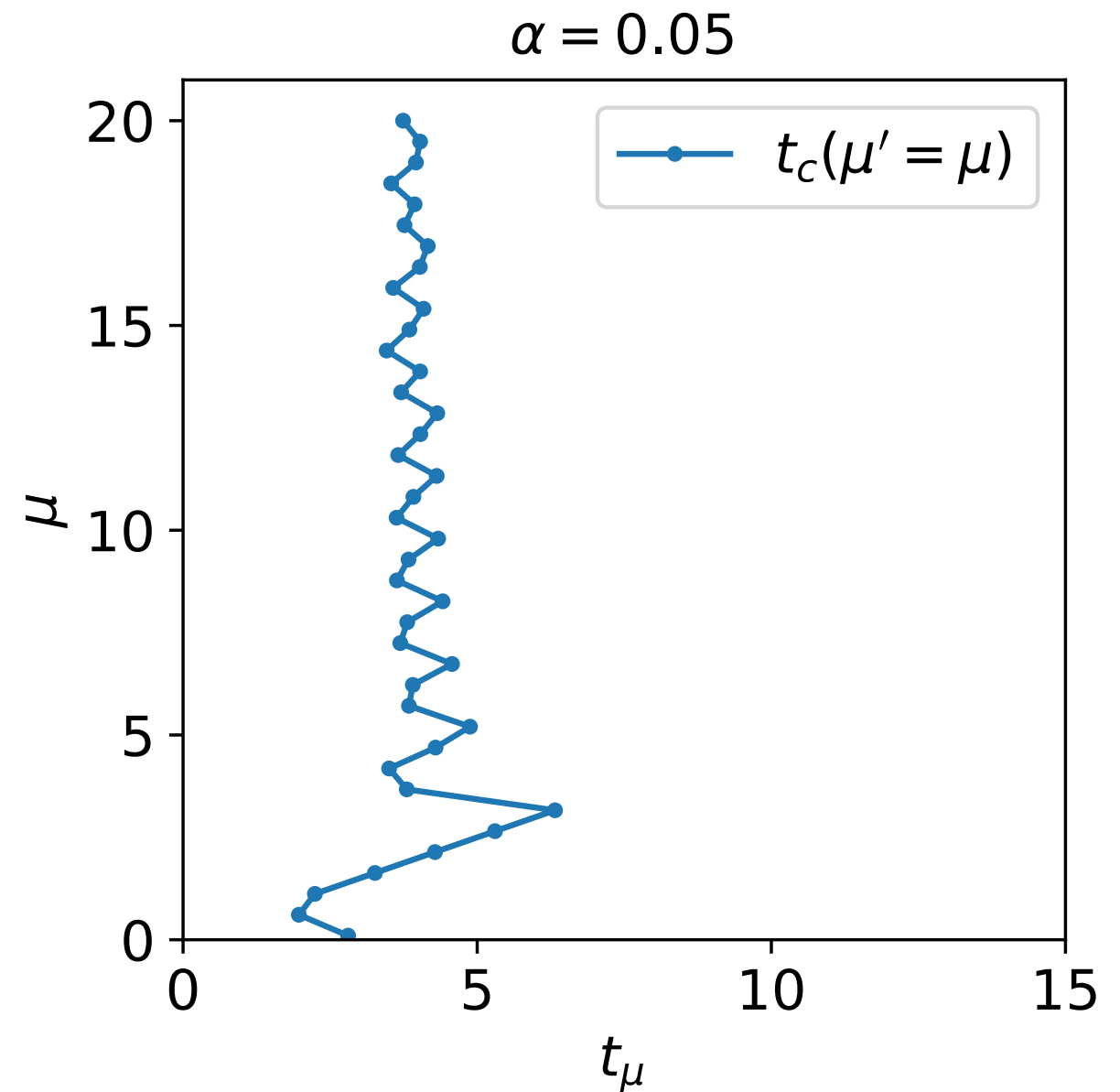
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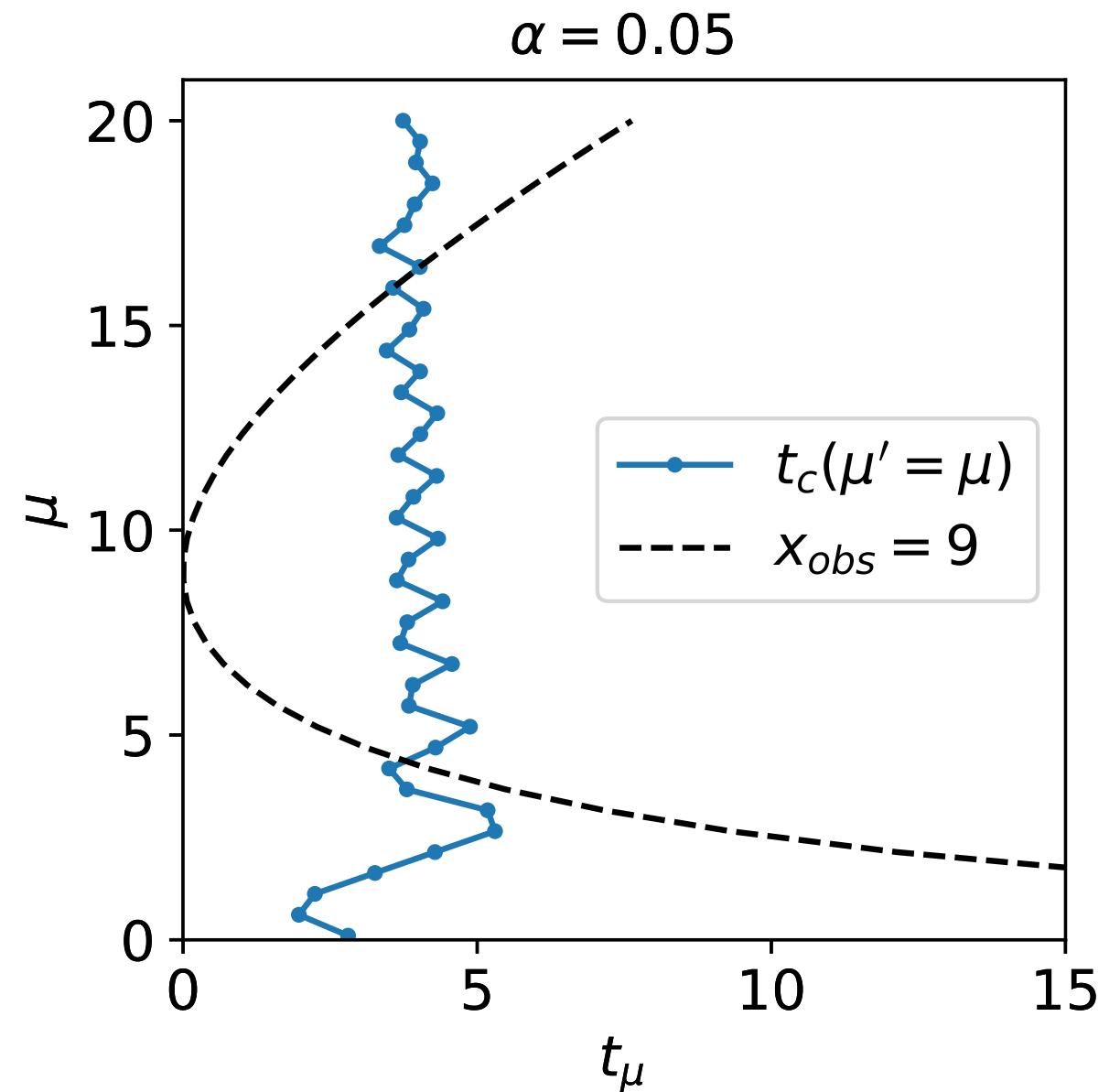
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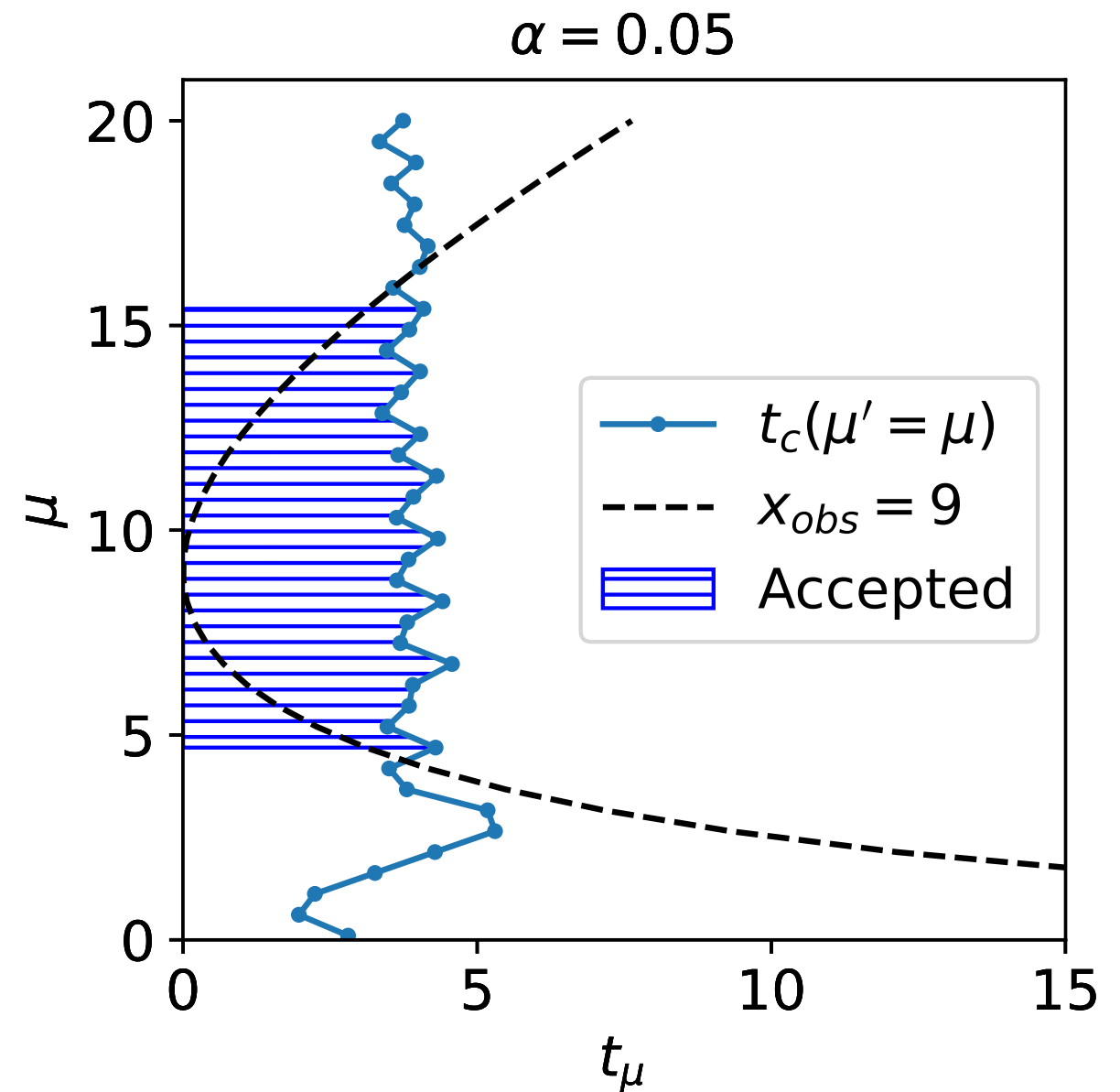
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- Accept $t_\mu(x_{obs}) < t_{\mu,c}$



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Likelihood ratio examples

- Poisson with background example

- $$P(x | \mu s + b) = \frac{(\mu s + b)^x e^{-(\mu s + b)}}{x!}$$

- s=5, b=10 fixed, x=20

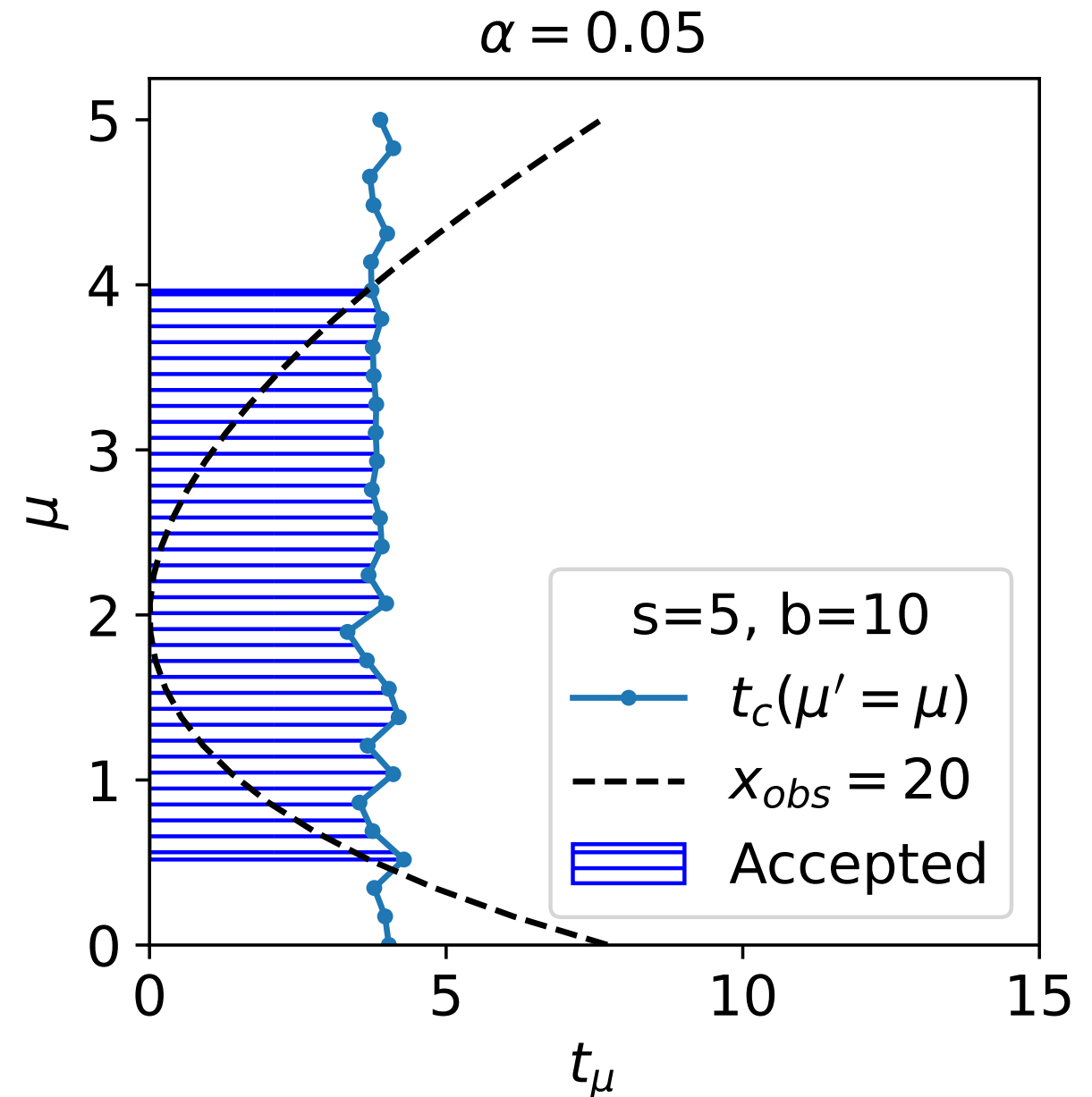
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- Plan: set upper limit on μ



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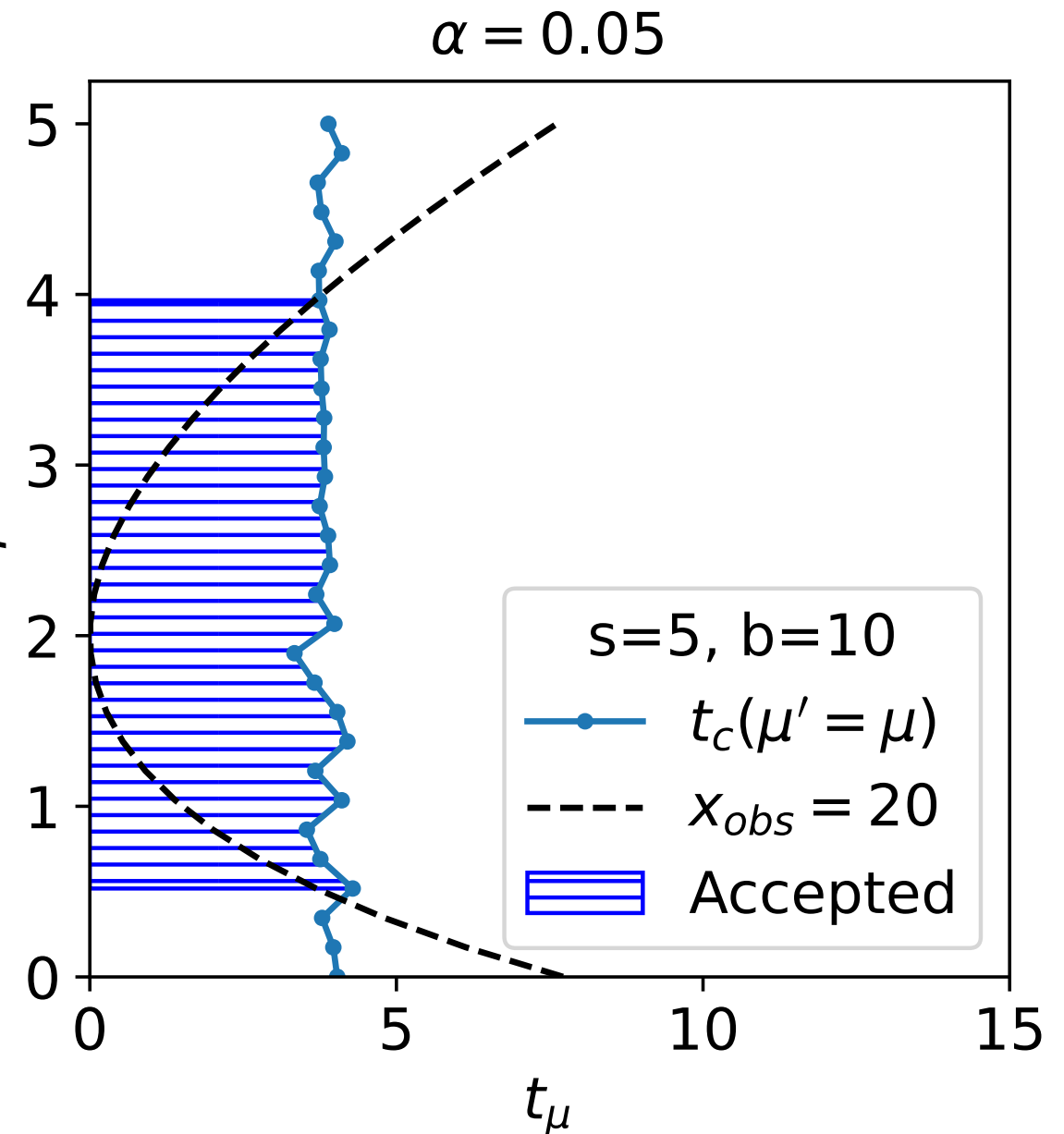
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- We should not consider $\hat{\mu} > \mu$ to indicate less compatibility with a model that assumes a rate μ .



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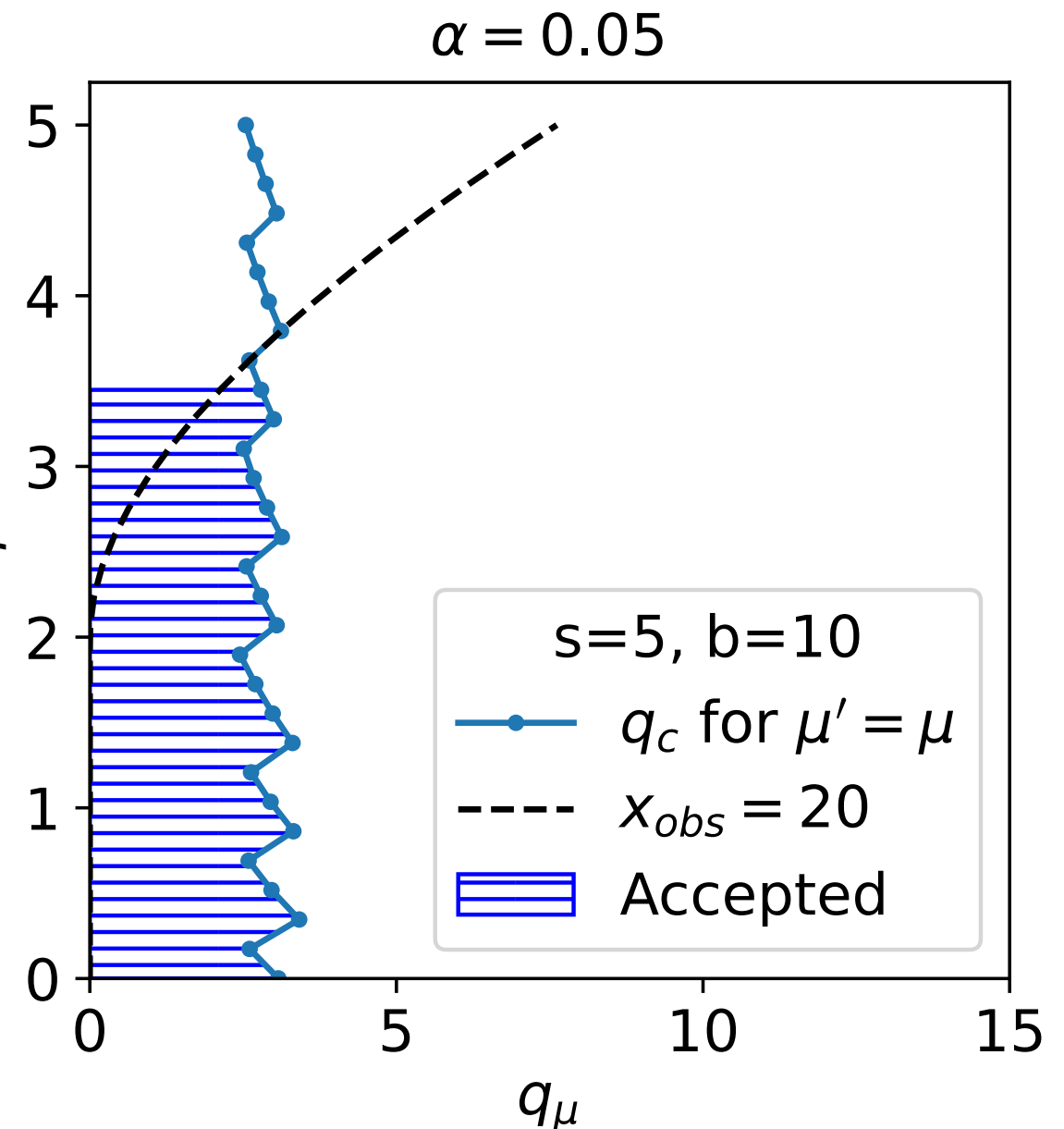
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- Solution: modify test statistic

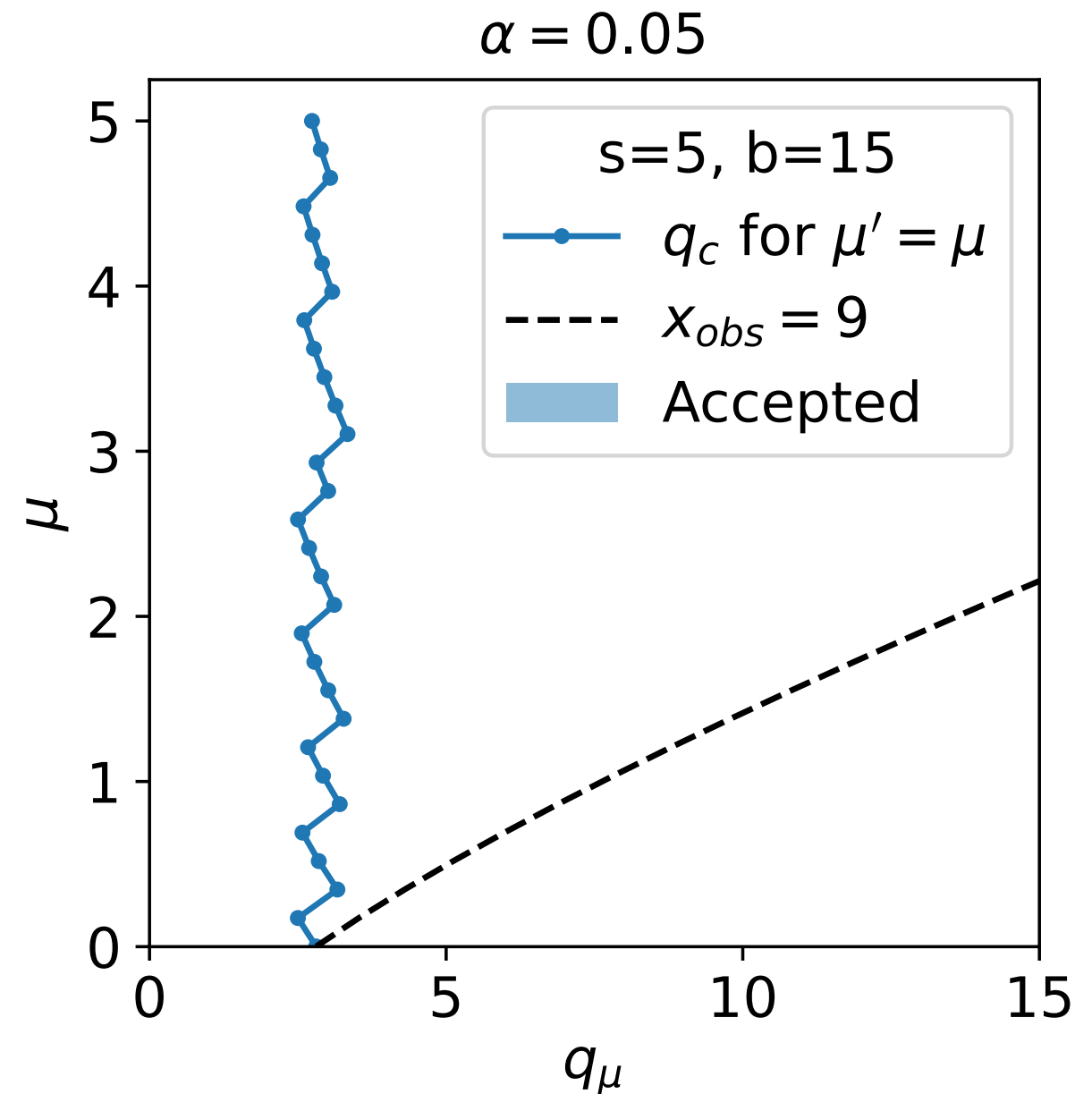
$$\text{Define } q_\mu = -2 \ln \frac{\mathcal{L}(\mu)}{\mathcal{L}(\min(\mu, \hat{\mu}))}$$

- i.e. over-fluctuations are “not extreme”



Likelihood ratio examples

- Same example as before, but $b=15$, $x=9$
- Problem: under fluctuation
 - No values accepted!
 - Possible but unsatisfying outcome of frequentist test

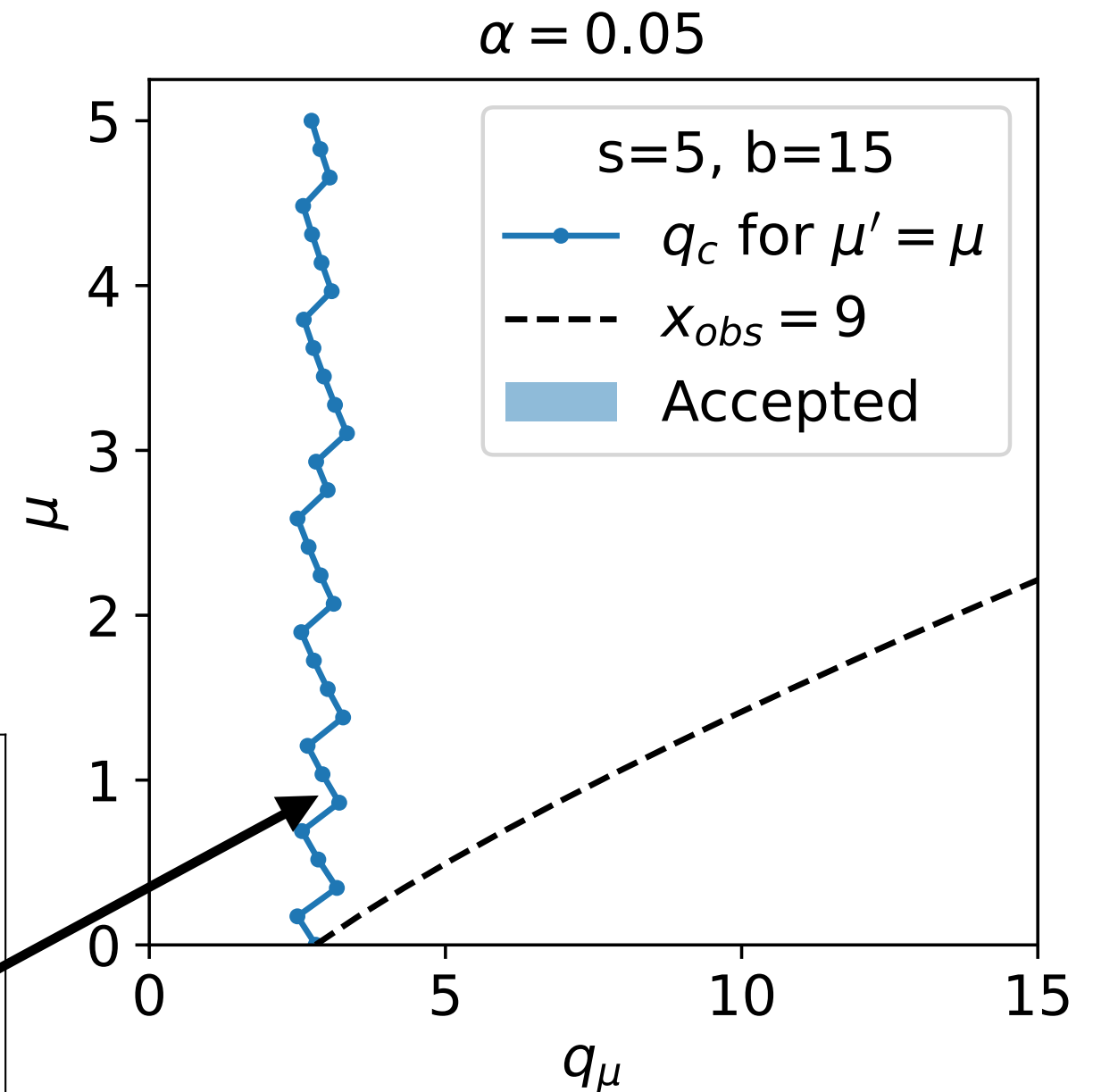
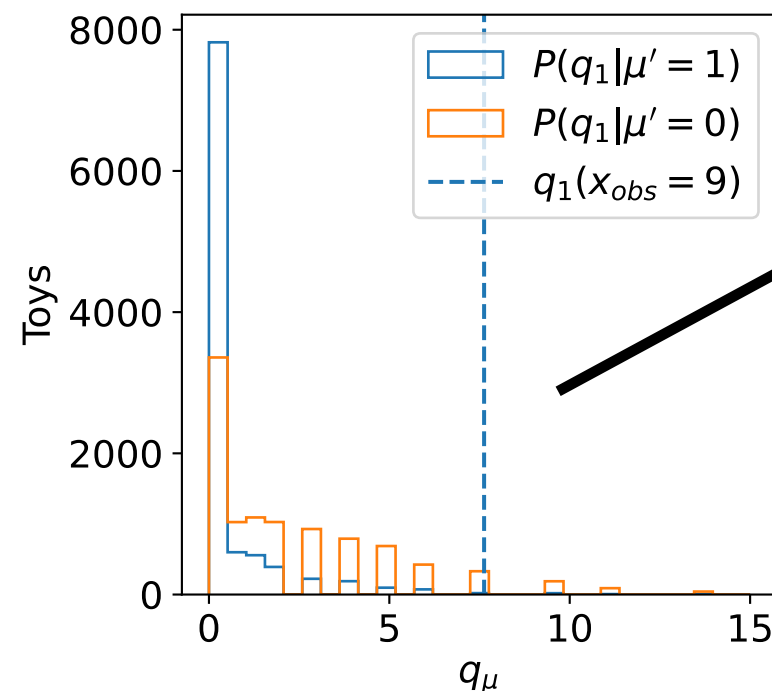


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The result $x=9$ is rare for both S+B and B-only hypotheses.



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CLs

- CL_s criterion departs from purely frequentist CL to ameliorate the null set problem (among others)

- Original expositions by [A. Read](#), [T. Junk](#)
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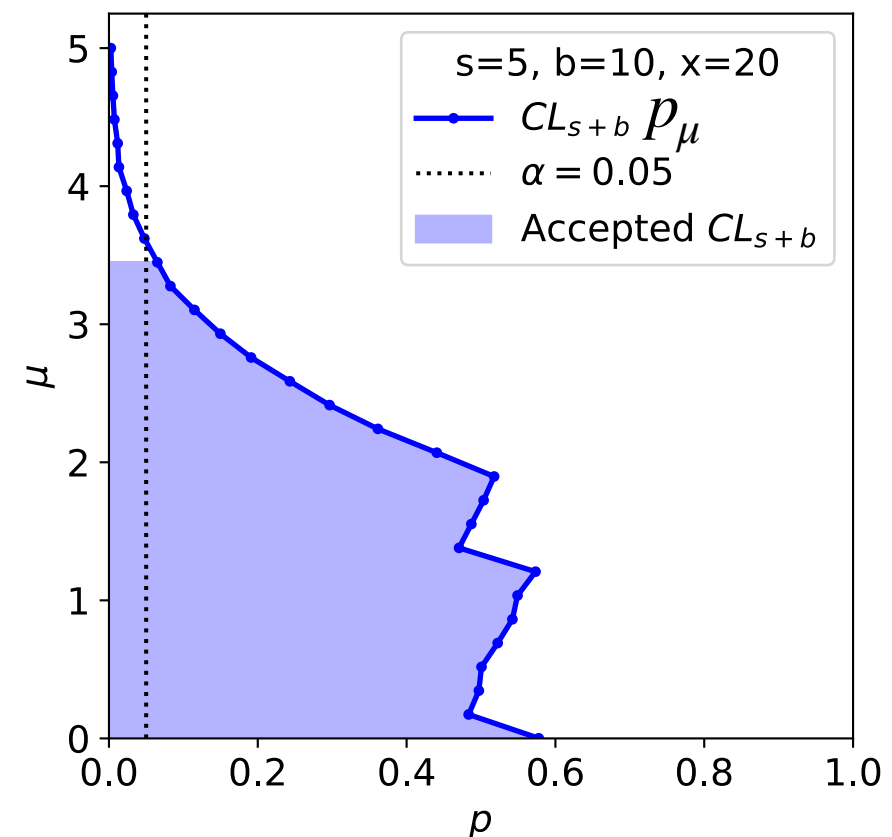
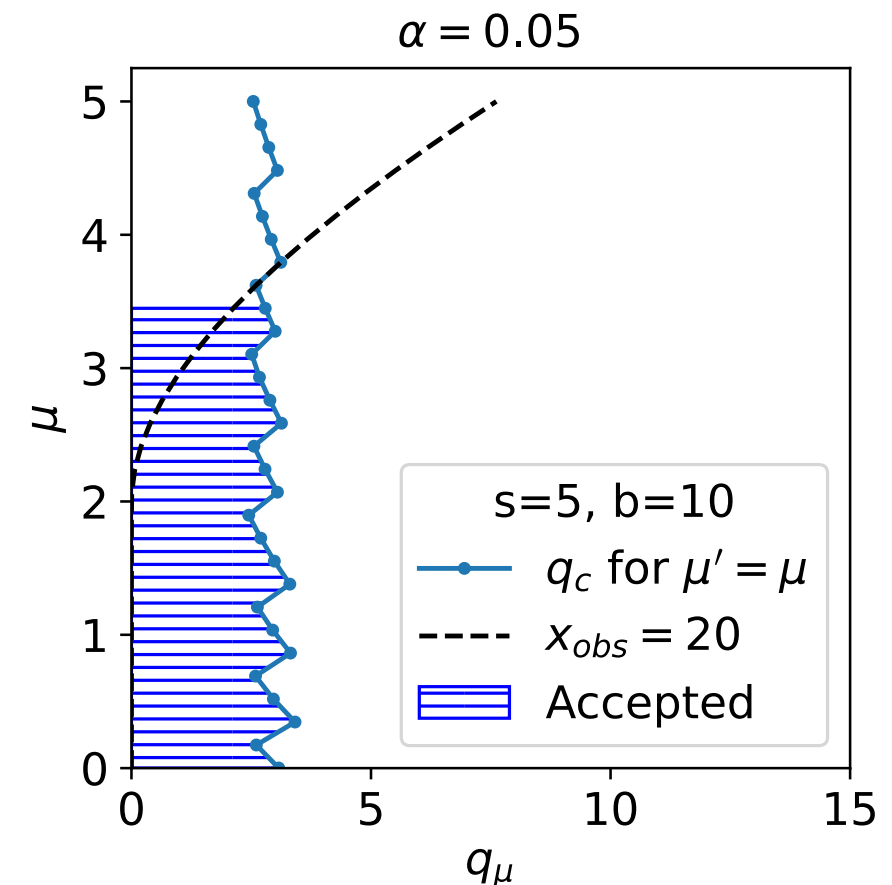
- First we reformulate our old test:

- Define $p_\mu = \int_{t_\mu(x_{obs})}^{\infty} P(t_\mu | \mu' = \mu) dt_\mu$

- This is a p-value

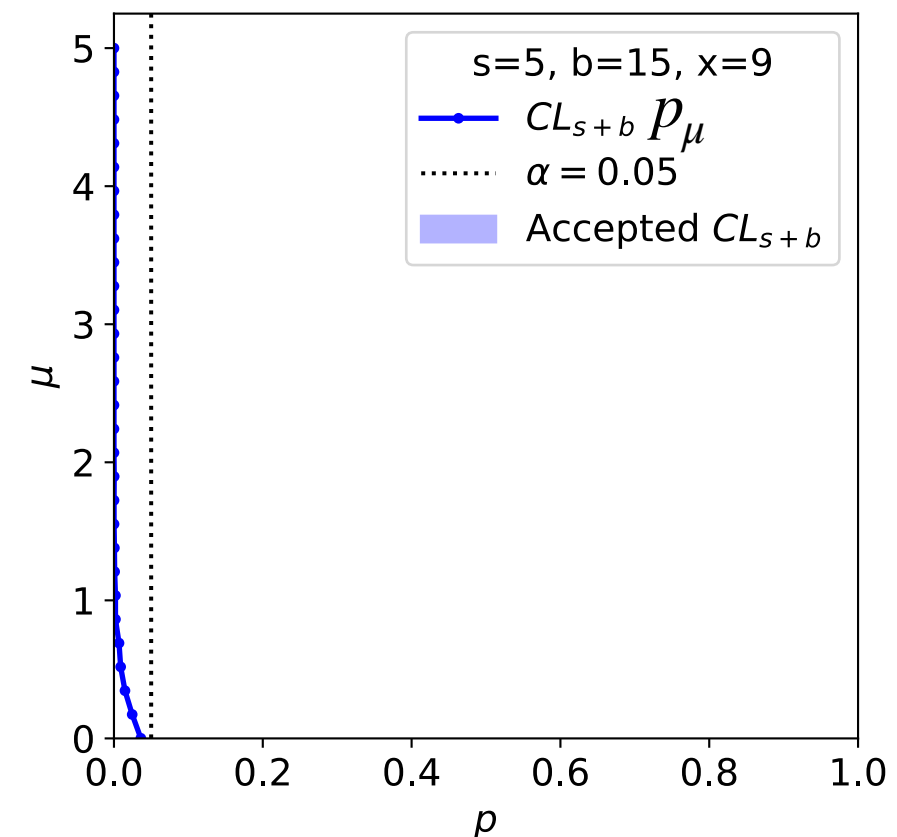
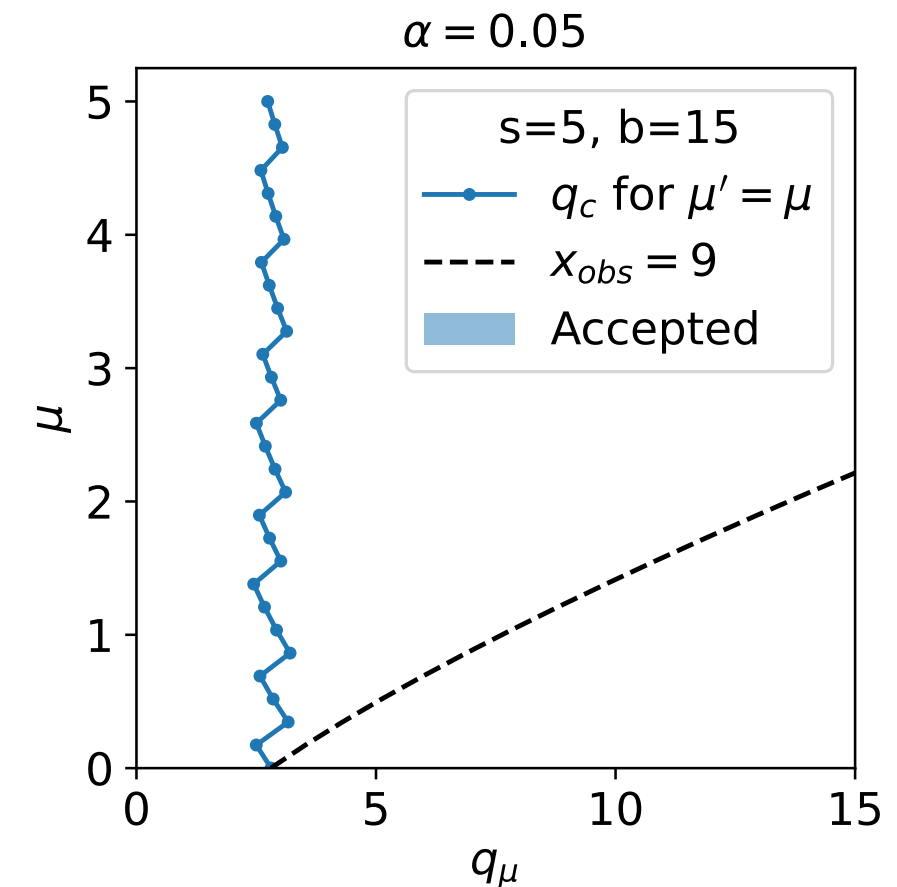
- Then we accept the region $p_\mu > \alpha$

- Right: initial S+B example reformulated



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 - Right: under-fluctuation S+B example reformulated



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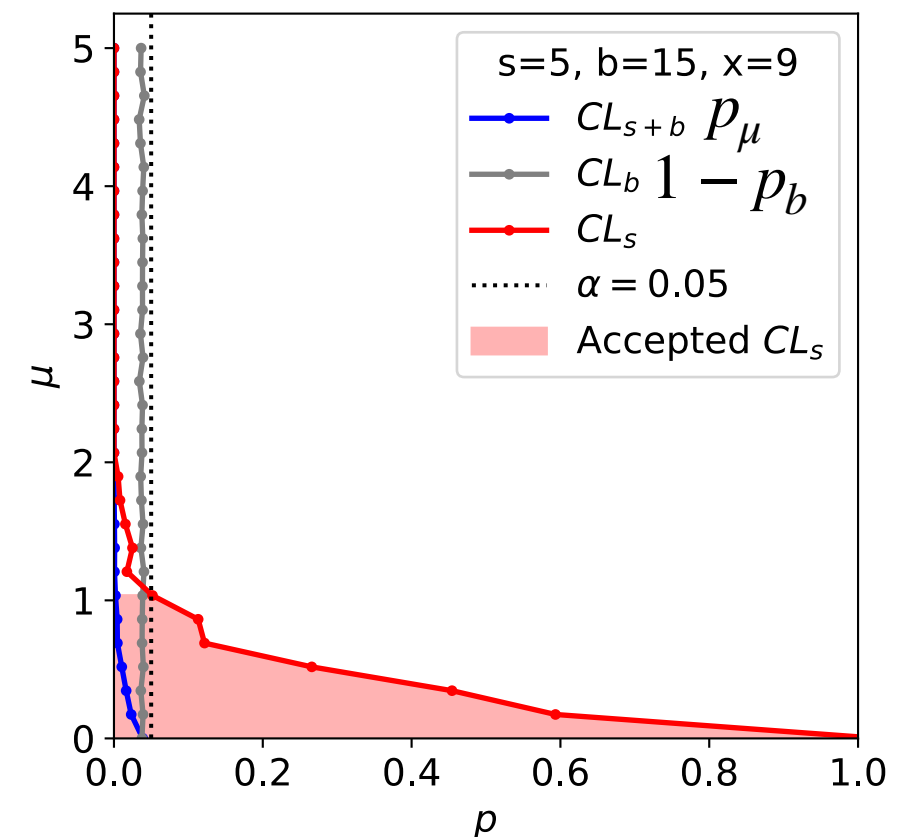
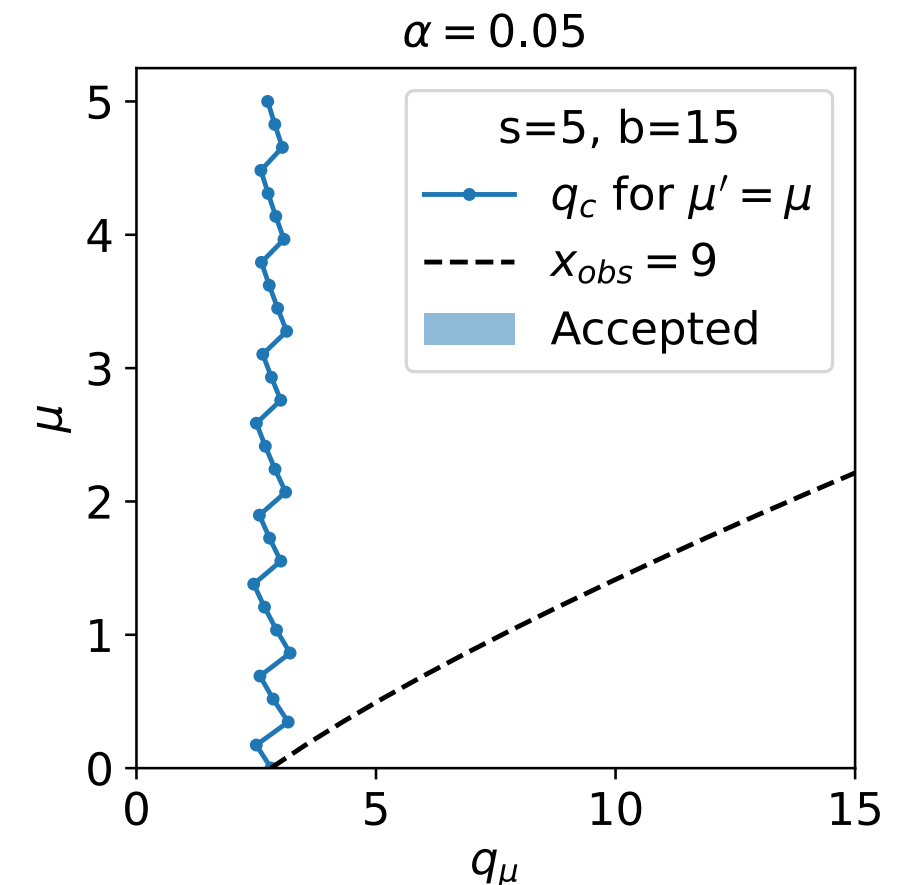
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- Now define background-only p-value

- $1 - p_b = \int_{t_\mu(x_{obs})}^{\infty} P(t_\mu | \mu' = 0) dt_\mu$

- Accept instead $CL_s = \frac{p_\mu}{1 - p_b} > \alpha$



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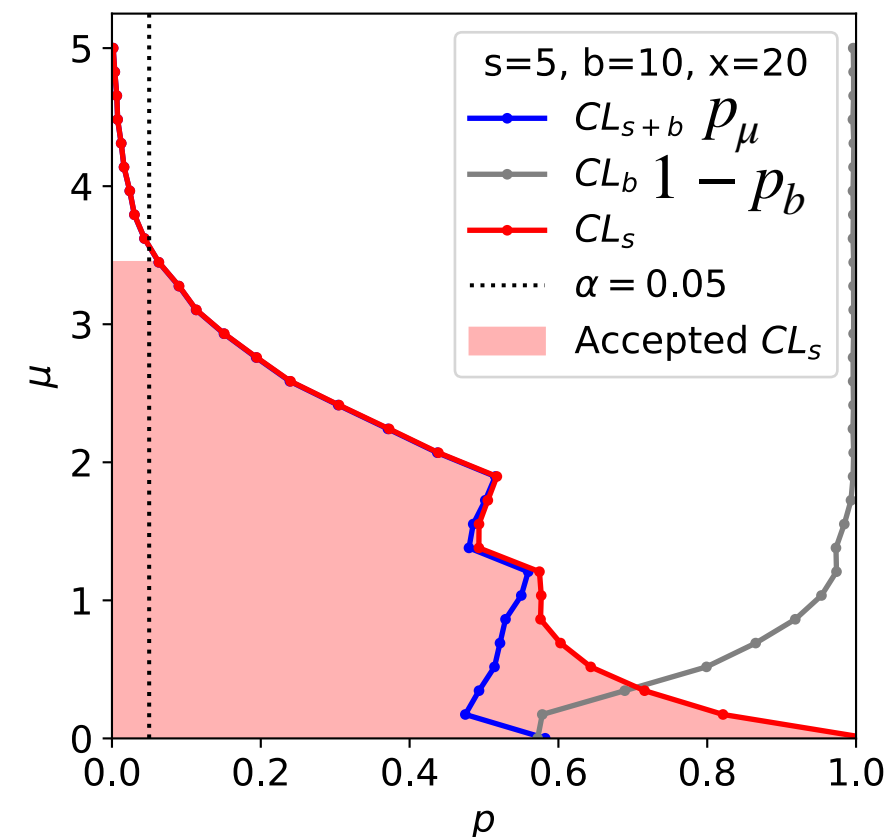
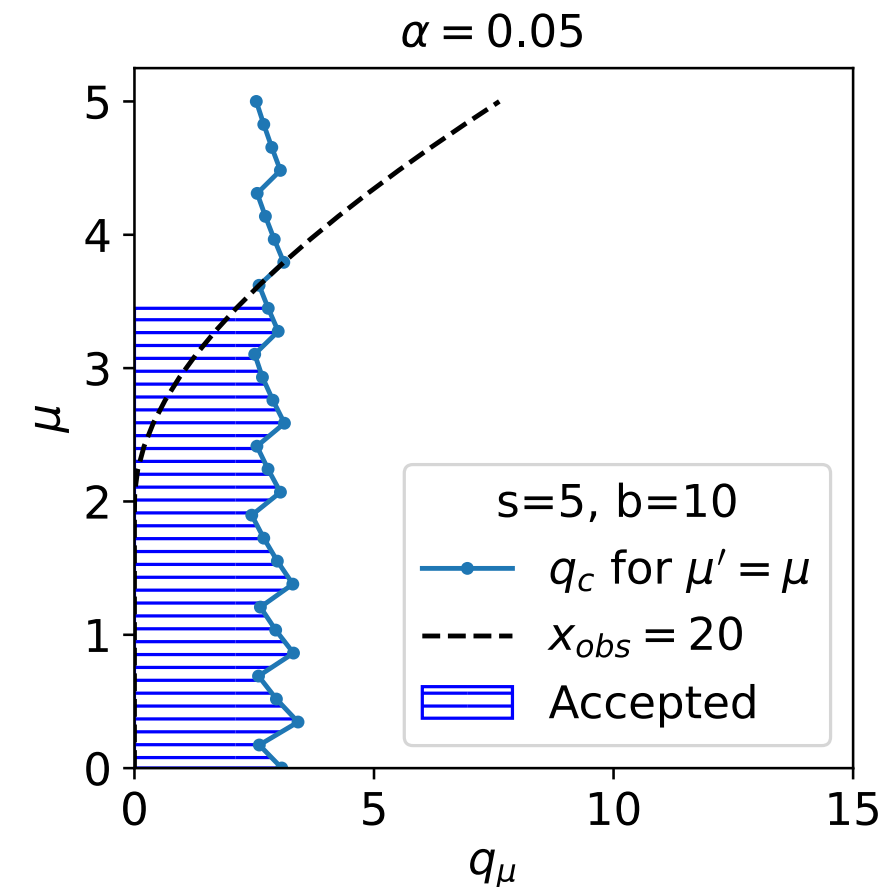
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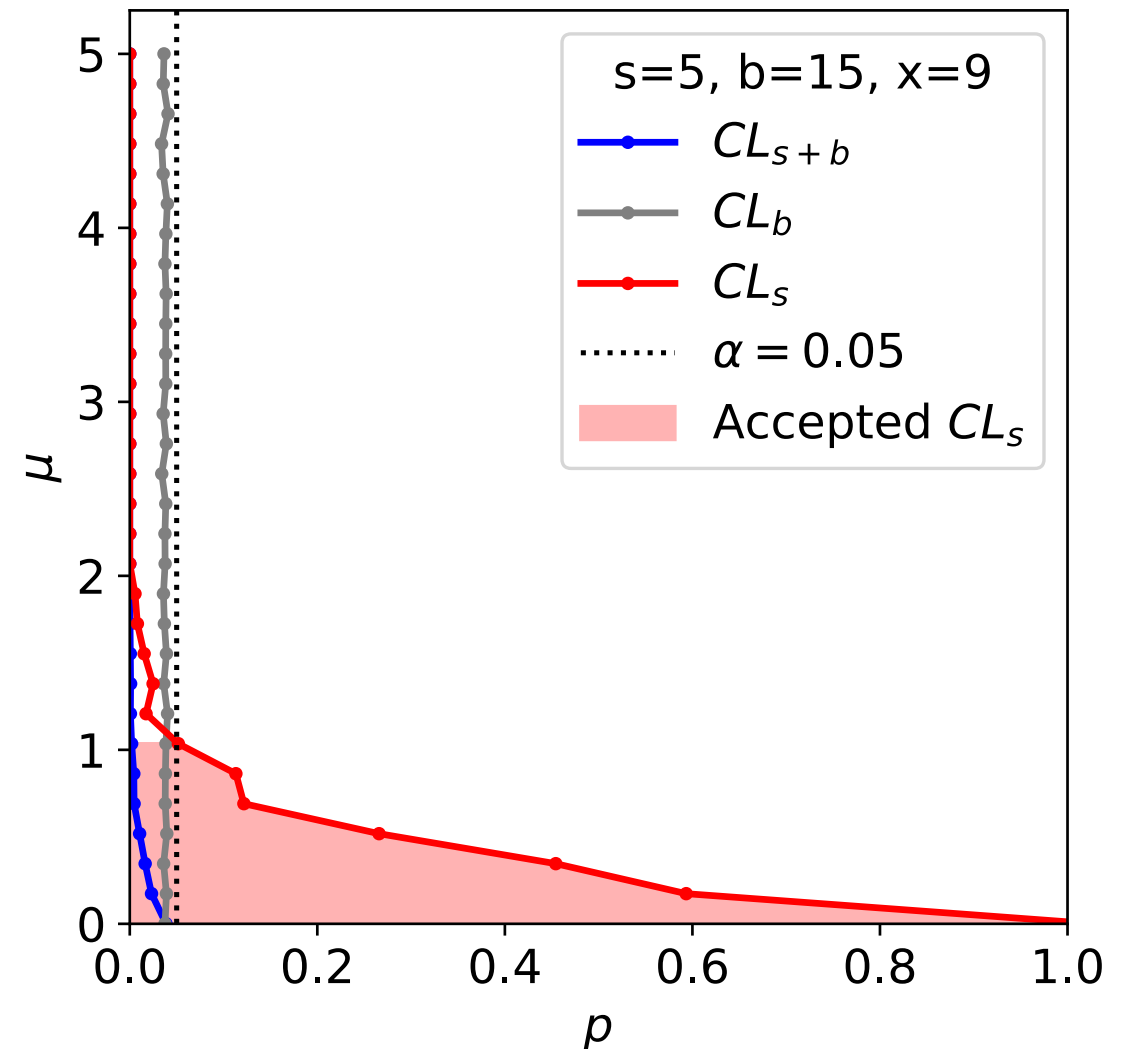
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- No effect in the first example



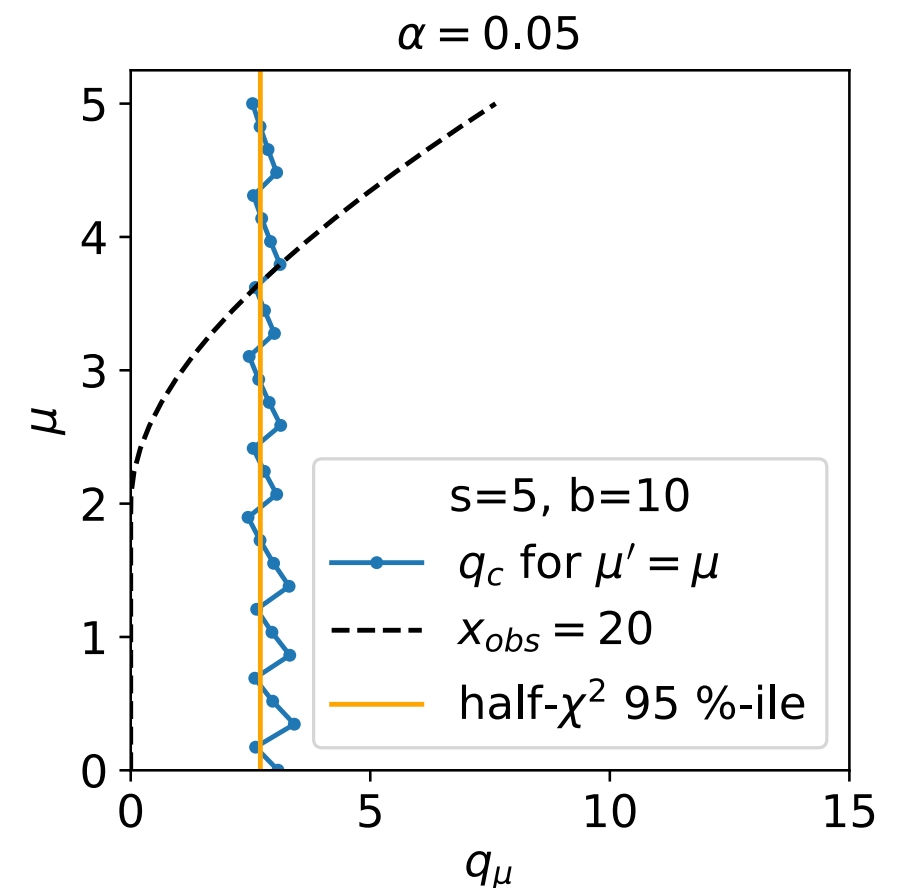
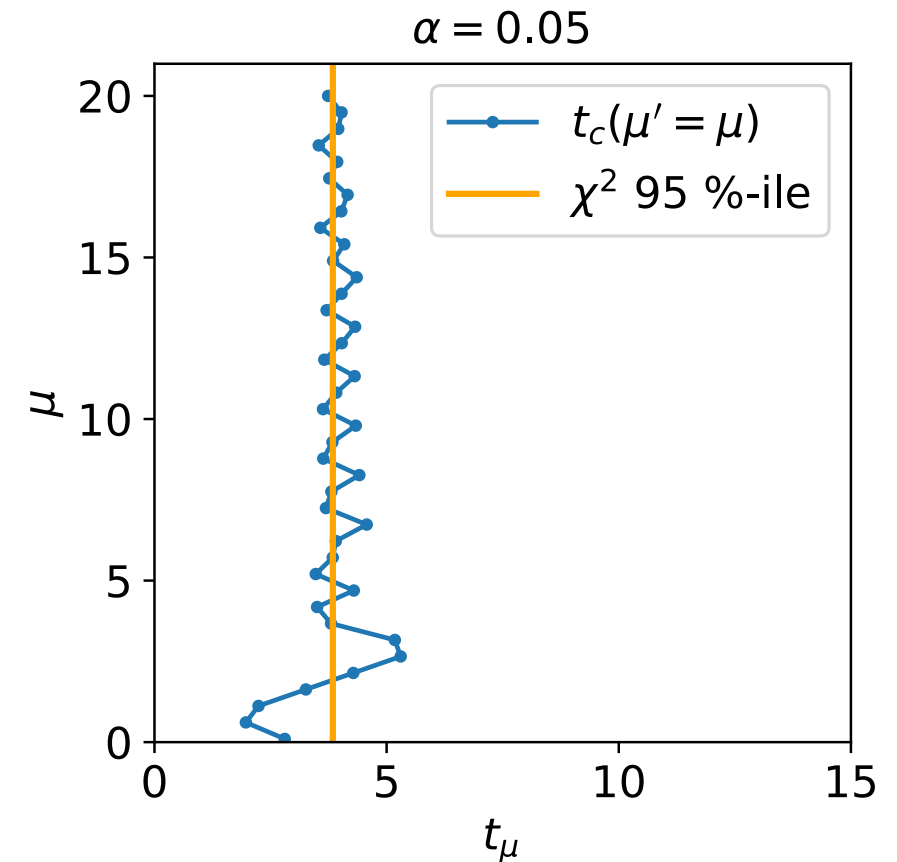
Finally, a combine command

```
$ cat card.txt
imax 1
jmax 1
kmax 0
-----
bin          bin1
observation    9
-----
bin          bin1  bin1
process      sig    bkg
process      0      1
rate         5      15
$ combine -M HybridNew --LHCmode LHC-limits card.txt
...
Limit: r < 1.10036 +/- 0.0189137 @ 95% CL
```



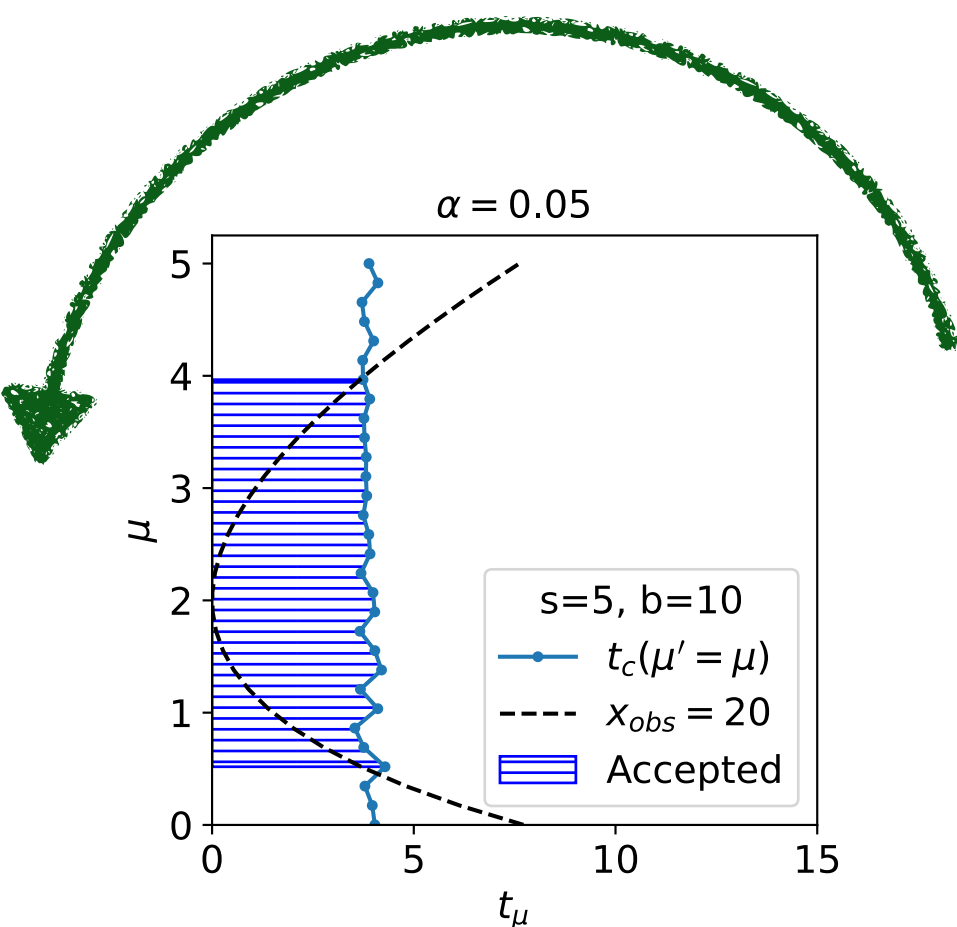
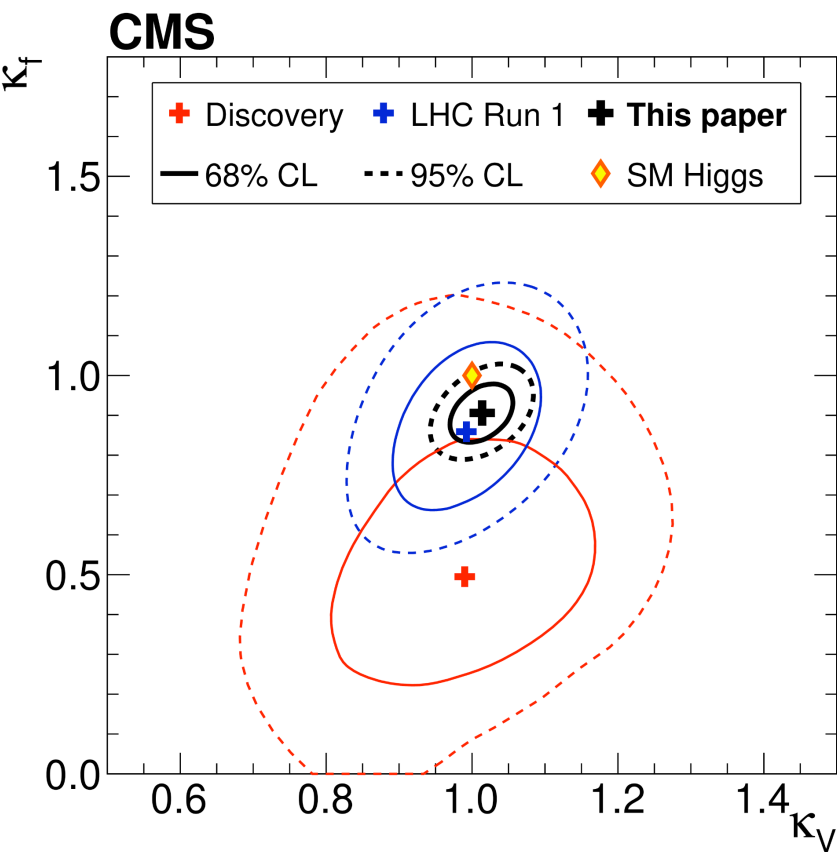
Asymptotic behavior

- Notice how $t_{\mu,c}$ and $q_{\mu,c}$ tend towards a constant?
- This is [Wilk's theorem](#) in action
 - Statement: as sample size grows, the distribution of the likelihood ratio $P(t_\theta|\theta')$ approaches a χ^2 distribution
 - With $df = \dim(\theta)$
 - Hence we can approximate by just evaluating $t_\theta(x_{\text{obs}})$!
- For q , formulas slightly more complex
 - [CCGV](#) provide the recipe: non-central half- χ^2
 - The non-centrality is found using the *Asimov* dataset
 - A special x_μ for a given μ such that $\hat{\mu}(x) = \mu$
 - Note for Poisson data, it may be non-integral!
 - This dataset produces the median expected limit



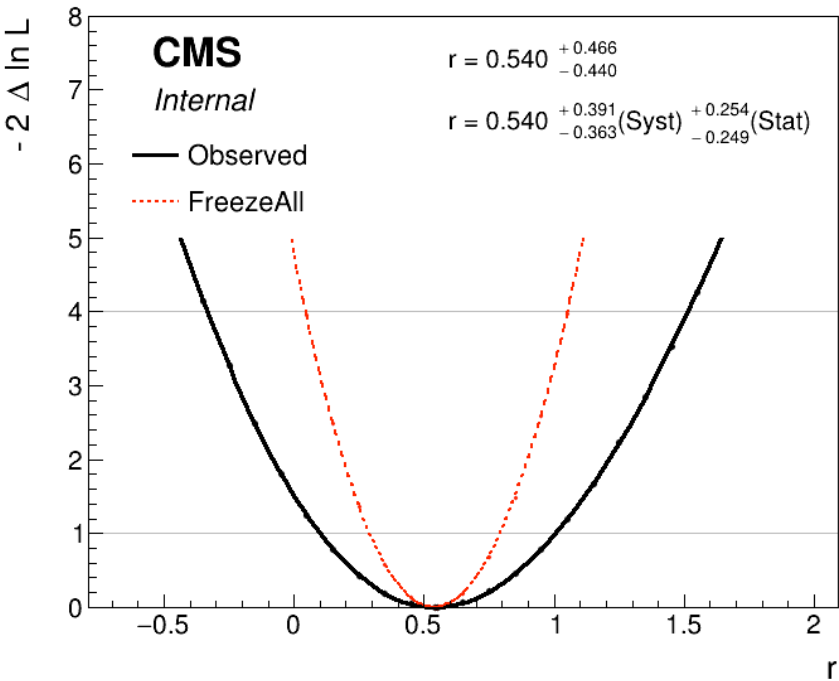
Asymptotic behavior

- This is how we make deltaNLL contours



```
scipy.stats.chi2.ppf(q, df)
```

Quantile	t_c	
	$df=1$	$df=2$
0.68	0.989	2.279
1σ (0.6827...)	1	2.296
0.95	3.841	5.991
2σ (0.9545...)	4	6.180



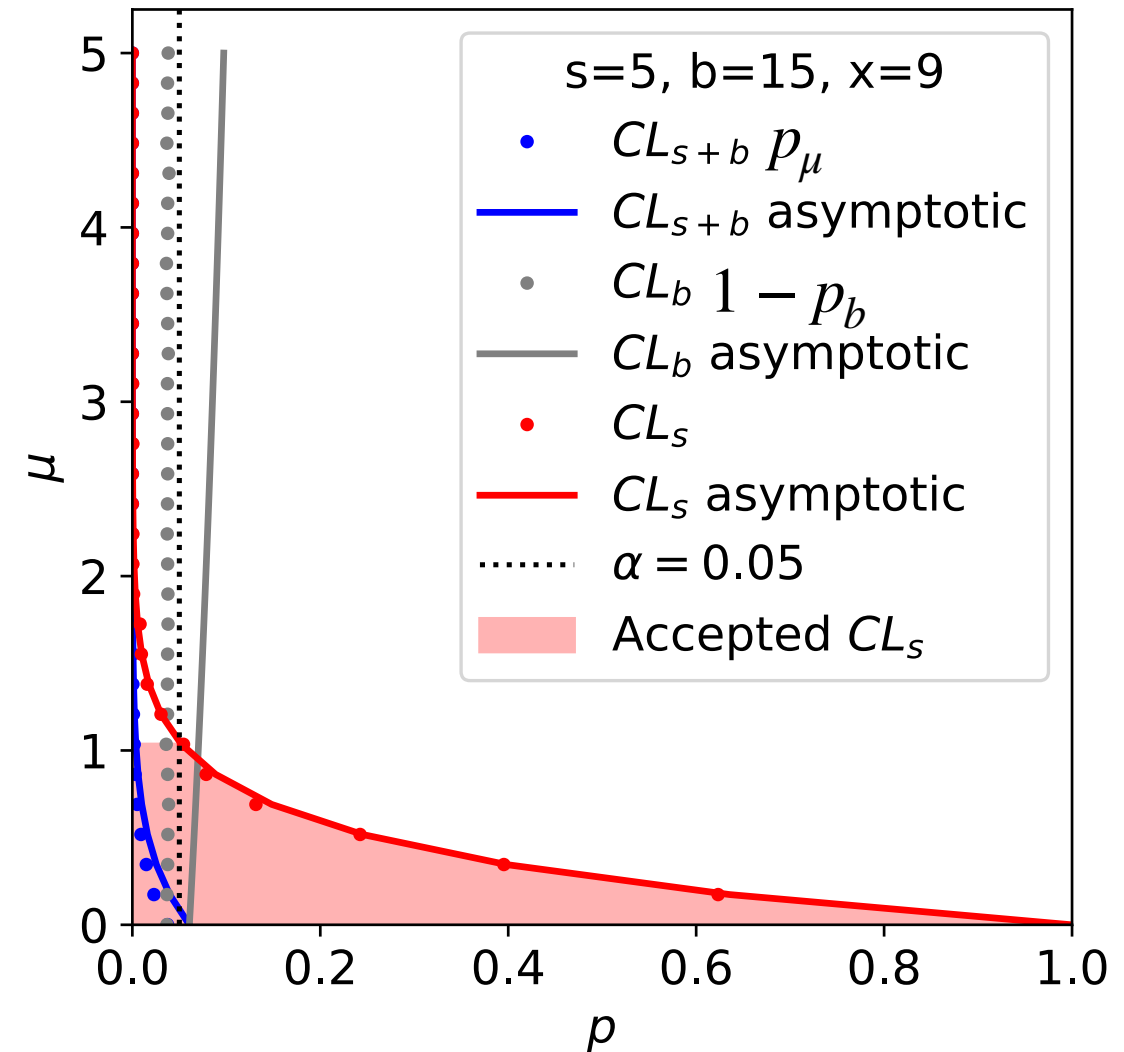
AsymptoticLimits in combine

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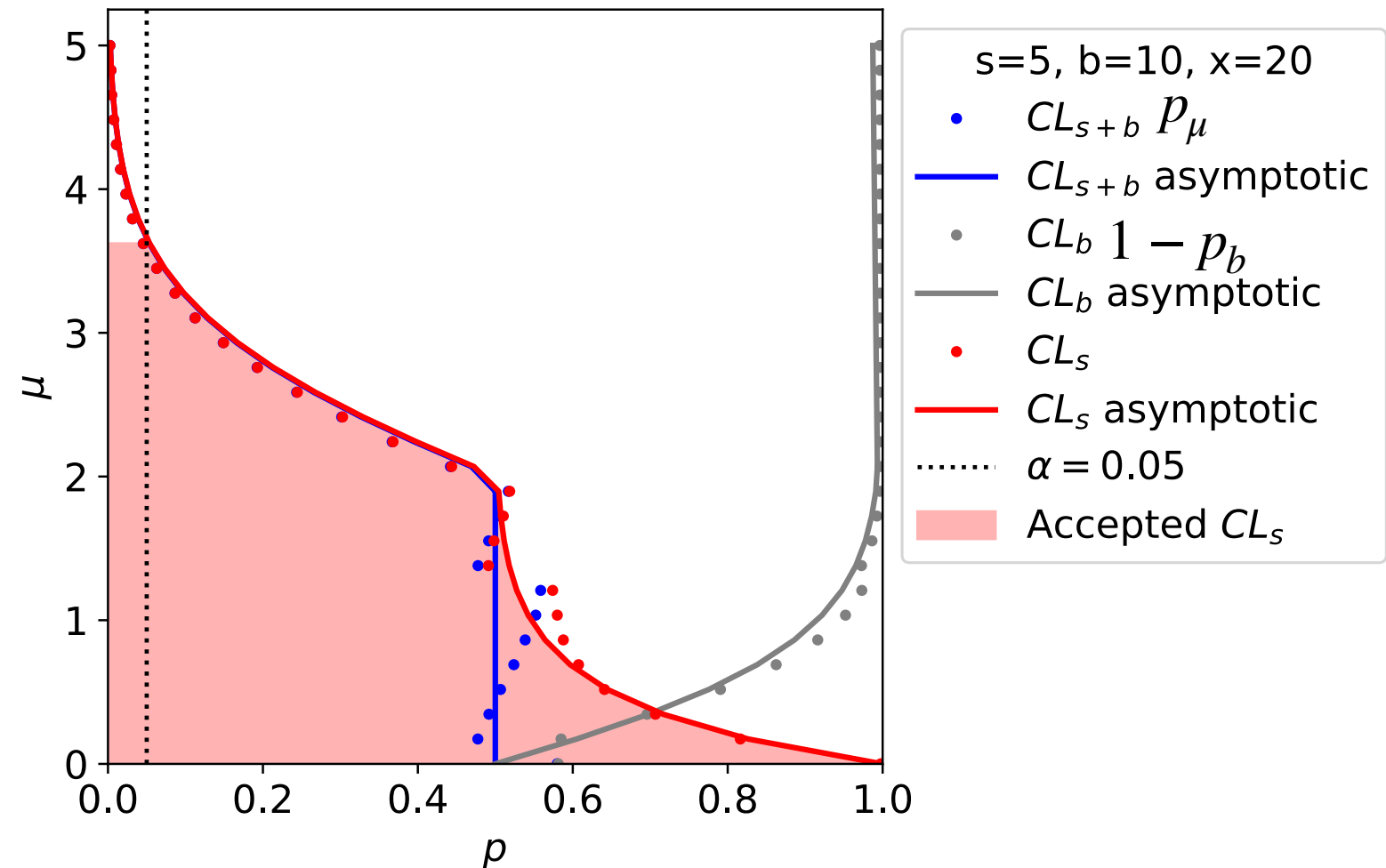
bin          bin1  bin1
process      sig   bkg
process            0    1
rate          5    15

$ combine -M AsymptoticLimits card.txt
...
-- AsymptoticLimits ( CLs ) --
Observed Limit: r < 1.0502
```



AsymptoticLimits in combine

```
$ cat card.txt
imax 1
jmax 1
kmax 0
-----
bin          bin1
observation   20
-----
bin          bin1  bin1
process      sig    bkg
process      0      1
rate         5      10
$ combine -M AsymptoticLimits card.txt
...
-- AsymptoticLimits ( CLs ) --
Observed Limit: r < 3.6595
```



Test statistic for discovery

- Poisson with background example

$$P(x | \mu s + b) = \frac{(\mu s + b)^x e^{-(\mu s + b)}}{x!}$$

- $s=20$, $b=15$ fixed, $x=39$

- Cannot use t_μ :

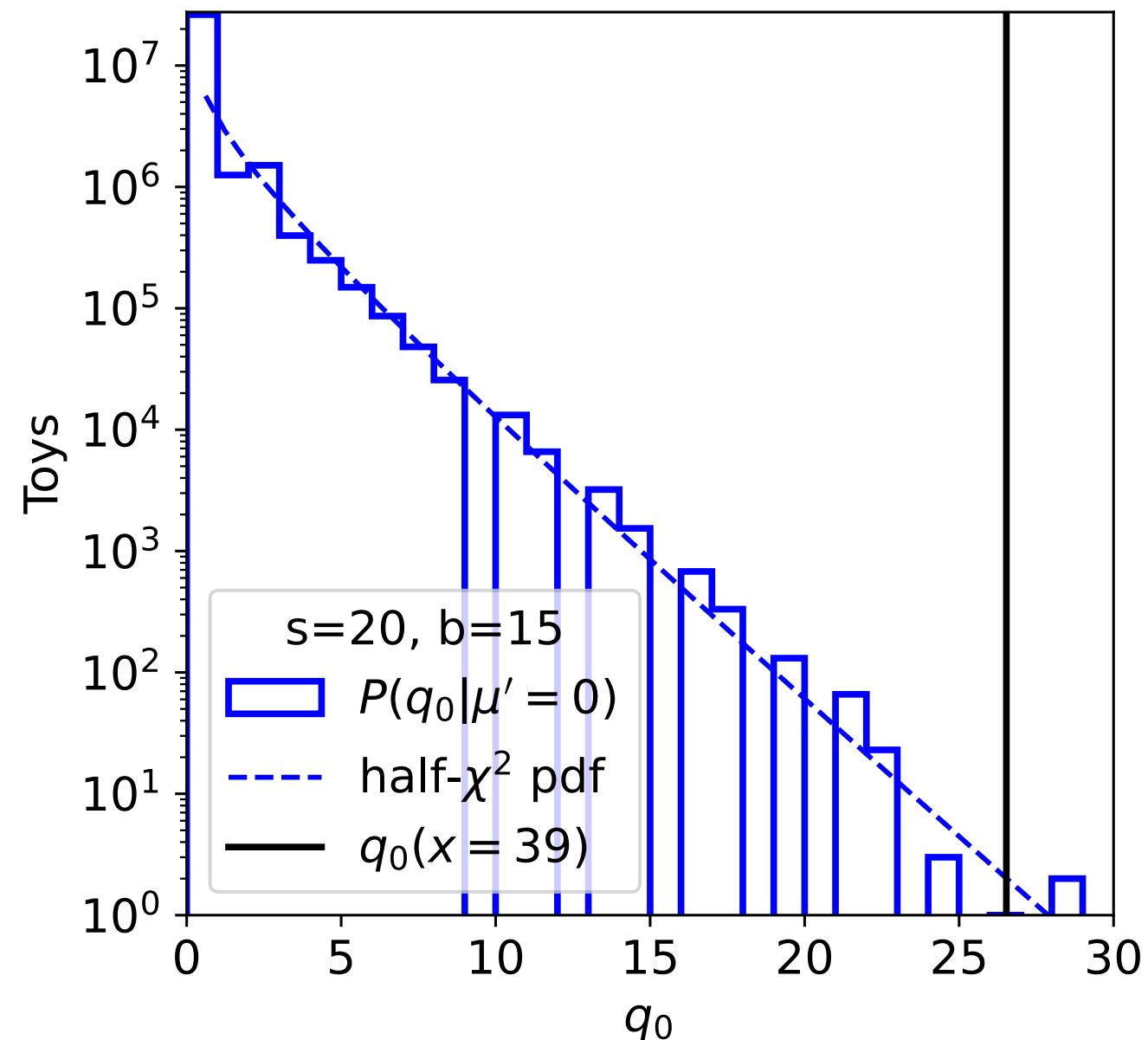
- Severe under-fluctuation would count as discovery! Certainly something was discovered, but not an excess over background. Disallow in test statistic:

$$\text{Define } q_0 = -2 \ln \frac{\mathcal{L}(0)}{\mathcal{L}(\max(0, \hat{\mu}))}$$

- i.e. under-fluctuations are “not extreme”

- Deceptively simple result: $Z = \sqrt{q_0(x_{obs})}$

- Only true if one POI



Adding uncertainties

Full statistical model

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- Renewed interest in publishing full statistical models: [arxiv:2109.04981](https://arxiv.org/abs/2109.04981)

Full statistical model

- Split likelihood parameters into *parameters of interest* (POIs) θ and nuisance parameters ν , and define *auxiliary measurements* y that target the latter
 - Then the joint pdf factorizes $P(x, y; \theta, \nu) = P(x; \theta, \nu)P(y; \nu)$
 - N.B. y are *global observables* in RooFit
- To frequentists, $P(x, y; \theta, \nu)$ can be used as a likelihood in (θ, ν)
 - Profile away ν -dependence: $\mathcal{L}_p(\theta; x, y) = \max_{\nu} \mathcal{L}(\theta, \nu)$
 - With $\mathcal{L}_p$ we can do all of what was shown before (in approximation)
- Bayesians can insert a *ur-prior* $\pi(\nu)$ and use Bayes' theorem to get $P(\nu | y)$
 - Then *marginalize* out ν : $P(x | \theta) = \int P(x | \theta, \nu)P(\nu | y) d\nu = \int P(x | \theta, \nu) \frac{P(y | \nu)\pi(\nu)}{P(y)} d\nu$
 - Proceed as before with $P(x | \theta)$
- Renewed interest in publishing full statistical models: [arxiv:2109.04981](https://arxiv.org/abs/2109.04981)
 - Enables recasting in either language

An uncertain background

- Adding some background uncertainty to our counting model
 - Let $f_P(n; \mu s + b)$ for some fixed s , b , and POI μ be a Poisson pdf
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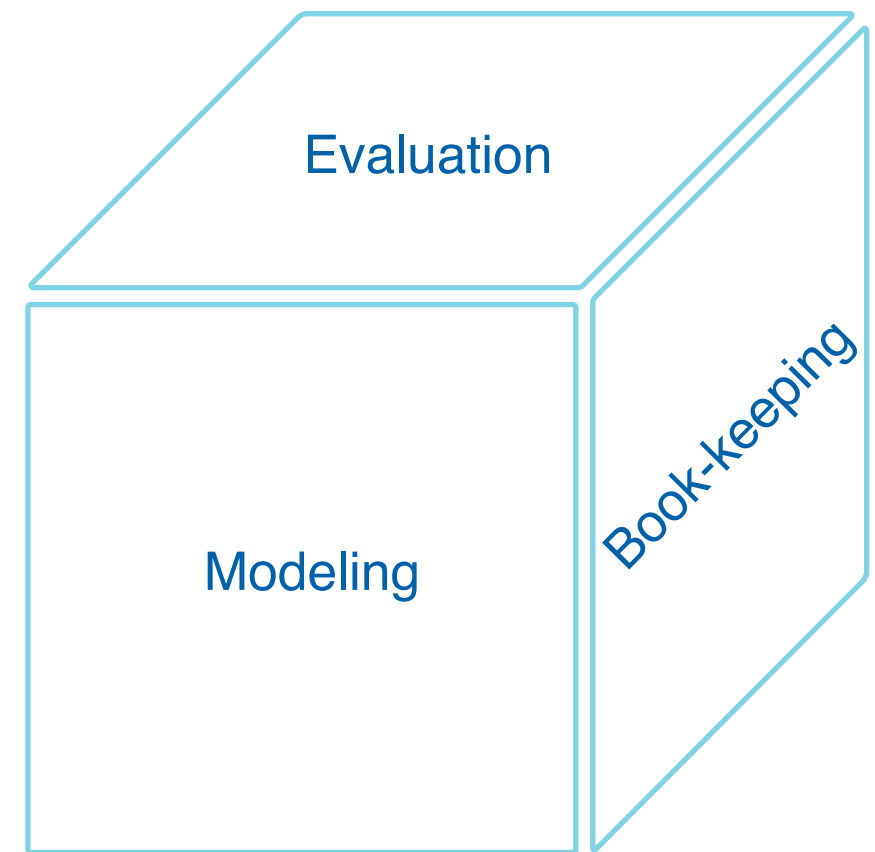
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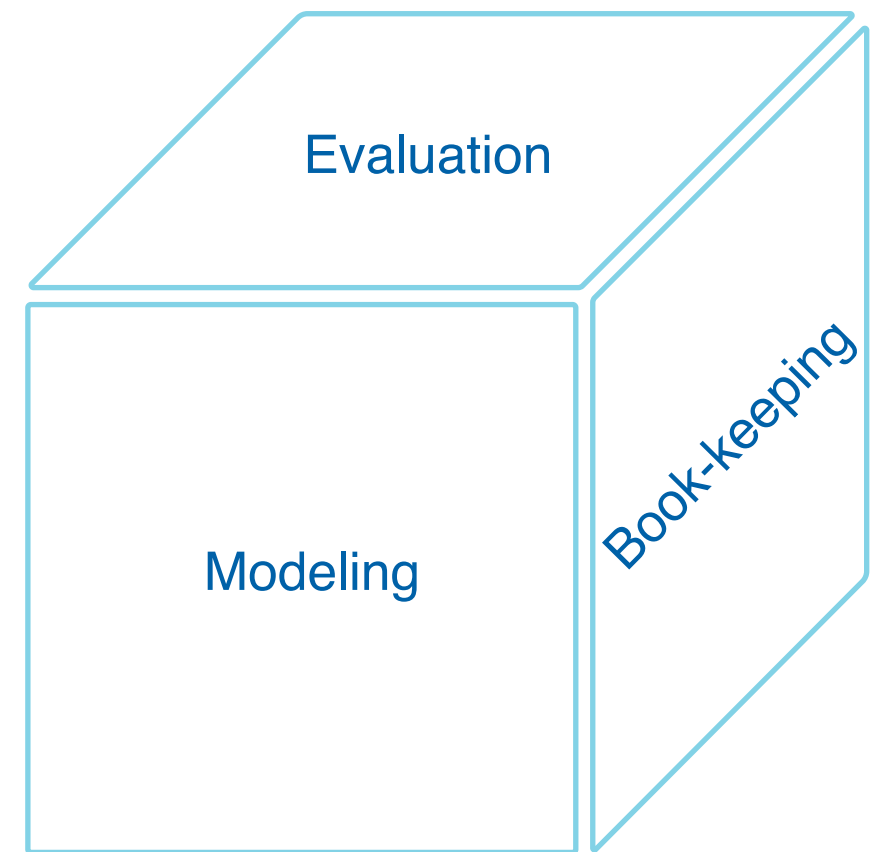
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- ...and many more
 - In all cases we now have a new observable (ν_{b0}, n_{cr}) , a new nuisance parameter ν_b , and several new constants (δ, κ, b_{cr}) to compute (e.g. from simulation)

Typical tasks in building a model



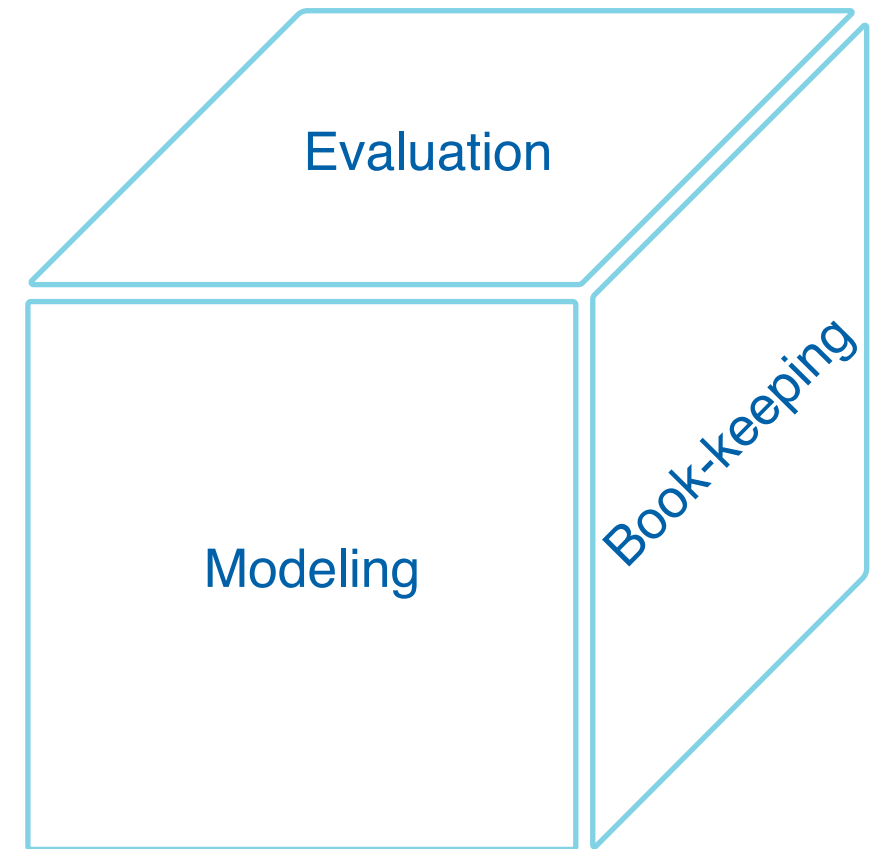
Typical tasks in building a model

- Enumerate effects to get dimension of ν
 - Don't forget anything! Unknown unknowns?



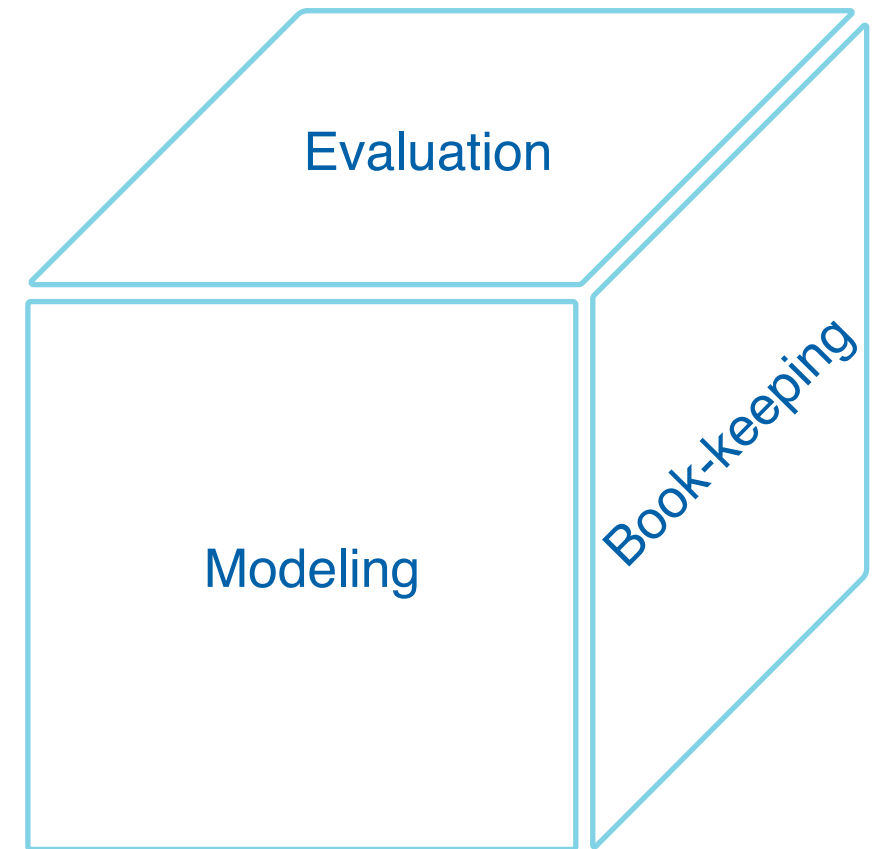
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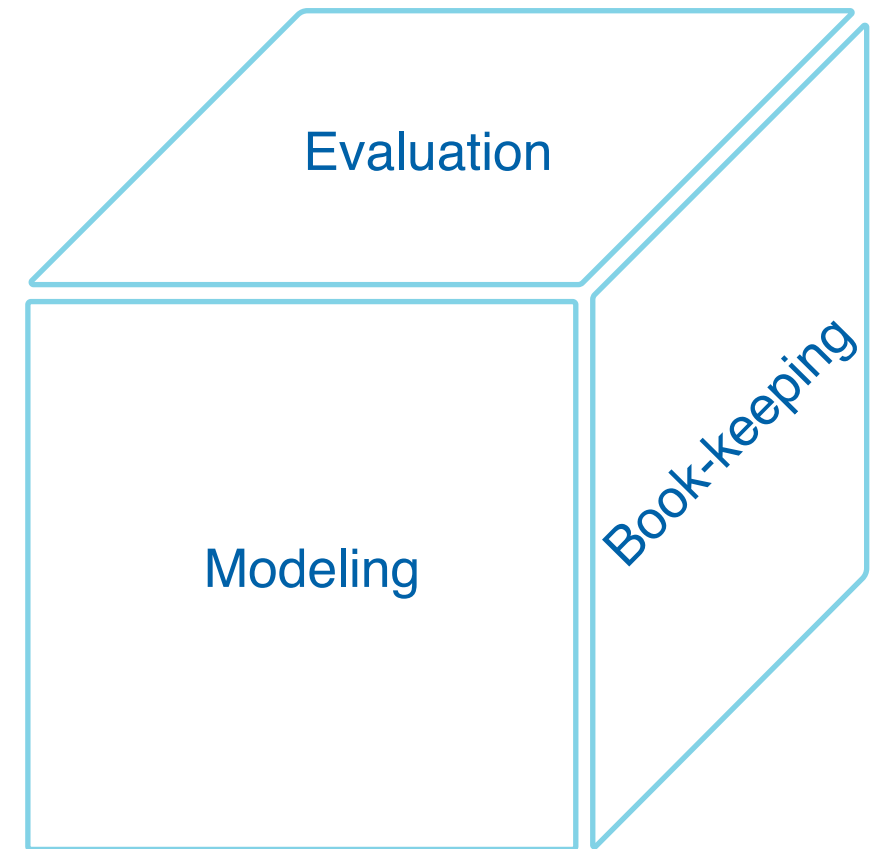
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 - e.g. the options 1-3 from before



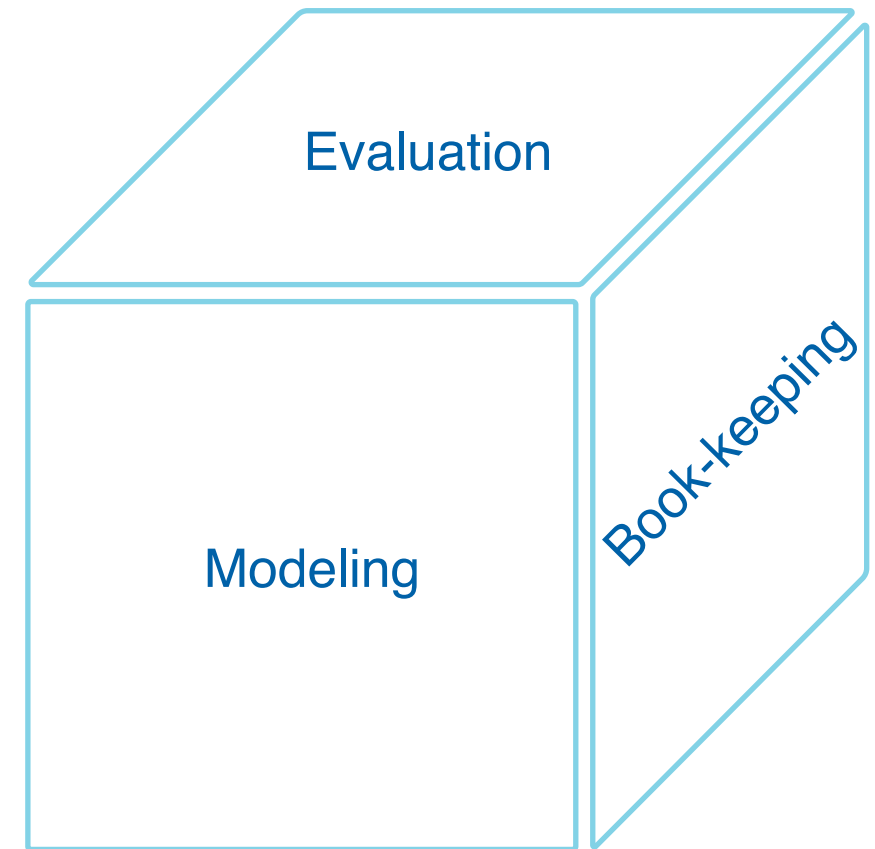
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 - In practice: interpolate between shifted or weighted MC



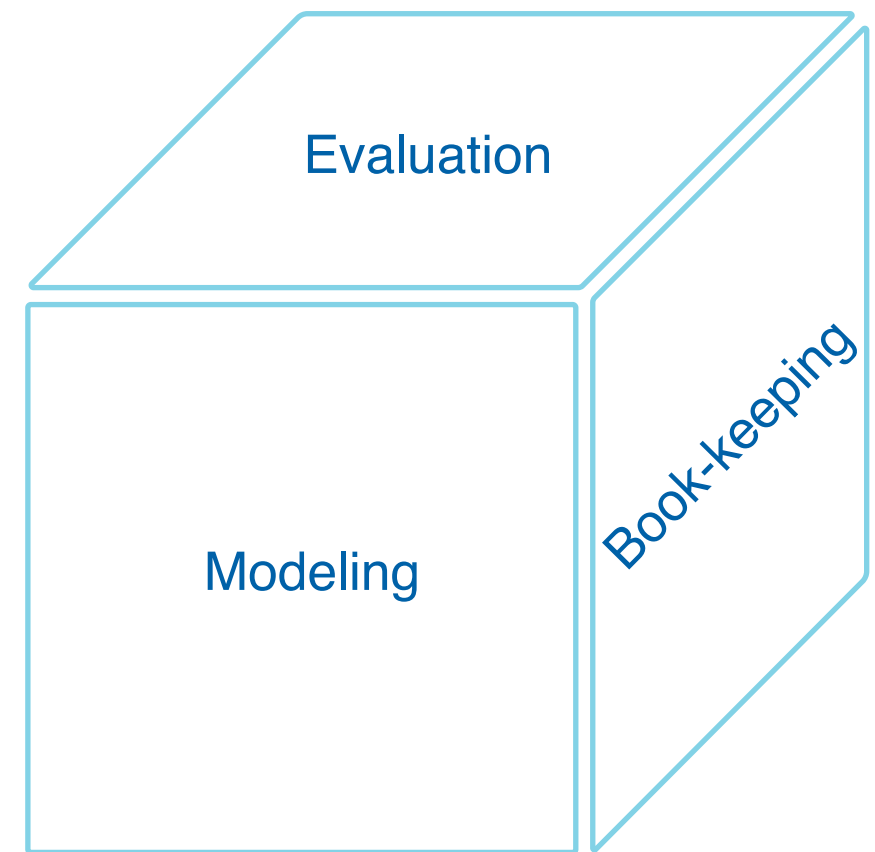
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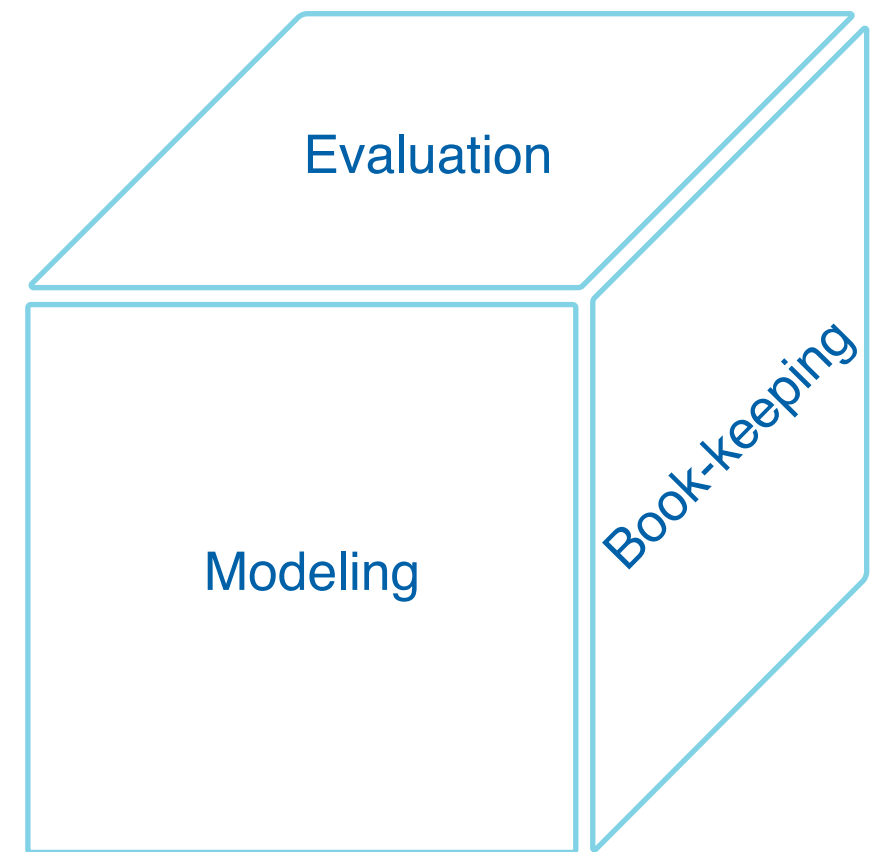
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- Iterate
 - Compromise: fidelity/computability/practicality
 - Prune low-impact effects
 - Initial model might not fit observed y well



Typical tasks in building a model

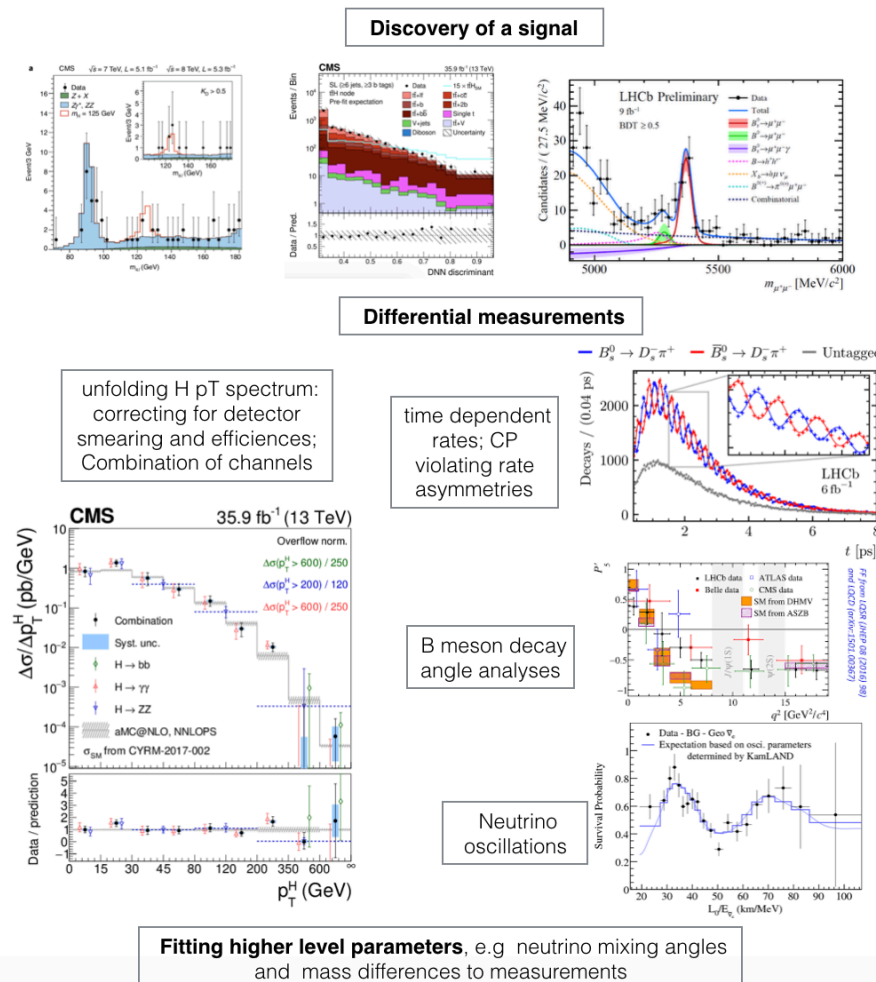
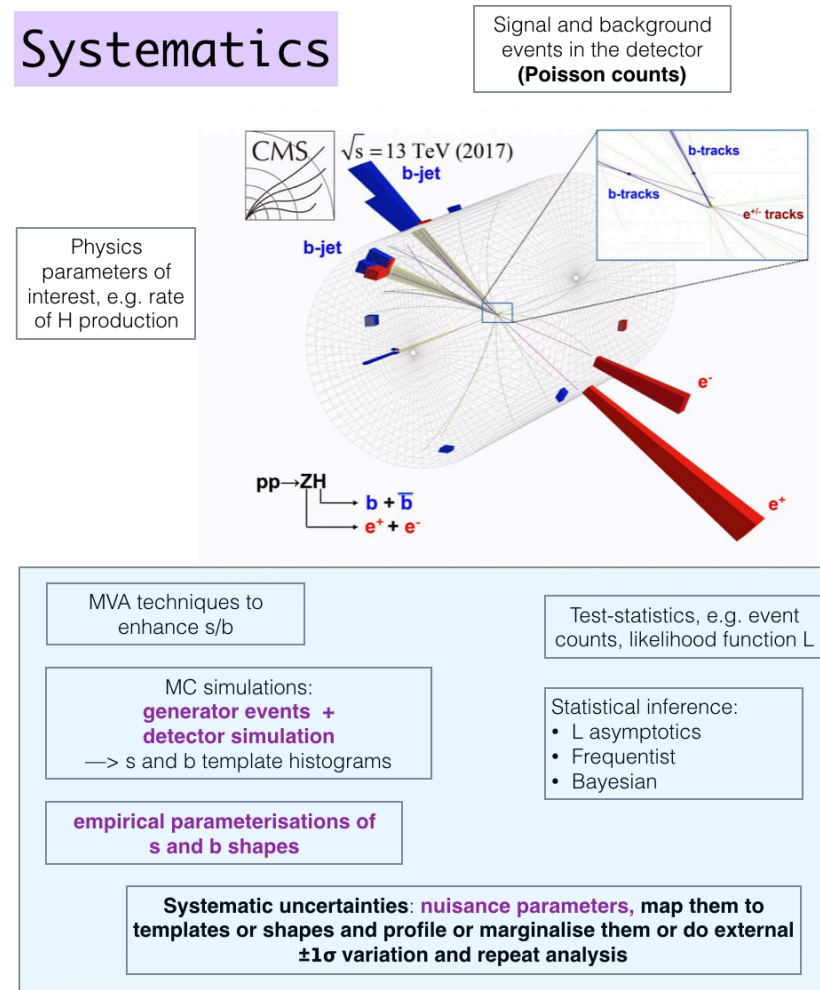
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 - Initial model might not fit observed y well
- A core feature of *Combine* is to simplify these tasks



Modeling techniques

- Can be a whole workshop
 - Was: [PHYSTAT-Systematics 2021](#)
 - Excellent presentations covering a wide range of techniques
 - A one-slide overview was [produced](#) (more on types than techniques)

Systematics



Nuisance pars:

Luminosity

Detector:

- Acceptance
- Efficiency for specific particles
- Energy scales
- Resolutions

Signal process template:

- Theory modelling uncertainties
- Limited MC statistics

Background processes template:

- Theory total cross section uncertainty
- Theory modelling uncertainties
- Limited MC statistics

Empirical s and b shape modelling;

- Parameterisations
- Non-parametric
- smoothing and morphing of MC templates

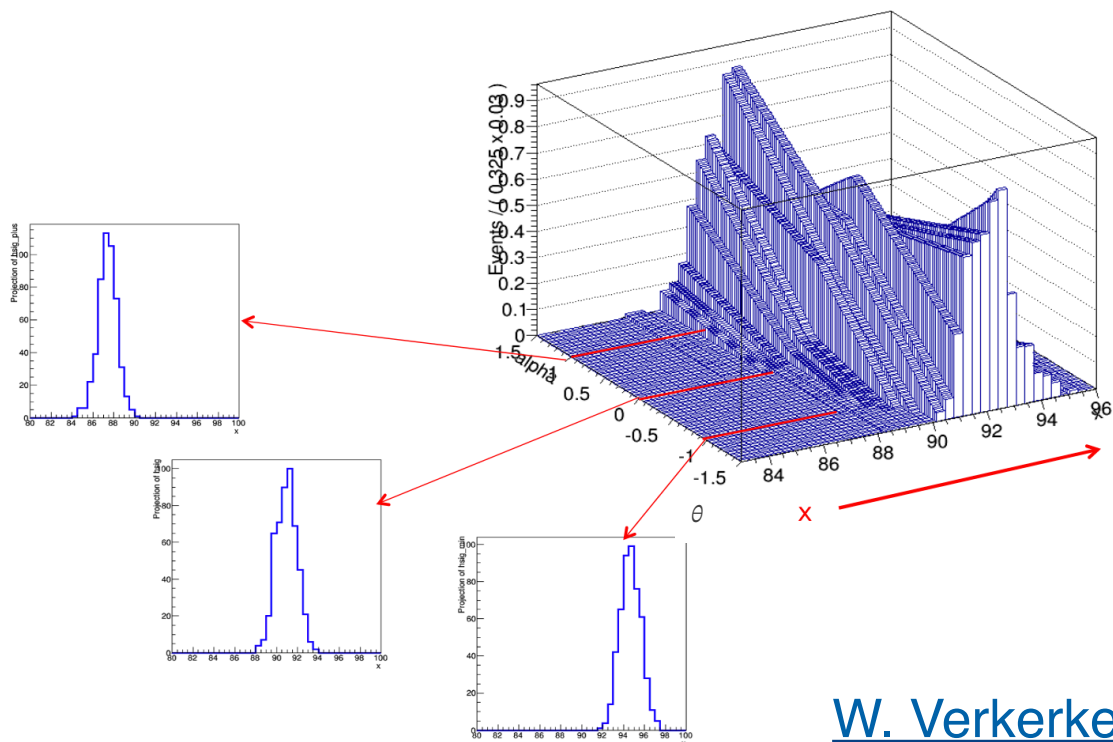
Nuisance parameters can be constrained from:

- Detector calibration data
- Control samples with different event selection
- from the data distributions
- measurements from other experiments
- theory calculations

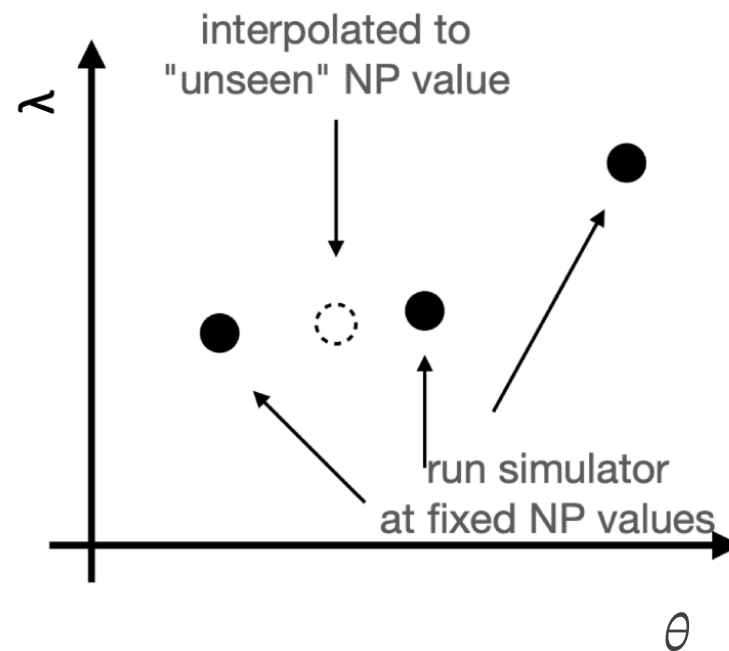
Modeling techniques

- **Rich** set of interpolation/extrapolation techniques at end-stage
 - Morphing: vertical, horizontal, moment; splines; gaussian process; asymmetric shift interpolation; additive/multiplicative effects; MC stat uncertainty, [BB-lite](#); ...
 - i.e. what is done in [RooFit](#)/[pyhf](#)/[zfit](#)/[iMinuit](#)/[combine](#)/etc.
 - What features do each of these tools offer? Nobody has it all!

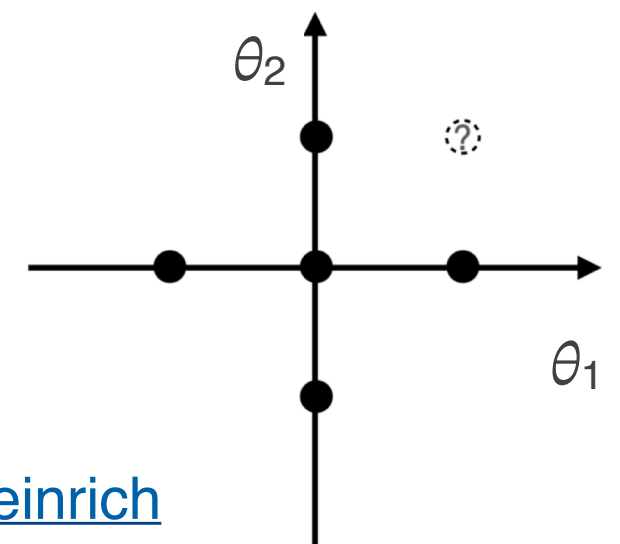
Visualization of bin-by-bin linear interpolation of distribution



[W. Verkerke](#)



Combining effects



[L. Heinrich](#)

The end

Hopefully you have some idea now what this means

“An observed (expected) upper limit is placed on the signal strength μ , using the profile likelihood ratio test statistic, following the CL_s criterion, under asymptotic assumptions, and found to be ...”

Additional references

- Procedure for LHC Higgs combination <http://cdsweb.cern.ch/record/1379837>
- R. Cousins, Statistics in Theory <https://arxiv.org/abs/1807.05996>
- Asymptotic formulae for likelihood-based tests “CCGV” <https://arxiv.org/abs/1007.1727>
- Publishing statistical models <https://arxiv.org/abs/2109.04981>